Technical guide

Universal XML Guide

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Date	Description of Change	Reference
09-04-2025	8.3.1.1 - Document Retrieval	WI00872692
19-Aug-2025	Update screen shots to CW Next and add '00' to times under Calculating Time Zones Where Required	WI00894875/ CS02039009
31-Oct-2025	Target Branch using Recipient Role Organization Address	WI00941179

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2. Universal XML Concepts

2.1 Overview

Universal XML schemas are CargoWise schemas that apply across multiple areas or modules in CargoWise. Unlike the Native XML schemas, which contain every piece of data available from the reference module only, Universal XML schemas contain information from multiple areas in CargoWise represented in a more generic and universal data structure.

The benefit of using a Universal XML schemas 'Universal' approach, is that through mapping your data to a single common schema, you can seamlessly import and export data across various modules or areas within CargoWise.

Universal XML schemas include all possible fields from all modules that have been mapped, so the schemas are quite comprehensive. At first glance, the schemas seem quite large, but we do not duplicate the same fields twice on any element, and we only include elements mapped to or used in a particular module when exporting data. Although the schemas are broad to cover all modules, the datasets are smaller than the schemas tend to indicate.

2.2 Universal Data types

- **Universal Shipment** - This can represent a Consolidation, Consolidation with Shipments, a Shipment, Shipment with Sub shipments, a Declaration, Shipment and Declaration pair, a Forwarding Booking, a Transport Booking, or any data type that can be related to a Freight Movement throughout CargoWise. The dataset includes Shipment data, Transport Legs, Commercial Invoices, Packing Information, and Customs Entries etc.
- **Universal Event** - These are used to record and publish events that occur in the life of a given job or entity. They contain an Event Type, the date the event occurred, or in the case of an estimated event, the date the event is estimated to occur on. They also contain a list of 'Context' values that identify what the event applies to and provide supplementary information about the event.
- **Universal Activity** – Represents the operational entities from the Productivity Tools modules including the Customer Service Ticket, Project, and Work Item modules.
- **Universal Transaction** - Represents any form of Accounting Transaction including Invoice, Credit Note, Payment, and Journal etc.
- **Universal Transaction Batch** - Is a batch of accounting transactions, contains a Batch Reference and a list of Accounting Transactions. Once generated can be requested by Batch Reference but not imported.
- **Universal Document Request** - This sends a query to return documents from jobs. A single UniversalEvent containing documents as Base64-encoded binary matching the parameters and filters enclosed in the message. Every module that supports exporting UniversalEvent XML works with the UniversalDocumentRequest.

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2.3 Code mapping

The Universal XML Engine has the capability to translate externally defined codes into the local codes used in your CargoWise system on the way in. It does this by allowing users to define a list of 'Code Mappings' from the codes the external system uses (aka: Foreign Codes) to the codes you use internally.

An example of where you could use this is where you have a partner sending you Universal Shipments containing Consols they want you to process. When they send you their XML messages, the Organizations they send would have their Organization Codes in them. They may be completely different to the codes you use for the same Organizations in your own system.

If you have entered Code Mappings from their codes to your codes before the file is imported, the correct Organizations will be saved in your CargoWise system even though the XML provided had their codes in it.

2.3.1 Functional overview

Once the benefits are fully understood, CargoWise users typically exchange electronic data with multiple partners. Each partner is known as a Data Provider as they provide data to be imported into CargoWise. Code mappings are configured per Data Provider along with other information you keep track of for your trading partners in the Organizations module.

Each Company in a CargoWise system will want the ability to determine their own set of Data have their own set of EDI Trading partners. For this reason, there are actually two stages of code mapping that occur. The first mapping is completed using the Target Company to map from the supplied Data Source Code to the Data Provider Organization, then all other codes are mapped using the mappings defined on the Data Provider Organization established in the previous step.

The following steps are followed:

1. Establish the Target Company for the XML being imported.
2. Check the XML to get the Data Provider Foreign Code and see if Code Mapping is enabled for the message. (Stop at this point if not enabled.)
3. Use the Code Mappings from the Target Company Organization Proxy to map from the Data Provider Foreign Code to find the Data Provider Organization.
4. Use the Code Mappings from the Data Provider Organization to map all Foreign Codes in the supplied to XML to the codes used in the target CargoWise system.

These steps are covered in more detail in the following sections. The process for entering Code Mappings against an Organization is the same for Steps Three and Four, so it is only described once using examples suitable for Step Four.

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2.3.2 Step One - Establishing the Target Company

The first step for Code Mapping is to establish the Company that is importing the XML data. There are two ways that can be used to determine the Target Company, channel specific and XML message specific. If the Target Company is specified in the XML message, that is what is used falling back to the channel specific Target Company.

2.3.2.1 Channel specific Target Company

Multiple channels can be used to import Universal XML into CargoWise, and each channel has a different way of determining the Company you are importing into.

2.3.2.1.1 Import from EDI Message

Universal XML provides System Administrators with the ability to queue XML files for processing via the EDI Messages module, using the 'Action' menu. This feature is intended solely for debugging purposes and should not be used in production workflows. When importing from a file, the target company is determined by the company the user is currently logged into.

2.3.2.1.2 Message via eHub

The target company is determined by extracting the middle three characters from the recipient's eHub ID. For example, if the eHub ID is WUTDAU1F2, the target company would be DAU.

2.3.2.1.3 Message via eAdaptor

When UniversalInterchange is used, the target company is identified by extracting the middle three characters from the RecipientID in UniversalInterchange header. If UniversalInterchange is not used, the target company is determined by the Company.Code specified within the DataContext section of the Universal XML.

For Universal XML, the DataContext is included in the first top level element as follows.

```
<UniversalShipment xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
  <Shipment>
    <DataContext>
      <DataTargetCollection>
        <DataTarget>
          <Type>AsycudaManifest</Type>
          <Key>MAN0000066</Key>
        </DataTarget>
      </DataTargetCollection>

      <Company>
        <Code>DAU</Code>
      </Company>
      <EnterpriseID>HYE</EnterpriseID>
      <ServerID>CMT</ServerID>
      <DataProvider>HYECMTDAU</DataProvider>
    </DataContext>

    <!-- content removed for the sake of brevity -->
  </Shipment>
</UniversalShipment>
```


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The Target Company will also be used for any other functionality where Company context is required during the import process.

Refer to section 2.4 DataContext Element for more information on how to use the data context element when importing and exporting Universal XMLs

2.3.3 Step Two - Reading the Data Provider Foreign Code

Universal XML Messages allow the sending party to include an element with a Code in it used to identify the Data Provider Organization. This is called the Data Provider Foreign Code.

All forms of Universal XML have a DataContext element directly contained in the top-level element. In the DataContext element, there are two elements that are used for Code Mapping.

- DataProvider - Contains the Data Provider Foreign Code for Code Mapping.
- CodesMappedToTarget – Optional Boolean element that can be used to disable Code Mapping in Universal XML. If 'true', no code mapping will occur.

You can see the DataProvider element in the sample XML provided on the previous page. CodesMappedToTarget is included at the same level as the DataProvider element where required.

2.3.4 Step Three - Mapping the Data Provider Foreign Code to an Organization

If a Data Provider Foreign Code is included in the XML, and Code Mapping is enabled according to the elements in the provided XML, CargoWise finds the Company Organization Proxy for the Target Company established in the previous step and reads the Code Mappings defined on that Organization. If there is a Code Mapping present with a Foreign Code matching the Data Provider Foreign Code which points towards an organization, then that becomes the Data Provider Organization, and Code Mapping will proceed.

2.3.5 Step Four - Mapping from Foreign Codes to Local Codes

Once the Data Provider Organization has been established in the previous step, the EDI Code Mappings are read from that Organization and used to translate the Foreign Codes to Local Codes as the rest of the XML is imported. Instructions on how to configure code mappings follow.

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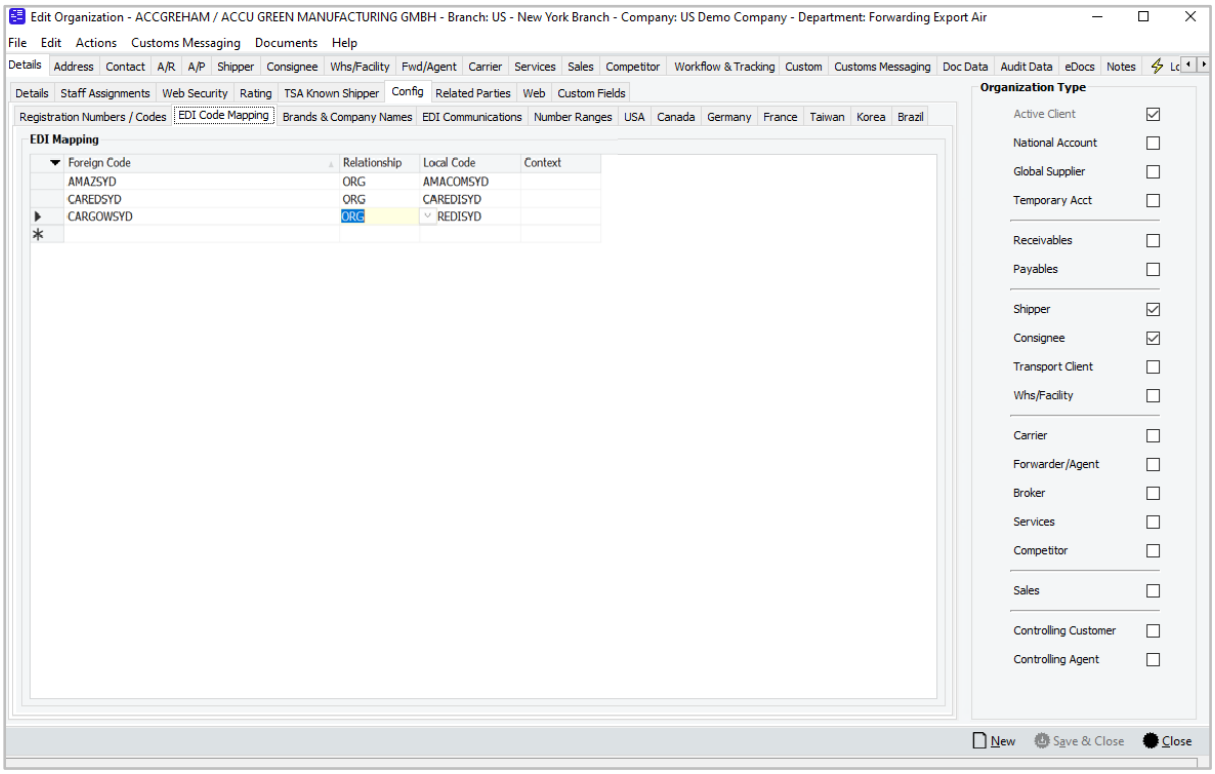
2.3.6 Configuring Mappings for an Organization

Go to the **Maintain > Master Data > Organization** module and open an organization. Navigate to the **Details > Config > EDI Code Mapping** sub tab where you will see the EDI Mapping grid. The Foreign Code field represents the codes as supplied by the Data Provider Organization. The Relationship drop-down menu allows you to select the type of code being mapped, which in turn identifies the related module. The Local Code field is the internal code that the Foreign Code will be mapped to in your CargoWise system.



As shown in the screenshot above, if the local organization code for the data provider in the imported XML is AUSBEAMEL, then organization codes like AMAZSYD, CAREDSYD, and CARGOWSYD would be translated to AMACOMSYD, CAREDISYD, and CAREDISYD respectively during import. CargoWise supports many types of coded list translations during XML import. To view the full list of available mapping types, click the dropdown menu in the Relationship column.

Refer to our learning unit: 1COR224 - How do set up EDI Code Mapping?

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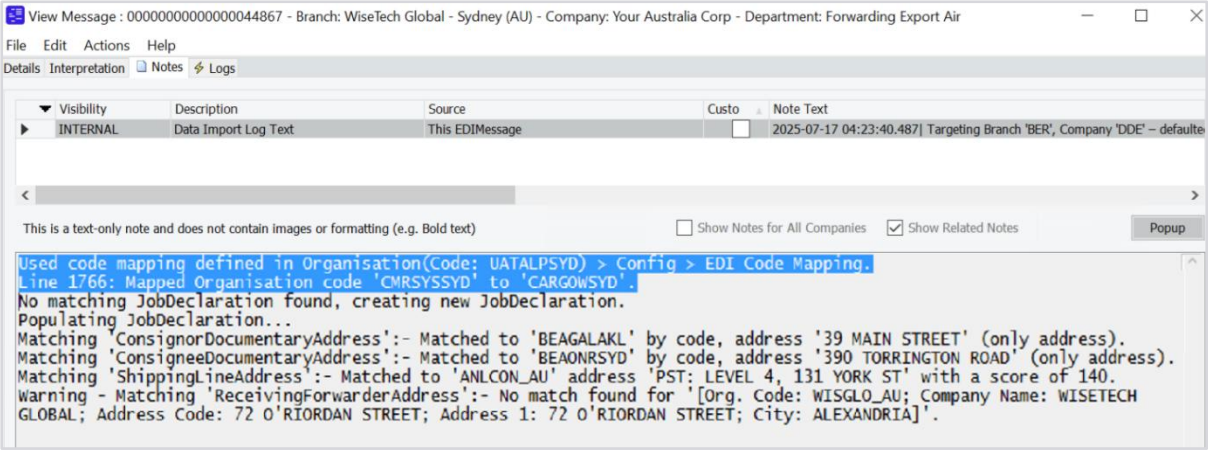
2.3.7 Evidence Code Mapping has taken place.

With Universal XML, you can tell if Code Mapping occurred on an individual message as codes that are successfully mapped are recorded in the Data Import Log. You can see this log by opening the EDI Message that the Universal XML was imported from and clicking on the Notes tab to show the Data Import Log Note as shown below.



You can see that the Data Provider Organization used for code mapping is identified, and all codes mapped are shown as well.

Refer to our learning unit: 1COR302 – How do I use the EDI Messages Notes tab to understand how a job was created?

2.4 DataContext element

Universal XML schemas have a DataContext element at the top. This is used to describe where the Universal Data came from when exporting XML and is used when importing XML to describe how the data should be applied within CargoWise. This element is optional for data coming into CargoWise but is always present when data is exported from CargoWise. Shown below is the DataContext element from a Universal Event exported from a Forwarding Shipment:

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```xml
<DataContext>
  <DataSourceCollection>
    <DataSource>
      <Type>ForwardingShipment</Type>
      <Key>S00001006</Key>
    </DataSource>
  </DataSourceCollection>

  <ActionPurpose>
    <Code>APP</Code>
    <Description>As Per Payload</Description>
  </ActionPurpose>
  <Company>
    <Code>SAM</Code>
    <Name>Sample Company Inc.</Name>
  </Company>
  <EnterpriseID>CGW</EnterpriseID>
  <EventType>
    <Code>AED</Code>
    <Description>All Export Documents Received</Description>
  </EventType>
  <ServerID>SYD</ServerID>
  <TriggerCount>1</TriggerCount>
  <TriggerDate>2011-06-07T09:46:17.433</TriggerDate>
  <TriggerDescription>EXPORT DOCS RECEIVED</TriggerDescription>
  <TriggerType>Trigger</TriggerType>
</DataContext>
```

The elements shown above are broken into functional groups as described in the following sections.

2.4.1 DataContext when exporting XML

When Universal Data is sent from a CargoWise system, the DataContext element contains enough information to uniquely identify the entity being exported globally. To do this, you use a combination of elements included in the DataContext to make up a globally unique key as follows:

- **EnterpriseID** - Three-character code identifying a group of companies globally.
- **ServerID** - Three-character code identifying a server used by a group of companies. The combination of EnterpriseID + ServerID gives you a globally unique code for a CargoWise installation.
- **Company.Code** -Three-character code identifying a company registration on a CargoWise installation. For modules where there are separate job number streams per company, you would need to include this code as part of the key. For global modules like Forwarding, this would technically not be required.
- **DataSourceCollection** - Elements in the DataSourceCollection contain DataSource Types and Keys identifying which job or entity the Universal Data was generated from. The Type identifies the CargoWise module; the Key is a job or entity reference unique to the CargoWise installation.

The combination of these elements can be used to globally identify the source of any Universal XML. The Enterprise ID and ServerID uniquely identify a particular system, and the Company Code plus DataSource identify the source entity within that system.

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2.4.1.1 DataSourceCollection elements

In XML exported from CargoWise, the first element in the DataSourceCollection always identifies the source of the data for the Top Level Universal Shipment. This first element is always added by CargoWise on any Universal Shipment or Event.

If there is only one DataSource in the collection, then the Universal Shipment contains data from that data source in CargoWise including all child and related Universal Shipments. A good example of this would be a Forwarding Consolidation with a collection of all its child Forwarding Shipments in the SubShipmentCollection.

Additional DataSource elements can be added on the Top Level DataContext element. (i.e., not in DataContexts on Universal Shipments in the SubShipmentCollection) When this happens, it means that a subset of data has been exported from CargoWise, not a complete set. The additional DataSource elements identify the DataSource (or DataSources) of SubShipments that contain complete data within the set.

An example of where this is used is for Forwarding Shipments where there is a parent Consolidation. When a Forwarding Shipment is exported, the Top Level Universal Shipment will be created using data from the Consolidation. The Forwarding Shipment itself will be included as a Universal Shipment in the SubShipmentCollection. Any other Forwarding Shipments on the Consolidation will not be included in the SubShipmentCollection.

In this case, the DataSourceCollection on the Top-Level Universal Shipment contain a reference to the Consolidation and the Forwarding Shipment, and the DataSourceCollection on the Universal Shipment in the SubShipmentCollection will contain a reference to the Forwarding Shipment. This setup allows you to tell which Shipments, Sub-Shipments, or even Sub-Sub-Shipments contain complete data including all children.

The advantage of working this way is that when processing the incoming data, you know that you always have all parent information. It doesn't matter if you export from a Consolidation, a Shipment, or a Sub-Shipment, you always work with the same top-down structure starting at the top and including everything relevant to the level you are exporting. In the words of our Architect, “It's Shipments all the way down”.

2.4.1.2 Workflow Trigger Action Information

As well as showing what CargoWise system and which Job/Entity in CargoWise data comes from, the DataContext also has information about the Workflow Trigger Action that caused the export of Universal XML. The following elements are completed from the Trigger Action configured to send Universal XML and its parent Milestone, Trigger, or Exception:

- **<ActionPurpose>** - This is taken from the Purpose field on the Trigger Action. This element includes sub elements containing the Purpose Code and its associated description. New Purpose Codes can be defined in the Registry under **Workflow Manager > Trigger Purpose** and allow you to include an Intended Use or Purpose for the data you are sending.

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- **<EventType>** - On the Milestone, Trigger, or Exception, there is an Event field where you can enter an Event Code. This element includes sub elements containing the Event Code and its associated description.
- **<EventUser>** - The user who added the Event that tripped the Trigger or Milestone. Allows you to know who performed the action that resulted in the Universal XML being sent.
- **<TriggerCount>** - Contains the number of times the Milestone, Trigger, or Exception has been fired so far. The count starts at '1' and increments for every export so you have a Send Count or Data Version Number in the receiving system.
- **<TriggerDate>** - Event time of the Triggering Event.
- **<TriggerDescription>** - The Description field from the associated Milestone, Trigger, or Exception.
- **<TriggerType>** - Contains either Milestone, Trigger, or Exception depending on the type of Workflow entity the Trigger Action is defined against. TriggerType will be "Manual" when the XML get generated by a user action using "Send Universal XML" option in CW.
- **<Company>** - Is the Company from the Trigger or Milestone that caused the Universal XML to be sent. Each Trigger or Milestone in CargoWise has a Company Code against it in the grid. For more information on Triggers and Milestones.

Refer to Workflow Trigger FAQ and our other Workflow learning units for more information on the Workflow entities mentioned above.

2.4.2 DataContext when importing XML

When importing Universal XML into CargoWise, the DataContext is used to describe how the data should be applied within CargoWise. The content of the DataContext is used to determine:

- The Company to import the XML under
- The Module(s) to import the XML into
- Which Job/Entity to Update
- Code Mapping options

2.4.2.1 Targeting a Company

To target a Company, you need to identify the Company in the DataContext using the following elements:

- **EnterpriseID** – Three-character code identifying a group of companies globally.
- **ServerID** – Three-character code identifying a server used by a group of companies.
- **Company.Code** – Three-character code identifying the company you want to target within a CargoWise installation.code

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The combination of EnterpriseID + ServerID + Company Code provides a globally unique code for a particular Company within the target CargoWise installation. The Company Code is ignored unless all three values are provided, and they match to a valid Company registered within the target system.

To view your EnterpriseID, ServerID and Company.Code, from within the Jump screen, click **Help > About.** These three codes will display in the above order of EnterpriseID & Company.Code & ServerID

2.4.2.2 Targeting a Branch

2.4.2.2.1 Universal Shipment

When importing a Universal Shipment, there are a series of fallbacks used to select the branch used when processing the XML content.

1. Match using the Branch.Code element from the Universal Shipment.
2. Match the 'Recipient' OrganizationAddress to a Branch Org Proxy.
3. Match the 'Recipient' OrganizationAddress Port to a Branch Port/Additional Related Port.
4. Match the recipient role's OrganizationAddress to a Branch Org Proxy.
5. Match the recipient role's OrganizationAddress to a Branch Port/Additional Related Port.
6. Match using a Recipient Role and Port Code.
7. Default to the first active branch of the target Company.

When using port matching options, the port must exist within the company context for the match to be valid.

Options 1–3 are best suited for sending Universal Shipments from middleware applications via eAdaptor to CargoWise when you have control over which elements are included in the XML.

Options 4–6 provide the most comprehensive functionality for the CargoWise-to-CargoWise (E2E) integration method, as the recipient has no control over the XML content.

Option 7 serves as a fallback when none of the first six options can be applied.

The selected branch impacts processes such as Job Number generation, particularly when the system is configured to incorporate the creating branch into the Job Number.

See How do I customize my Job Numbers?

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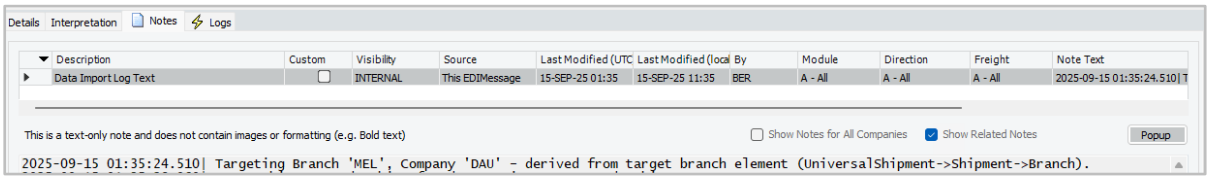
Option 1 – Match using the Branch.Code element from the Universal Shipment

If the target Branch.Code is included in the Universal Shipment XML, and the Company.Code, ServerID, and EnterpriseID in the DataContext element match the target system, and the Branch.Code corresponds to an active branch for the Company.Code, then that branch will be used during XML import, otherwise it will fallback to the next option.

For example,

```
<UniversalShipment xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
    <Shipment>
        <DataContext>
            <DataTargetCollection>
                <DataTarget>
                    <Type>ForwardingBooking</Type>
                </DataTarget>
            </DataTargetCollection>
            <Company>
                <Code>DAU</Code>
            </Company>
            <EnterpriseID>WUT</Enterprise>
            <ServerID>A6F</ServerID>
        </DataContext>
        <Branch>
            <Code>MEL</Code>
        </Branch>
        <!-- content removed for the sake of brevity -->
    </Shipment>
</UniversalShipment>
```

EDI Message Data Import Log shows it is Targeting Branch 'MEL'



For the function to work correctly, Branch.Code must be placed outside the DataContext element in the Universal Shipment XML.

Option 2 – Match the 'Recipient' OrganizationAddress to a Branch Org Proxy

CargoWise looks for an OrganizationAddress with an AddressType of 'Recipient' within the Shipment.OrganizationAddressCollection element. The 'Recipient' AddressType is not generated in messages sent from CargoWise but can be added using middleware. This allows the targeted CargoWise system to know the intended recipient branch.

When importing the Universal Shipment, CargoWise attempts to match the 'Recipient' OrganizationAddress to an organization within the target system. If a match is found via Code Mapping or Address Matching, and the matched organization is configured on a branch as that branch's Org Proxy, that branch will be used when importing the Universal Shipment, otherwise it will fallback to the next option. If multiple or no match found, the system will fallback to the next option.

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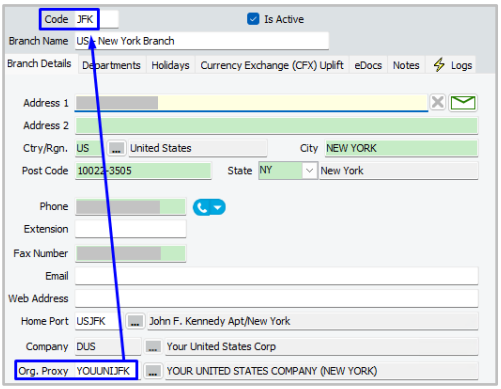
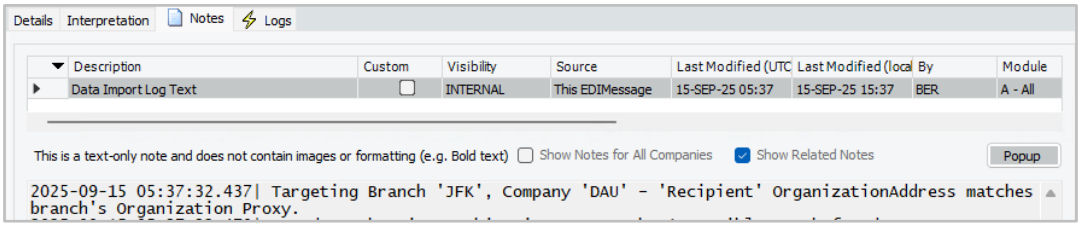
For Example,

```
<UniversalShipment xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
    <Shipment>
        <DataContext>
            <DataTargetCollection>
                <DataTarget>
                    <Type>ForwardingBooking</Type>
                </DataTarget>
            </DataTargetCollection>
            <Company>
                <Code>DAU</Code>
            </Company>
            <EnterpriseID>WUT</EnterpriseID>
            <ServerID>A6F</ServerID>
        </DataContext>
        <!-- content removed for the sake of brevity -->
        <OrganizationAddressCollection>
            <OrganizationAddress>
                <AddressType>Recipient</AddressType>
                <OrganizationCode>YOUUNIJFK</OrganizationCode>
            </OrganizationAddress>
            <!-- content removed for the sake of brevity -->
        </OrganizationAddressCollection>
    </Shipment>
</UniversalShipment>
```

YOUUNIJFK is the Org.Proxy for Branch JFK



EDI Message Data Import Log shows it is Targeting Branch 'JFK'



Option 3 – Match the 'Recipient' OrganizationAddress Port to a Branch Port/Additional Related Port

When the 'Recipient' OrganizationAddress is included using middleware as described in Option 2, and there is a UNLOCO specified, that UNLOCO will be used to try and locate a branch with a matching 'Home Port'. Branches can also have 'Additional Related Ports' entered on them indicating

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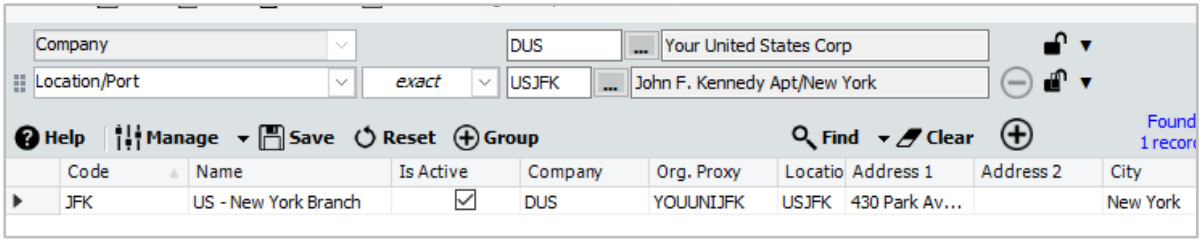
other ports also handled by that Branch. These are also used when trying to locate a matching Branch. If multiple matches found, system will randomly pick one of the matched branches; if none found, it will fallback to the next option.

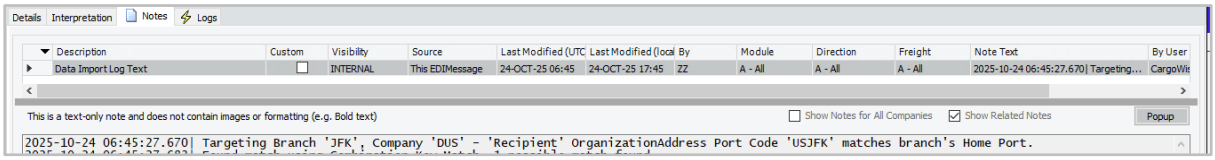
For Example,

```
<UniversalShipment xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
    <Shipment>
        <DataContext>
            <DataTargetCollection>
                <DataTarget>
                    <Type>ForwardingBooking</Type>
                </DataTarget>
            </DataTargetCollection>
            <Company>
                <Code>DUS</Code>
            </Company>
            <EnterpriseID>WUT</EnterpriseID>
            <ServerID>A6F</ServerID>
        </DataContext>
        <!-- content removed for the sake of brevity -->
        <OrganizationAddressCollection>
            <OrganizationAddress>
                <AddressType>Recipient</AddressType>
                <Port>
                    <Code>USJFK</Code>
                </Port>
            </OrganizationAddress>
            <!-- content removed for the sake of brevity -->
        </OrganizationAddressCollection>
    </Shipment>
</UniversalShipment>
```

USJFK is the home port of Branch JFK



EDI Message Data Import Log shows it is Targeting Branch 'JFK'



Option 4 – Match the recipient role's OrganizationAddress to a Branch Org Proxy

When importing a Universal Shipment, CargoWise searches the OrganizationAddressCollection for an OrganizationAddress whose AddressType matches the recipient role. If a match is found—either through Code Mapping or Address Matching—and the matched organization is configured as an

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Org Proxy on a branch, that branch will be used for the import. If multiple or no match found, CargoWise will fall back to the next option.

Recipient Role	OrganizationAddressType used for Match
Sending Agent (SAG)	SendingForwarderAddress
Receiving Agent (RAG)	ReceivingForwarderAddress
Export Broker (BRE)	ExportBroker
Import Broker (BRI)	ImportBroker
Departure CFS (DCF)	DepartureCFSAddress
Arrival CFS (ACF)	ArrivalCFSAddress
Departure CTO (DCT)	DepartureCTOAddress
Arrival CTO (ACT)	ArrivalCTOAddress
Departure Container Yard (DCY)	ContainerYardEmptyPickupAddress
Arrival Container Yard (ACY)	ContainerYardEmptyReturnAddress
Delivery Agent (DAG)	DeliveryAgent

For Example,

```
<UniversalShipment xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
    <Shipment>
        <DataContext>
            <DataSourceCollection>
                <DataSource>
                    <Type>ForwardingShipment</Type>
                    <Key>SCOMAE00003177</Key>
                </DataSource>
            </DataSourceCollection>
            <!-- content removed for the sake of brevity -->
            <RecipientRoleCollection>
                <RecipientRole>
                    <Code>DAG</Code>
                    <Description>Delivery Agent</Description>
                </RecipientRole>
            </RecipientRoleCollection>
        </DataContext>
        <!-- content removed for the sake of brevity -->
        <OrganizationAddressCollection>
            <!-- content removed for the sake of brevity -->
            <OrganizationAddress>
                <AddressType>DeliveryAgent</AddressType>
                <Address1>CORSO FERDINANDO MARIA PERRONE 11</Address1>
                <Address2></Address2>
                <AddressOverride>false</AddressOverride>
                <AddressShortCode>CORSO PERRONE</AddressShortCode>
                <City>GENOVA</City>
                <CompanyName>ITALY COMPANY</CompanyName>
                <Country>
                    <Code>IT</Code>
                    <Name>Italy</Name>
                </Country>
                <Email></Email>
```


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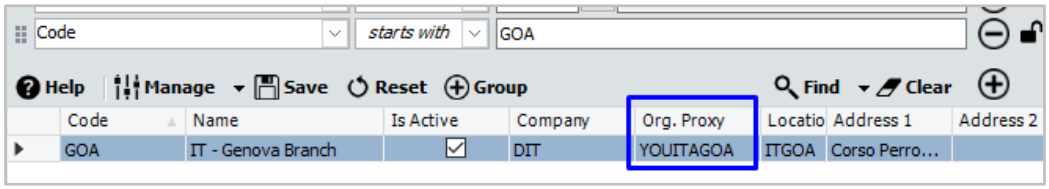
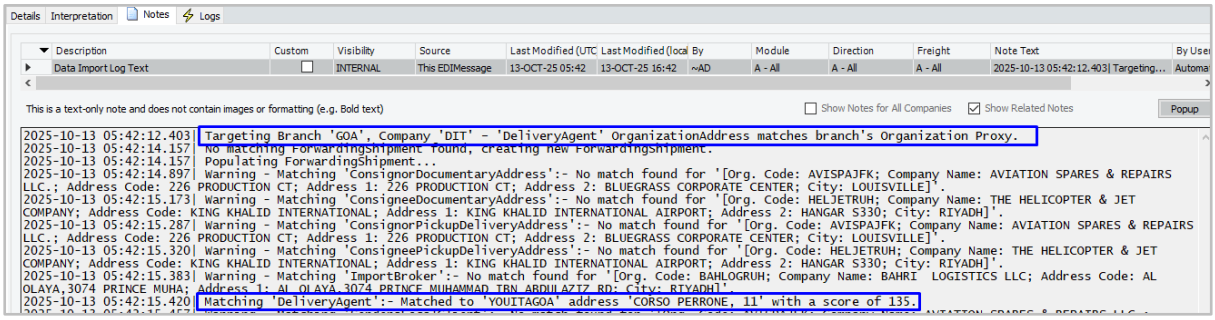
```
                <Fax></Fax>
                <OrganizationCode>ITYOU</OrganizationCode>
                <Phone></Phone>
                <Port>
                    <Code>ITGOA</Code>
                    <Name>Genova</Name>
                </Port>
                <Postcode>16152</Postcode>
                <ScreeningStatus>
                    <Code>UNK</Code>
                    <Description>Unknown</Description>
                </ScreeningStatus>
                <State></State>
            </OrganizationAddress>
        </OrganizationAddressCollection>
    </Shipment>
</UniversalShipment>
```


YOUITAGOA is the organization proxy for Branch GOA



When recipient role DeliveryAgent organization matches to YOUITAGOA, branch GOA is used to process the message



Option 5 – Match Recipient role's OrganizationAddress to a Branch home port/additional port

If no match found using above option, the recipient role organization port UNLOCO matches to a 'Home Port' or 'Additional Related Ports' on a branch, that branch will be used. If multiple matches found, system will randomly pick one; if none found, it will fallback to the next option.

For Example,


```
<UniversalShipment xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
    <Shipment>
        <DataContext>
            <DataSourceCollection>
                <DataSource>
                    <Type>ForwardingShipment</Type>
                    <Key>SCOMAE00003177</Key>
                </DataSource>
            </DataSourceCollection>
```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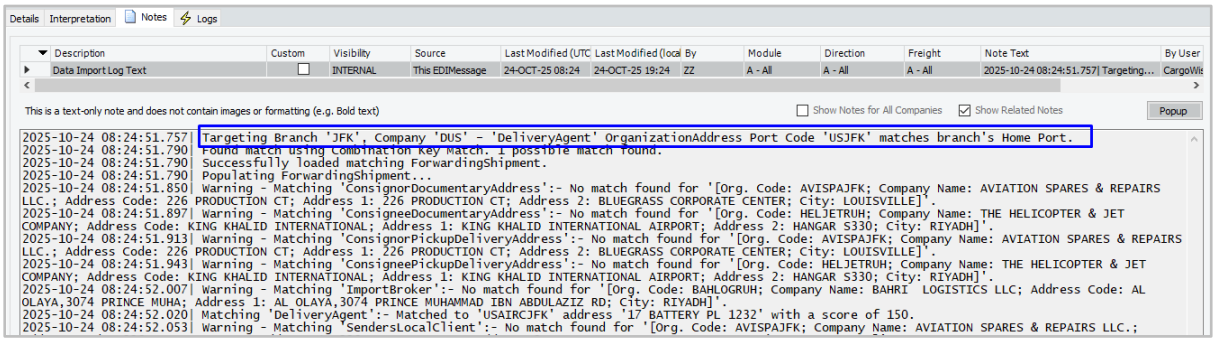
```
            <!-- content removed for the sake of brevity -->
            <RecipientRoleCollection>
                <RecipientRole>
                    <Code>DAG</Code>
                    <Description>Delivery Agent</Description>
                </RecipientRole>
            </RecipientRoleCollection>
        </DataContext>
        <!-- content removed for the sake of brevity -->
        <OrganizationAddressCollection>
            <!-- content removed for the sake of brevity -->
            <OrganizationAddress>
                <AddressType>DeliveryAgent</AddressType>
                <Address1>17 BATTERY PL STE 1232</Address1>
                <Address2></Address2>
                <AddressOverride>false</AddressOverride>
                <AddressShortCode>CORSO PERRONE</AddressShortCode>
                <City>New York</City>
                <CompanyName>US COMPANY</CompanyName>
                <Country>
                    <Code>US</Code>
                </Country>
                <Email></Email>
                <Fax></Fax>
                <OrganizationCode>abc</OrganizationCode>
                <Phone></Phone>
                <Port>
                    <Code>USJFK</Code>
                    <Name>John F. Kennedy Apt/New York</Name>
                </Port>
                <Postcode>10004</Postcode>
                <ScreeningStatus>
                    <Code>UNK</Code>
                    <Description>Unknown</Description>
                </ScreeningStatus>
                <State></State>
            </OrganizationAddress>
        </OrganizationAddressCollection>
    </Shipment>
</UniversalShipment>
```


When recipient role DeliveryAgent organization doesn't match to any branch's organization proxy, branch JFK is used to process the message because the organization port matches the branch home port



Option 6 – Match using a Recipient Role and Port Code

When no branch is found using above options, then a branch will be calculated using a combination of Recipient Role elements from the Data Context collection, and the Ports present in the Universal XML.

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Recipient Role	Ports used for Match
Sending Agent (SAG), Export Broker (BRE)	Port of Loading then Port of Origin
Receiving Agent (RAG), Import Broker (BRI)	Port of Discharge then Port of Destination
Departure CFS (DCF) Departure Carrier (DCR) Departure CTO (DCT) Departure Container Yard (DCY)	Port of Loading
Arrival CFS (ACF) Arrival Carrier (ACR) Arrival CTO (ACT) Arrival Container Yard (ACY)	Port of Discharge

Option 7 - Default to the first active branch of the target Company

If all other branch matching methods fail, the system defaults to the first active branch of the recipient company. This fallback exists solely to ensure that the data can still be processed, even when no suitable match is found. Rejecting the message entirely is not considered a viable option, as it would prevent the data from being processed in the system altogether.

2.4.2.2.2 Universal Event

Refer to Universal Event XML – Branch and Company

2.4.2.3 Targeting a module

Incoming Universal XML could potentially apply to many modules throughout CargoWise as the same schemas are mapped to multiple modules. There are two ways to target a module. One is to specify a RecipientRole that is recognized by a given module, the other is to specify a DataTarget in the DataTargetCollection which forces the module to process the data.

The DataTarget in the DataTargetCollection is not used when sending data from CargoWise, but is used when importing Universal Data into CargoWise.

DataTargets allow you to specify which module or modules should be updated by the incoming data. Each DataTarget has a mandatory Type, which, like elements in the DataSourceCollection, relates to a Module within CargoWise.

You can optionally include a Key in a DataTarget, but that is not required unless you want to match to existing entities in that module by entity reference (e.g., job number).

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2.4.2.4 Targeting a Job by Entity Reference (for example, Job Number)

If you include a DataTarget element, and you also provide a Key as part of that DataTarget, that Key will be used to match to an existing entity in the module. If an entity does not exist with that key, the XML Import will be rejected as no entity was found.

More information on the matching process follows below in 'Context Key Matching on Import'. There are more forms of matching that can be used when updating existing data, information on which keys are used in which module is detailed in the Module Mapping Guides.

> Module Mapping Guides describe the keys used for matching, give an overview of how the schema is mapped, and provide a list of element level mappings describing each field's location in the GUI. Refer to Module Mapping Guides.

2.4.2.5 Code Mapping options

The DataProvider element is an optional element that can be used to identify the source or provider of the data. Code Mapping looks at this element and tries to find an Organization within the target CargoWise system it can use to draw Code Mapping configuration from.

First, Code Mapping looks up the Company Org Proxy for the Company selected as described in Targeting a Company above. If the DataProvider element is present, and the value is present in the code mapping against the Company Org Proxy pointing towards an Organization, then that Organization is used as a Code Mapping configuration source. If no Organization is found, then the Company Org Proxy itself is used as a Code Mapping configuration source.

The CodesMappedToTarget element is used to indicate that Code Mapping is not required as the codes provided already match those used in the receiving Company's system.

2.5 Link Elements

XML is a good data type for representing a tree based data structure. Related elements can easily be represented by child elements, and one-to-many parent-child relationships can be represented as a sequence of typed elements within a 'Collection' sub element.

XML does not intrinsically describe cross relations between child elements. For this reason, where relations between branches in the tree are required, we have used our own standard built around integer-based Link elements across Universal XML.

Cross relations utilize a Link element on the parent element with an integer value used to identify that parent element uniquely within the file. These are not added to every element type, just to those where there are other elements relating to these.

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The child element in the relationship will have an element on it with the name made up of the parent type being related to plus the suffix Link. Where the same integer value is present in a parent's Link element, and a child's *[ParentType]*Link element, a relationship is implied.

As an example, the following relationship exists within Universal Shipment XML:

Parent: OrderLine.Link

Child: CommercialInvoiceLine.OrderLineLink

The relationship implied may be one-to-one, it may be many to many. There is no change in notation for either of these cases. The values used in Link elements are not persisted between sends, but they are guaranteed unique per Parent Type within each serialized Universal XML message.

Right now, types have only been required in the Universal Shipment XML Dataset. The following table is a list of Links currently implemented for the Universal Shipment.

Parent	Children
OrderLine.Link	CommercialInvoiceLine.OrderLineLink PackedItem.OrderLineLink
AdditionalBill.Link	InBondMoveDetail.AdditionalBillLink
Container.Link	InstructionContainerLink.ContainerLink PackingLine.ContainerLink ContainerLink.Link * see **Note 1** below
CommercialInvoiceLine.Link	PackedItem.CommercialInvoiceLineLink
EntryInstruction.Link	CommercialInvoiceLine.EntryInstructionLink EntryHeader.EntryInstructionLink
PackingLine.Link	InstructionPackingLineLink.PackingLineLink
TransportLeg.Link	Confirmation.Leg.Link * see **Note 2** below

Note 1: This particular case does not include the Parent Type as part of the Link element name as this is a many to many relationship where the parent type itself is already named ContainerLink. In usage, the ContainerLink.Link element is used in a collection on the InBondMoveDetail describing the many to many relationship that exists between these two entities.

Note 2: This was an early use of the Link functionality which was implemented before the standard naming convention was settled on. The only kind of 'Leg' in the schema is the TransportLeg, so it was deemed better to accept this exception to the rule rather than break existing interfaces relying on the current state of affairs.

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2.6 CollectionContent attributes and importing Collections

The default behaviour of collections within Universal XML is that where they are included, they are seen as a complete replacement for any rows existing in the target module. New rows are added, existing rows are updated, and rows not included in the supplied XML are removed or deleted. This is known as a 'complete' replacement.

There is one notable exception to this default behaviour which is the SubShipmentCollection. The Shipments in this collection are only ever used to add or update existing rows, never remove them. Partial SubShipment information is commonly provided in a Universal Shipment. For that reason, this collection defaults to a 'partial' replacement where rows not included in the XML are left as is.

Although all collections have a default behaviour (either complete or partial update), some collections allow you to specify on import whether the data you are providing is a complete representation of the collection, or only a partial representation of the collection. This is done using the CollectionContent attribute, which can have a value of either Complete or Partial.

By stating that the collection contains partial information, you are saying there may be other data in this collection that you have not included, so rows you don't include in the XML will be left alone. If you define it as a collection with complete information, any rows present in the database that are not included in the collection will be deleted.

The CollectionContent attribute is defined in the XSD on all collections that allow its use.

2.6.1 'Complete' CollectionContent update behavior

There are two methods that can be used for a complete replacement. The method used depends on whether or not there is a natural key for a given collection. A natural key is a value that uniquely identifies the context of an incoming row, like a container number for containers, or line number for order lines.

Where there is a natural key, elements supplied are matched to existing rows. Rows are updated when a match is found, added when not found. Where rows exist with no match in the elements from the collection, they are deleted.

Where there is no natural key, there is no way to match to existing data, so any existing rows are deleted, then the elements in the collection are used to build new rows in their place.

Both complete replacement methods bring the data in CargoWise in line with the data provided in the XML.

2.6.2 'Partial' CollectionContent update behavior

Similar to the complete replacement, there are two methods that can be used depending on whether or not there is a natural key for a given collection. Both methods are identical, except

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that when executing a partial replacement, rows not included in the XML are left alone instead of being deleted.

Where there is a natural key, elements supplied are matched to existing rows. Rows are updated when a match is found and added when not found. Where rows exist with no match in the elements from the collection, they are left as is.

Where there is no natural key, there is no way to match to existing data. As there is no way to match to existing rows to update, and unmatched rows will not be deleted, in this case all elements in the collection are used to create new rows. This is not an option to be used lightly, as multiple sends with the same data will cause the same rows to be added multiple times.

2.6.3 Collections Supporting the ContentCollection Attribute

Following is a list of all collections supporting the ContentCollection attribute, along with the default natural key. All of these are from the Universal Shipment schema. This functionality is generic and can be supported in any Universal schema but is only used in the Universal Shipment XML at this point.

Collection	Default Natural Key
Shipment.AdditionalReferenceCollection	Type + ReferenceNumber
Shipment.ContainerCollection	ContainerNumber
Shipment.PackingLineCollection	Module specific, refer to following note
Shipment.NoteCollection	Description + NoteContext
Shipment.Order.OrderLineCollection	LineNumber
Shipment.InBondMoveHeaderCollection. InBondMoveHeader.AdditionalReferenceCollection	Type + ReferenceNumber
Shipment.TransportLeg	PortOfLoading + PortOfDischarge

The default natural keys shown above are used in most modules importing Universal XML, however each module can override the natural key to suit its own business case.

For more specific descriptions of natural keys used for collections, refer to Module Mapping Guides related to that module.

In the case of Packing Lines, there is no default behaviour. The Forwarding Shipment module implements some very specific matching behavior which is described in the Update Note: Updating multiple pack lines through Universal Shipment XML.

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2.7 Customized Fields

CargoWise allows users to define Customized Fields as an extension of most operational modules within the product. These can be configured in three different ways for three different business cases.

1. Using the Custom Fields tab on the Workflow Template configuration form (All modules at the Shipment level).
2. For an Organization by defining a Customized Label configured from the *Custom > Customization* tab. (Specific modules at various levels)
3. From the Custom Attribute settings in the Registry (Specific modules at various levels).

For more information on how to configure Custom Fields, Customized Labels, or Custom Attributes, refer to the relevant guides from WiseTech Academy.

All three types of Customized Fields are represented the same way in Universal XML. These are represented as collections of CustomizedField elements, each containing a Field Name (Key) with the Value and DataType elements to describe the content contained within.

CustomizedFieldCollections are found as sub elements of the following element types:

- Shipment
- Container
- PackingLine
- TransportLeg
- OrderLine
- CommercialInvoiceHeader
- CommercialInvoiceLine
- Instruction

2.7.1 Exporting Customized Fields

All types of Customized Fields available on a given entity are exported as Universal XML. They are added to the outbound collection one after the other, order is not guaranteed. If two different Customized Field mechanisms define the same Key (label), the first value found with that key will be exported starting with the Customs Attributes then the Customized Labels, then the Workflow Custom Fields.

2.7.2 Importing Customized Fields

When importing Customized Fields, there is one collection of incoming values, but there are three different mechanisms used for configuring and in fact storing those values. Different mechanisms

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are used in different locations across CargoWise. Their usages are covered in more detail in the content available for each module from our myaccount Web Service Portal.

Importing follows the same logical sequence no matter which mechanisms are used. For each Key and Value provided as part of a CustomizedField element, the steps below are followed:

1. Where Custom Attributes are available and there is an Attribute defined with the Name matching the Key provided, the Value is imported as a Custom Attribute and nothing else.
2. Where Customized Labels are available and there is a Customized Label defined with the Label matching the Key provided, the Value is imported as a Customized Label and nothing else.
3. If Workflow Custom Fields are available, the Value is imported as a Custom Field regardless of whether there is a Custom Field defined with the Name matching the Key provided or not.

Workflow Custom Fields allow for values that are note configured to be stored, displayed, and edited. For this reason, where Workflow Custom Fields are available, non-configured Customized Field values can be imported. When values are removed in the GUI from non-configured Customized Fields, the fields will no longer appear after the parent entity is saved.

2.8 Dates and Timezones

Most date and time fields in CargoWise do not have a time zone directly recorded against them. When they are recorded in the database, there is no way to set and record the time zone. This is consistent across the vast majority of date and time fields in CargoWise.

This may seem strange considering CargoWise is a globally scaled logistics package with a comprehensive International Freight Forwarding module built in. This was a design decision made to facilitate interoperability with global data feeds and messaging systems.

The Universal XML interface is consistent with the application. Datetime elements now include a time zone offset on export. Any time zone offset included within a datetime element in Universal XML being imported is accurately stored.

2.8.1 Global Data Feeds

There are many data feeds within the global logistics ecosphere where no time zone is provided. If we built CargoWise requiring a time zone, we would not be able to accurately process these feeds unless we had a location for each date/time provided so we could calculate the time zone. This is often not provided in those feeds.

In the airfreight domain, many types of FSU status messages do not have the time zone. Airline departure and arrival schedules are published in the local time of each airport, or “port time” as are Seafreight, Road and Rail schedules. Cutoff times, Availability times and Slot dates are all published with no time zone as they are all tied to a port which infers the time zone.

If we required a time zone for all date and time based fields in CargoWise, we would not have been able to integrate as many data feeds as we do. The question then arises, “How do we deal

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with time zones for transit time calculations?"

2.8.2 Calculating Time Zones Where Required

Because different external interfaces have different date, time, and time zone requirements, datetime translation is done within the middleware.

In all International modules within CargoWise, for each date/time entered, there is a corresponding location that is provided. The location relating to each date/time field infers the time zone for that field. This means that the Time of Departure is entered assuming the time zone of the Port of Departure, and Time of Delivery is entered assuming the time zone of the Port of Delivery.

We use UNLOCOs to describe locations within CargoWise. If you need a UTC time, or a time including the time zone, you can use the associated UNLOCO to calculate the time zone offset. This offset can then be appended to the datetime to give a datetime with time zone offset or subtracted from the datetime value to give a UTC time.

As an example, if the Port of Arrival was AUSYD (Sydney, Australia), the time zone offset would be +10 and would be applied as follows.

Datetime from CargoWise:	`<ArrivalDate>2012-04-13T11:00:00</ArrivalDate>`
With time zone +10 added:	`<ArrivalDate>2012-04-13T11:00:00+10</ArrivalDate>`
Or converted to UTC:	`<ArrivalDate>2012-04-13T01:00:00Z</ArrivalDate>`

2.8.3 Universal Event XML supports Event Time Offset

The Universal Event XML generated from CargoWise is enhanced to contain the Date Time Offset. The incoming Universal Event XMLs can also include the Date Time offset in the Event Time.

Refer to Event Time Offset available in Universal Event XML

2.9 Message Number Collection

Universal response messages will contain a <MessageNumberCollection> for identification purposes.

The Universal Response message from CargoWise will include the collection as seen in the below sample:

```
<UniversalResponse version="1.1" xmlns="http://www.cargowise.com/Schemas/Universal/2011/11">
    <Status>PRS</Status>
    <MessageNumberCollection>
        <MessageNumber Type="TrackingID">c2036869-ccd2-5chd-82b0-8155b456c8bc</MessageNumber>
        <MessageNumber Type="InterchangeNumber">00000000000000019874</MessageNumber>
        <MessageNumber Type="MessageNumber">00000000000000000220</MessageNumber>
    </MessageNumberCollection>
    <ProcessingLog />
</UniversalResponse>
```


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The below Types are used withing the Message Number Collection:

- **Tracking ID** - <MessageNumber Type="TrackingID">
  EDI Interchange Tracking ID
- **Interchange Number** - <MessageNumber Type="InterchangeNumber">
  EDI Interchange number
- **Message Number** - <MessageNumber Type="MessageNumber">
  EDI message number

2.10 Business Rule Engine

The Business Rule Engine Module provides users with the ability to define their own rules using CargoWise powerful macros to validate inbound Universal Shipment XML. This enables users to define as many rules as they need for all modules in CargoWise, which are supported by Universal Shipment XML.

Each rule should consist of a logical expression using macros, which includes data fields from CargoWise database and new macros to refer elements from Universal XML. Users can define the criteria for the rule to be evaluated in the Business Rule Engine module or they can include the rule/s to be evaluated on-demand via XML payload.

To learn more about the usage of the Business Rule Engine refer to Business Rule Engine

3. Sending Universal XML

3.1 Overview

Previous XML messaging systems implemented in CargoWise used transmission methods such as Email, FTP, or manually saving a file to a specified location. All of these methods are 'store and forward' transmission methods where data sits on various servers along the way waiting to be picked up by a polling service.

Rather than continuing using transmission methods that by their nature introduce delays and provide no guarantee of delivery or knowing when a delivery has failed, Universal XML is sent from CargoWise using our eAdaptor web service interface.

eAdaptor provides an easy connection to our eHub Gateway. Previously we provided customized data mappings for CargoWise systems by writing code which was then deployed as an upgrade. This meant that an upgrade had to be run every time we added or changed a mapping. Now those changes can be handled via eHub and updated without clients having to run an upgrade on their CargoWise systems allowing us to respond more quickly to client's needs.

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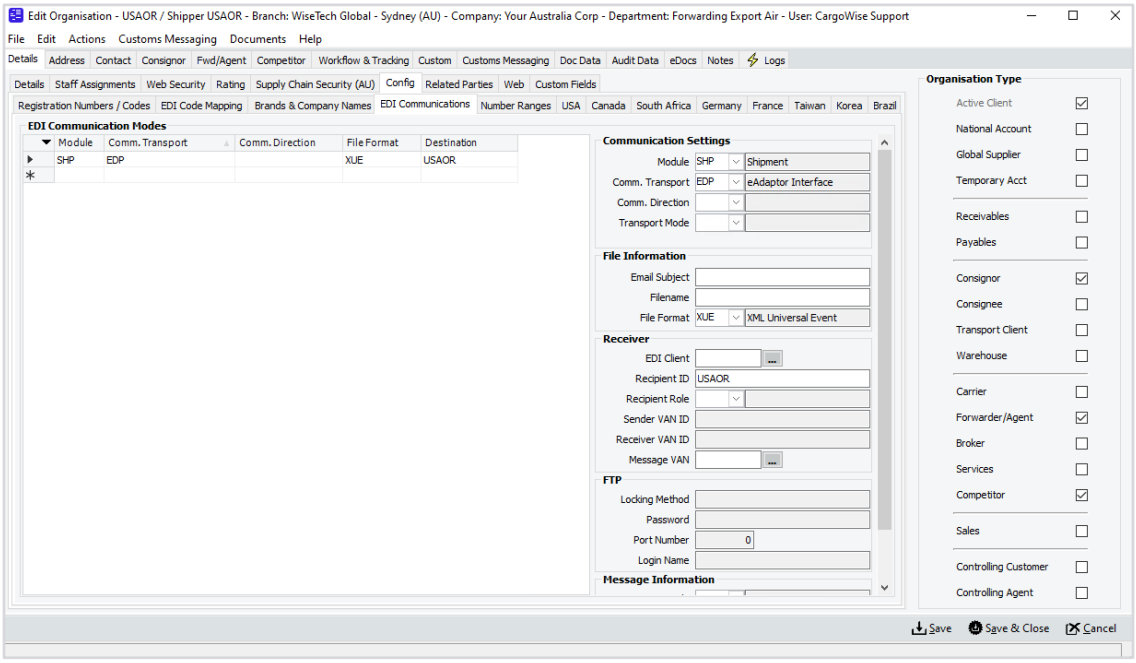
3.2 Configuring Communications

Before you can send Universal XML, you need to define which data you want to send to which Web Service. You do this on the EDI Communications sub tab of the Organization you want to send Universal Data to. If you were sending data internally, you would generally use the Organization linked to the Company you log into in CargoWise (otherwise known as your Company Organization Proxy).

> If the recipient is your Organization Proxy, EDI Messages will be generated as "Internal." These messages are meant for internal consumption by other modules within CW and are not sent via eAdaptor. To ensure these messages are sent externally, you can utilize the Purpose Code field, which allows you to enable "Disable Internal Messages."

First, go to the Organizations module within CargoWise and open the Organization you want to send Universal XML to. Make sure the Details tab is selected, then the Config sub tab, then the EDI Communications sub tab.

In the EDI Communication Modes grid, you need to define for each Module and Universal Data Type where the data is going (eHub or Local Web Service), and the Recipient Code that the target Web Service can use to route your message if required. You enter this information in the following fields as described below:



- **Module** – Must contain the Module Code the data is being sent.
- **Comms. Direction** – Always contains TRX for Transmit.
- **File Format** – This is where you enter the Universal Data Type. Commonly used options here are:

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a. XUE – XML Universal Event

b. XUS – XML Universal Shipment

- The options change depending on which Module you have selected to indicate the functionality available in that module; all Universal Data options have a description containing 'XML Universal' and the data type.
- **Message Purpose** – This is an optional field, used when you want finer control over the destination of data than Module and Data Type give. You can specify a Purpose Code on a Workflow Trigger Action. When a Trigger Action is configured to send Universal Data, CargoWise first looks for a match on Module + Data Type + Purpose Code, then falls back to a row with matching Module and Data Type, but no Purpose Code.
- **Comm. Transport** – The two options you can select here are HUB - eHub Gateway and EDP – eAdaptor Interface. If you try and select any other options, an error is displayed indicating you can only send Universal Data via eAdaptor to eHub or a Web Service implementing the eAdaptor Interface.
- **Destination Address** – When sending via the eAdaptor Interface or to eHub, you must specify a Destination Address. This can be anything you like when sending to your own eAdaptor Interface, but when sending via eHub must match the Destination Address Code supplied by our internal eServices team so that eHub can process your data appropriately.

To add all relevant fields, right click in the grid and select **Customize Columns**, then add the necessary columns.

For more information, refer to the following learning units:
How do I configure an organization to send job data electronically to another CargoWise system?
Where do I set up Message Trigger Purposes and how do they work?

3.3 Sending automatically via Workflow

The best way to send Universal Datasets is to use Workflow Trigger Actions to automatically push Universal Datasets out of CargoWise. Workflow is supported in all CargoWise modules where Universal Data is supported, and it's the timeliest and most reliable way of getting data published from your CargoWise system.

Rather than having to remember when to manually export operational data, using Workflow allows you to pre-configure what conditions must be met, or what event must occur before Universal Data will automatically be generated and sent. As timelines shorten, having operational data pushed to all concerned parties as soon as possible is a major concern.

Workflow driven data exports remove the operator delay and error from the process. For this reason, we recommend using Workflow wherever you can when sending Universal XML.

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3.4 Adding Workflow Triggers

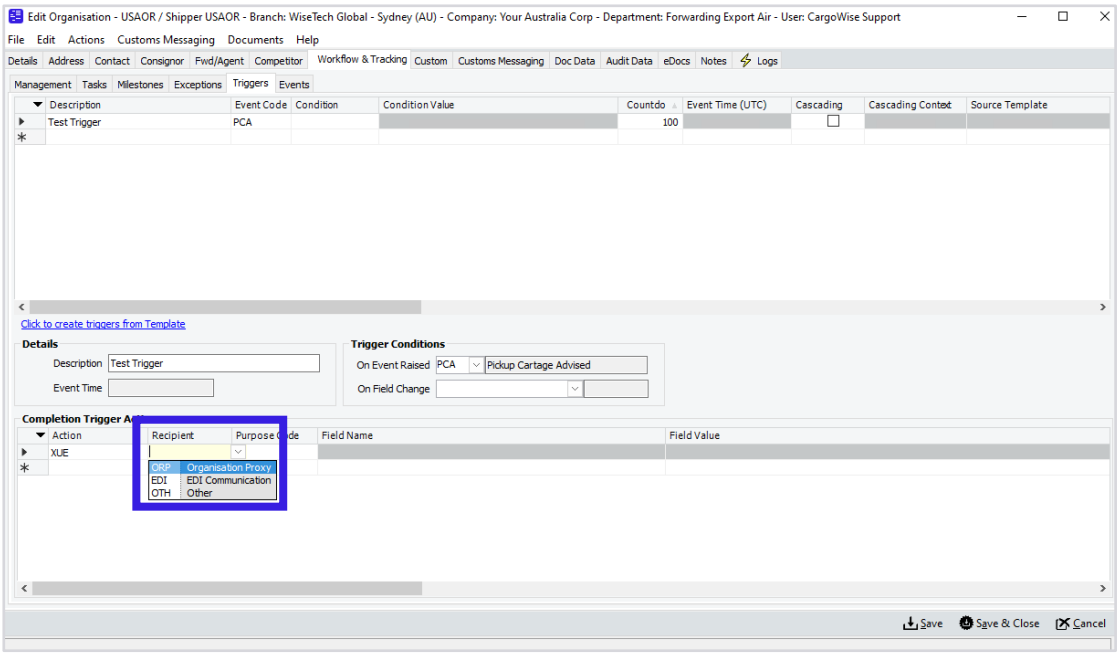
Assuming you have the destination web service configured correctly on the Organization on your Company Organization Proxy, you can setup Workflow Triggers to automatically send data when pre-defined events occur. If you open the Module you specified in the EDI Communication Mode (SHP meaning the Forwarding Shipment module) then open an Item from within the Module, you can setup a Trigger on the Triggers sub tab under the Workflow & Tracking tab on that Item.

To test this, start by creating a Shipment as you normally would, then select the Workflow & Tracking tab, then the Triggers sub tab. Add a Trigger in the Triggers grid at the top of the tab with a Description and an Event code.

> Refer to our learning unit 1COR172 - How can I send Universal XML for Testing Purposes?

Add an Action to your Trigger in the Completion Trigger Actions grid at the bottom of the tab. Set the Action code to XUE (Send XML Universal Event), the Recipient to ORP (Org Proxy) and the Purpose code to what you set the Message Purpose as in the Company Org Proxy Form.

> Refer to our learning unit 1PRO008 - How do I setup an XML file Trigger Action in a Workflow Template?



Next, go to the Events sub tab and create a new Event by right clicking within the grid and select the Add New Event option.

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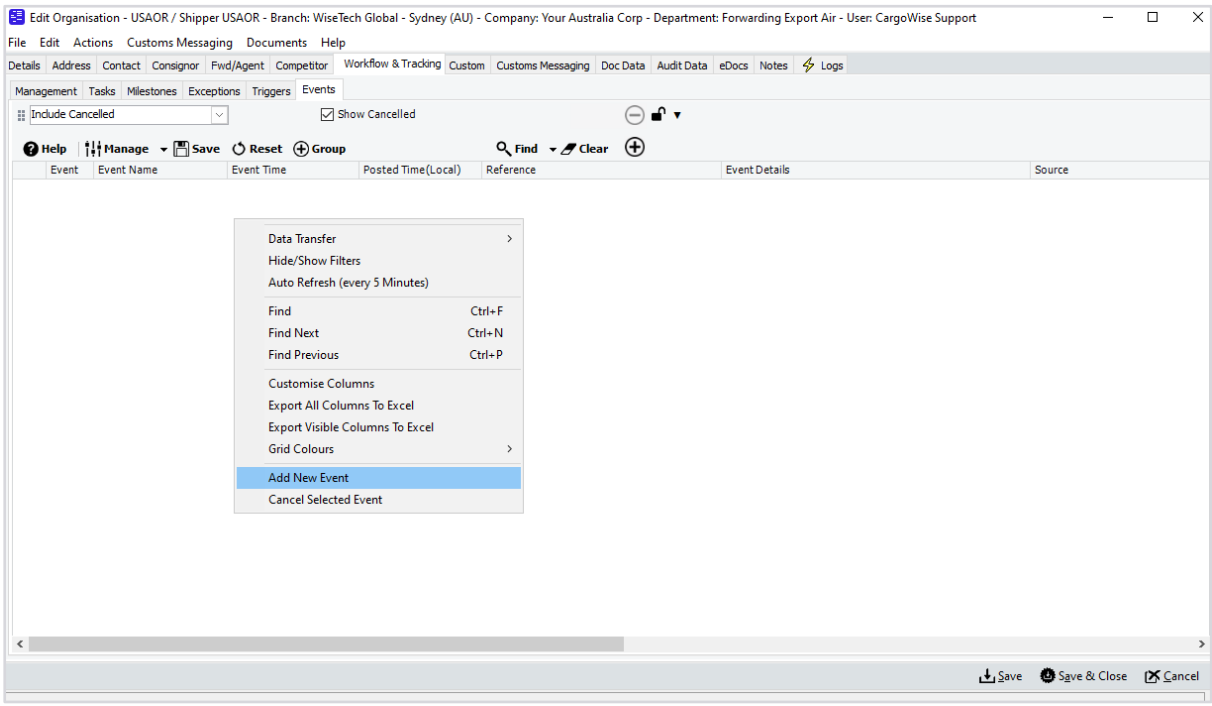
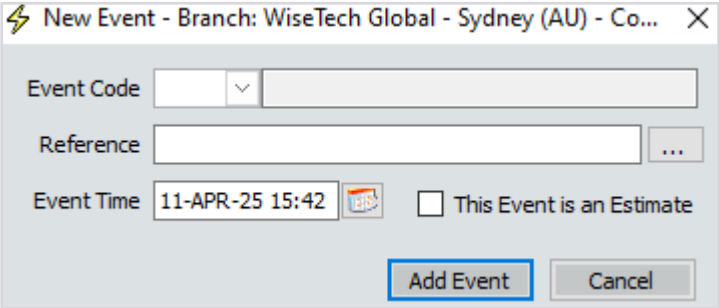


Set the Event Code as the Event Code in the Trigger you just created.



Click on the **Find** button and you will see the event you just created. Click on the **Save & Close** button to save the event.

You can export other forms of Universal XML from modules that support them by using different Trigger Action Codes.

- **Universal Activity** – XUA (Send XML Universal Activity)
- **Universal Shipment** - XUS (Send XML Universal Shipment)
- **Universal Transaction** – XUT (Send XML Universal Transaction)

As long as your Service Tasks are running, the Event Log Walker Service Task (LWK) will process the event and export it. Then either an EAM(eAdaptor Outbound) Service task, or in the case of using eHub the EHO(eHub Outbound) Service task, will send the message .

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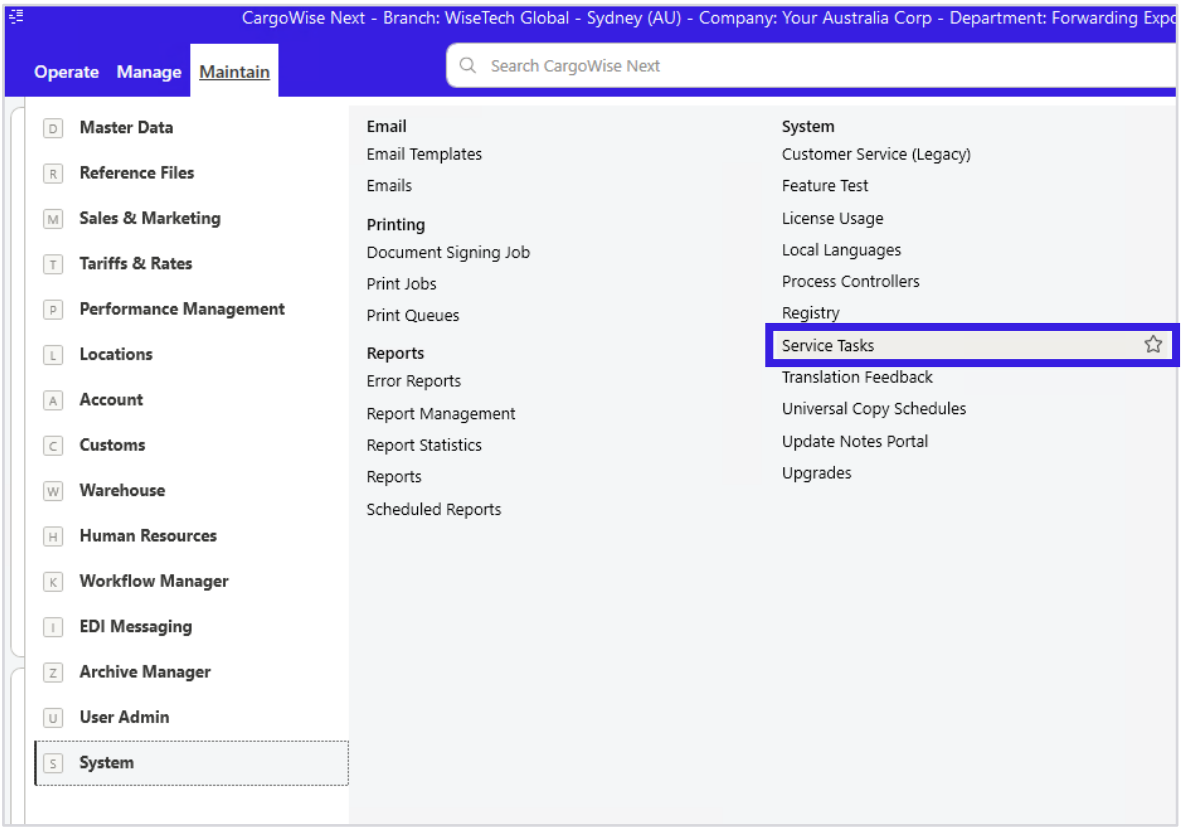
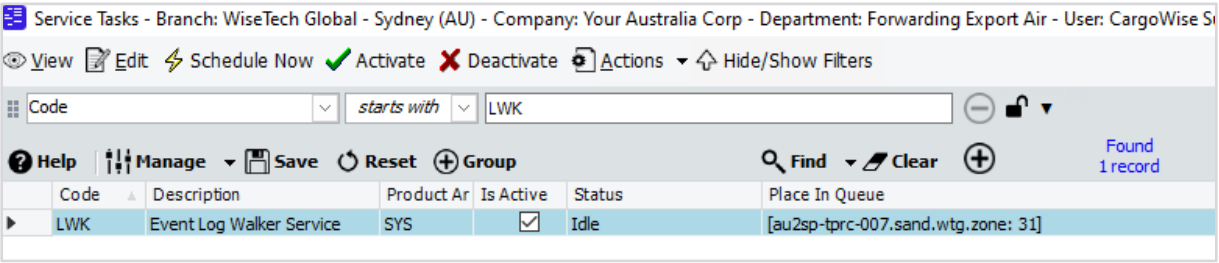
Refer to our learning unit 1PRO012 – How are Events automatically logged?

To check if your Service Tasks are running, go to **Maintain > System > Service tasks.**



Use filters to search for the LWK Service Task and then click on the **Find** button. As you can see below, the Event Log Walker Service Task is running, but currently idle.



Refer to our learning unit 1COR417 – Understanding Service Tasks

To find the resulting EDI Message, go to **Maintain > System > EDI Message**. Use the filters, you can see any events that have been exported with a status of QUE (Queued), which means they are now waiting to be picked up and processed by eAdaptor.

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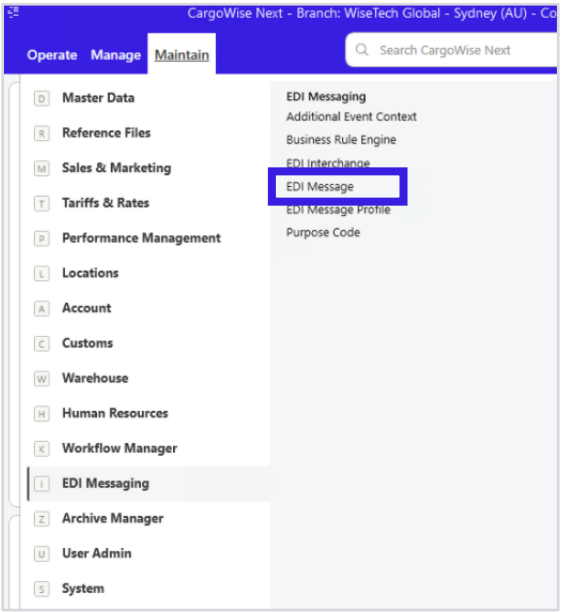
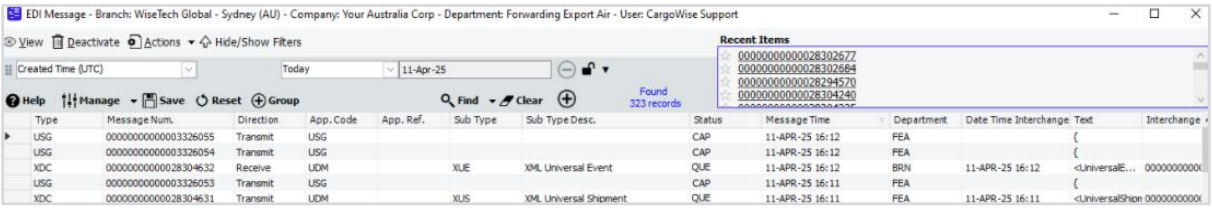
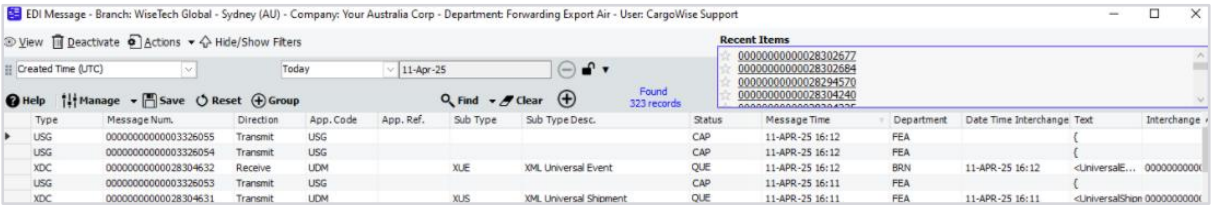




When you use a Trigger Action with Action Code XUS, the LWK Service Task will pick up the Universal Shipment and export it using the same process as with the Universal Event. It will show the same log types on the Service Task similar to the Universal Event.

If you do this, then add another Event, then go back to the EDI Messages module, you will see a Universal Shipment with a status of QUE waiting to be picked up and sent by eAdaptor.



Refer to our learning unit 1COR186 – How do I check an XML message has been sent from a job?

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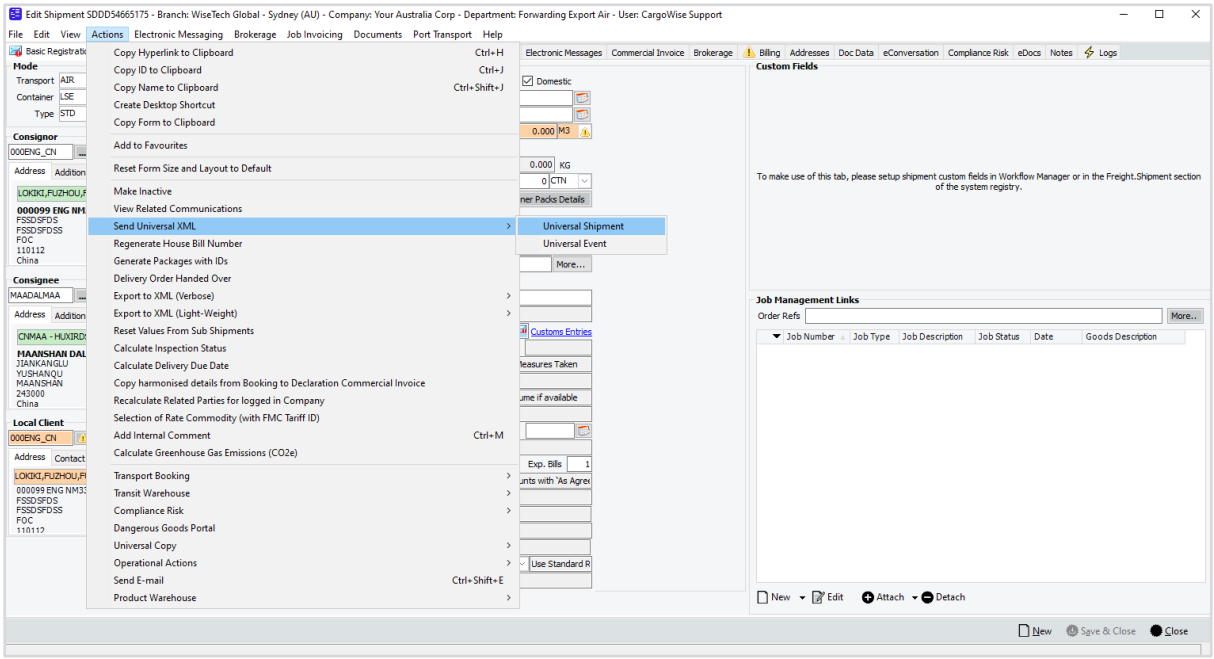
3.5 Automatically Populating Workflow Triggers

Workflow Triggers are generally setup in the Workflow Templates module so that Triggers for each new Shipment can have a default set of Triggers defined. If the Workflow Templates are setup properly, the user will never have to think about it. The Workflow Template functionality is explained further in the Workflow core documentation.

Refer to our Workflow Administration learning units in the WiseTech Academy.

3.6 Sending Manually via Actions Menu

If you need to send Universal XML on a one-off basis, from the Actions menu, select Send Universal XML without setting up Workflow Trigger Actions.



When you select this, the following window will appear. Manually enter the information required to send Universal Data that would normally send from the Workflow module.

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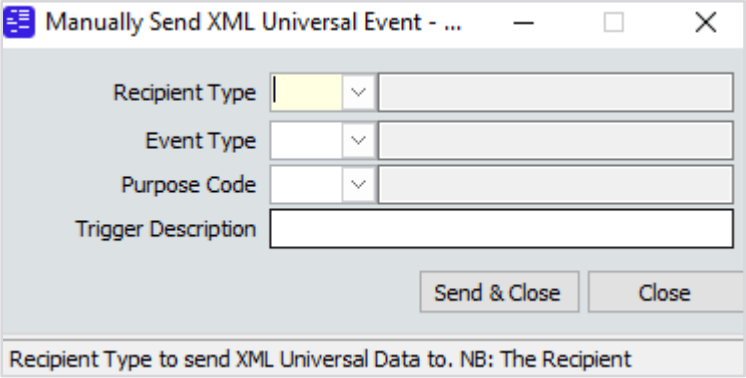
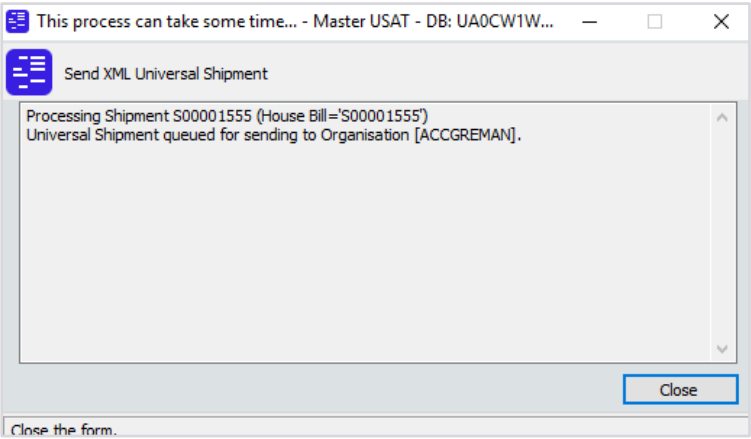


Click on the **Send & Close** button. Provided that EDI Communication Modes are setup correctly under the Recipient Organization, the following prompt will display indicating the XML has been successfully.



Refer to our learning unit [How can I manually send Universal XML files?]

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4. Receiving Universal XML

Universal XML can be sent into CargoWise via eAdaptor for Asynchronous queue-based processing. When sending in XML using asynchronous processing, the Universal Interchange is used as a wrapper to group multiple XML Messages into one Interchange.

The Universal Interchange also allows you to optionally request processing result responses so that you can monitor processing results in your middleware or send them back to external systems. When requesting processing responses, you can specify the circumstances under which you want a response (success/failure/both) and the channel/recipient you want the response to be sent to.

Processing result responses are optional as there are a number of other ways you can monitor message processing within CargoWise itself. These are described later in the following sections covering Message Processing Logs and Notification Emails.

4.1 Wrapping Universal XML into Universal Interchange

Each Universal XML message must be wrapped in a Universal Interchange before sending into CargoWise for asynchronous processing. You can include multiple Messages within the one Interchange, and they will be processed in the order in which they are included in the Interchange.

> The Universal Interchange is required if you are sending an XML to eAdaptor for Asynchronous processing, but not required if you are using eAdaptor to send in a Universal XML for synchronous processing.

Below is a sample Universal XML message that will add a CMI (Commercial Invoice Imported) event against Shipment S00043996.

```
<UniversalEvent xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
  <Event>
    <DataContext>
      <DataTargetCollection>
        <DataTarget>
          <Type>ForwardingShipment</Type>
          <Key>S00043996</Key>
        </DataTarget>
      </DataTargetCollection>
      <Company>
        <Code>DAU</Code>
      </Company>
      <DataProvider>HYECMTDAU</DataProvider>
      <EnterpriseID>HYE</EnterpriseID>
      <ServerID>CMT</ServerID>
    </DataContext>

    <EventTime>2016-11-30T13:52:03.153</EventTime>
    <EventType>CMI</EventType>
```


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```
    <CreatedTime>2016-11-30T02:52:14.963</CreatedTime>
    <IsEstimate>false</IsEstimate>
    <ContextCollection>
      <Context>
        <Type>MBOLNumber</Type>
        <Value>HBL85161TRN</Value>
      </Context>
      <Context>
        <Type>HBOLNumber</Type>
        <Value>423908</Value>
      </Context>
    </ContextCollection>
  </Event>
</UniversalEvent>
```

Taking the XML file, you would like to wrap, delete the "`<?xml version="1.0" encoding="utf-8"?>`" tag at the beginning of the file (where present) then copy the remaining file using CTRL+A to select all, then CTRL C to copy the content.

```
<UniversalInterchange xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.0">
  <Header>
    <SenderID>ABCDEFGHI</SenderID>
    <RecipientID>HYEDAUUAT</RecipientID>
  </Header>
  <Body>
    <!-- Insert Payload Here -->
  </Body>
</UniversalInterchange>
```

Paste the UniversalEvent XML content in place of the "`<!-- Insert Payload Here -->`" comment element of the universal interchange wrapper shown above. You should end up with a properly wrapped Universal Event as shown following.

```
<UniversalInterchange xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.0">
  <Header>
    <SenderID>ABCDEFGHI</SenderID>
    <RecipientID>HYEDAUUAT</RecipientID>
  </Header>
  <Body>
    <UniversalEvent xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
      <Event>
        <DataContext>
          <DataTargetCollection>
            <DataTarget>
              <Type>ForwardingShipment</Type>
              <Key>S00043996</Key>
            </DataTarget>
          </DataTargetCollection>
          <Company>
            <Code>DAU</Code>
          </Company>
          <DataProvider>HYECMTDAU</DataProvider>
          <EnterpriseID>HYE</EnterpriseID>
          <ServerID>CMT</ServerID>
        </DataContext>

        <EventTime>2016-11-30T13:52:03.153</EventTime>
        <EventType>CMI</EventType>
        <CreatedTime>2016-11-30T02:52:14.963</CreatedTime>
        <IsEstimate>false</IsEstimate>
        <ContextCollection>
          <Context>
            <Type>MBOLNumber</Type>
```


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```
            <Value>HBL85161TRN</Value>
          </Context>
          <Context>
            <Type>HBOLNumber</Type>
            <Value>423908</Value>
          </Context>
        </ContextCollection>
      </Event>
    </UniversalEvent>
  </Body>
</UniversalInterchange>
```

4.2 Requesting a Processing result response

When sending in Universal XML via eAdaptor, you can request an asynchronous acknowledgement message to be sent back when the contained XML is processed by the Universal Shipment/Event Messaging Inbound (UMI) Service Task. The acknowledgement will contain basic processing results represented as a Universal Event indicating either success or failure.

This allows you to create your own automated monitoring and alert system via your middleware rather than monitoring XML processing using the EDI Message module in CargoWise.

4.2.1 Requesting an Acknowledgement

Acknowledgements are requested using a new 'Acknowledgement' request element as shown in the following XML template.

```
<UniversalInterchange xmlns="http://www.cargowise.com/Schemas/Universal/2011/11">
  <Header>
    <SenderID>MIDDLEWARE_SND</SenderID>
    <RecipientID>HYEDAUUAT</RecipientID>
    <Acknowledgement>
      <Required>OnAll</Required>
      <Channel>eAdaptor</Channel>
      <RecipientID>MIDDLEWARE_RCV</RecipientID>
      <ContextCollection>
        <Context>
          <Type>MyJobReference</Type>
          <Value>BAT0000455347</Value>
        </Context>
      </ContextCollection>
    </Acknowledgement>
  </Header>
  <Body>
    <!-- Insert Universal XML Message Payload Here -->
  </Body>
</UniversalInterchange>
```

The parameters specified in the Acknowledgement request element will be used when processing each Universal XML message contained within that Universal Interchange. The Acknowledgement element contains the following sub elements:

- **Required** – Describes when an acknowledgement is required. Options are:
  - c. *OnAll* – For every message sent.
  - d. *OnError* – Only for those that error or are rejected.

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e. *OnSuccess* – Only for those that succeed.

- **Channel** – The channel used for sending the acknowledgement back. Options are *eHub*, or *eAdaptor*
- **RecipientID** – The RecipientID that will be used in the Universal Interchange when sending the Acknowledgement back via the designated channel. (The SenderID will be the EnterpriseID, ServerID and Company Code from the Company processing the Universal XML)
- **ContextCollection** – Any Context elements specified in the ContextCollection of the Acknowledgement Request will be returned in the ContextCollection of any Universal Events that are returned. This is where you can specify your own Job and/or Message identifiers that you can use to help identify which sent message the response relates to

4.2.2 Acknowledgements sent back

Acknowledgements are sent back in the form of Universal Events wrapped in a Universal Interchange with the following information.

- **SenderID** – The EnterpriseID, ServerID and Company Code from the processing CargoWise system
- **RecipientID** – The RecipientID that was specified in the inbound Acknowledgement Request element. This is typically used in middleware to target the response to a particular interface or pipeline
- **EventType** - The EventType will always be DIM (Data Imported) as that is the action that occurred, even on failure
- **ContextCollection** – The ContextCollection will contain all of the Context elements that were provided in the ContextCollection of Acknowledgement elements from the inbound XML. This will also contain the processing results for the submitted Universal XML in Context elements with the following Types
- **ProcessingResultStatus** – Three-character code indicating processing success or failure. This is the same code that appears in the Status column in the EDI Message Module. For a complete list of options, please see the eAdaptor Developer's Guide
- **DataImportLog** – Human readable processing log containing the same information you see on the Notes tab of the EDI Message in the Data Import Log Text note
- **DataContext** – This will contain supplementary information identifying the System and Company targeted when processing the inbound XML

If the sample Universal Interchange XML shown previously was used to send a Universal Event successfully, the Acknowledgement XML sent back would look like this.

```
<UniversalInterchange xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
  <Header>
    <SenderID>HYEDDEDAU</SenderID>
    <RecipientID>MIDDLEWARE_RCV</RecipientID>
  </Header>
  <Body>
    <UniversalEvent xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
      <Event>
```


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```
        <DataContext>
          <Company>
            <Code>DDE</Code>
          </Company>
          <DataProvider>HYEDAUDDE</DataProvider>
          <EnterpriseID>HYE</EnterpriseID>
          <ServerID>UAT</ServerID>
        </DataContext>
        <EventTime>2017-03-02T07:26:28.36</EventTime>
        <EventType>DIM</EventType>
        <ContextCollection>
          <Context>
            <Type>MyJobReference</Type>
            <Value>BAT0000455347</Value>
          </Context>
          <Context>
            <Type>DataImportLog</Type>
            <Value>Linked Event to Shipment S00001000 (House Bill='FRED235478923').</Value>
          </Context>
          <Context>
            <Type>ProcessingResultStatus</Type>
            <Value>PRS</Value>
          </Context>
        </ContextCollection>
      </Event>
    </UniversalEvent>
  </Body>
</UniversalInterchange>
```


4.3 Manual queuing for import (for test purposes only)

For testing purposes, you can manually queue Universal XML for processing by the UMI service task.

> This is not a facility that is intended for use in your Production system.

Start by creating your Universal XML file with valid Universal XML content.

Then go to the **Maintain > EDI Messaging > EDI Message** Module.

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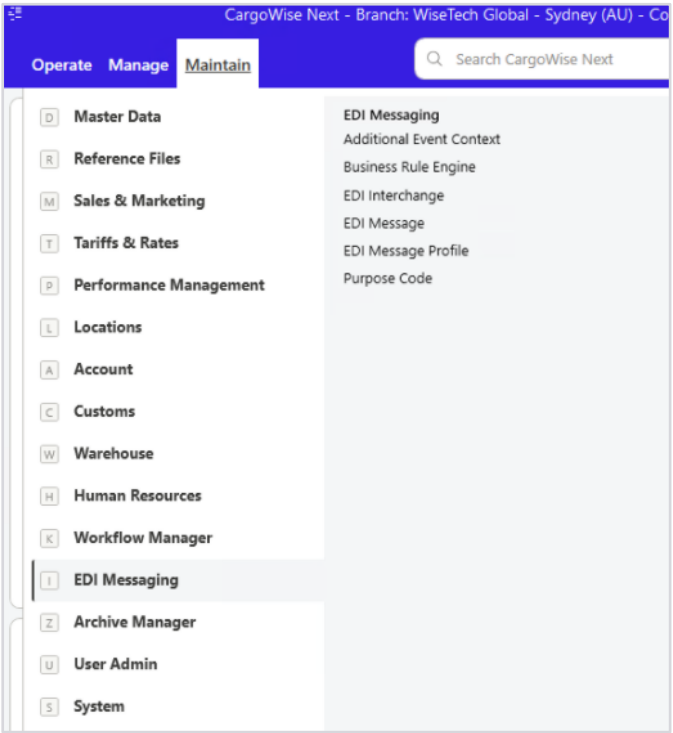
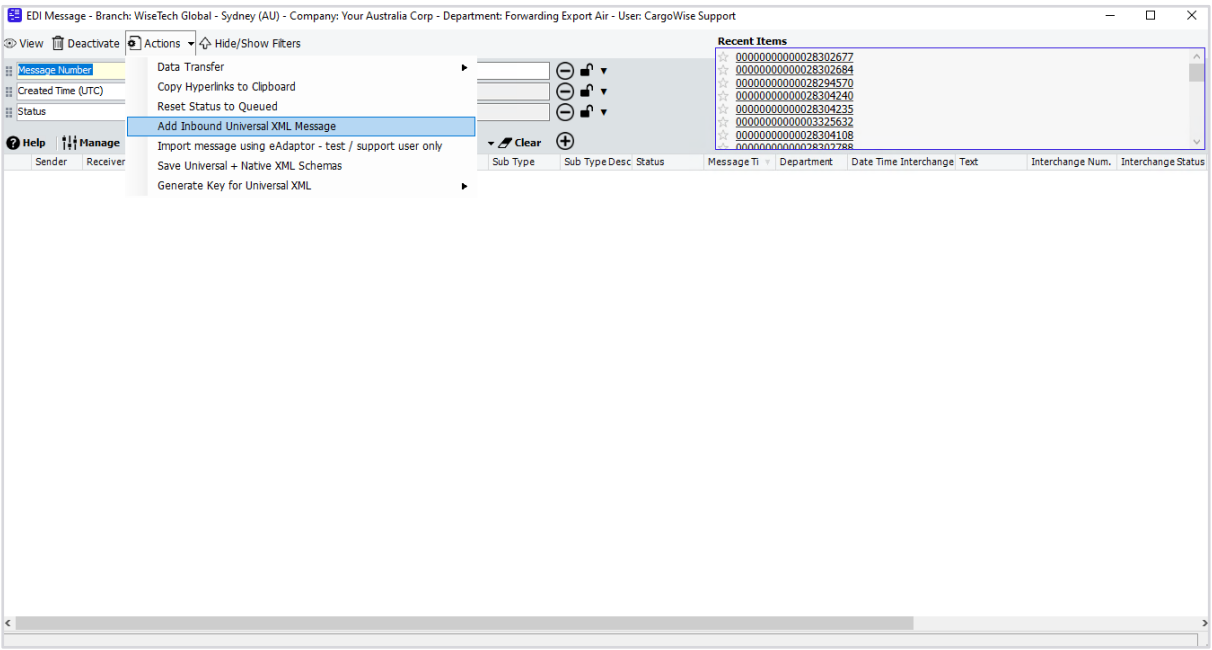


From the Actions drop down menu, select Add Inbound Universal XML Message.



This will open a dialog box.

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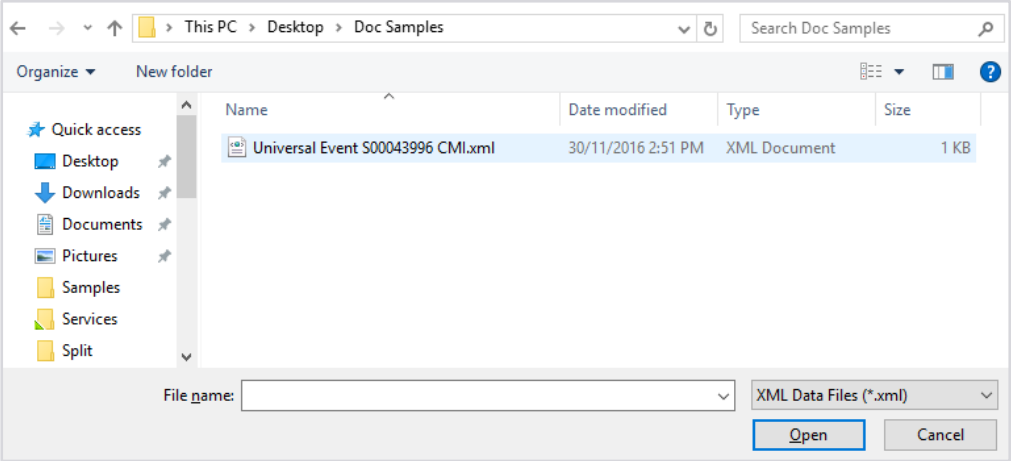
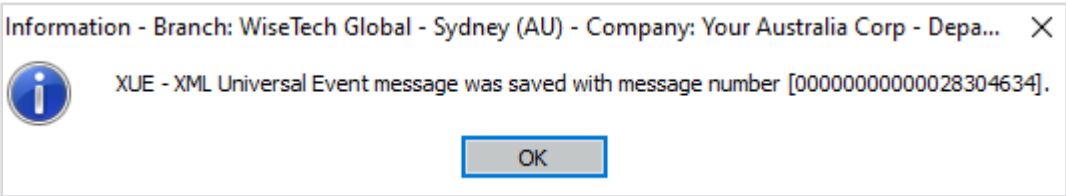
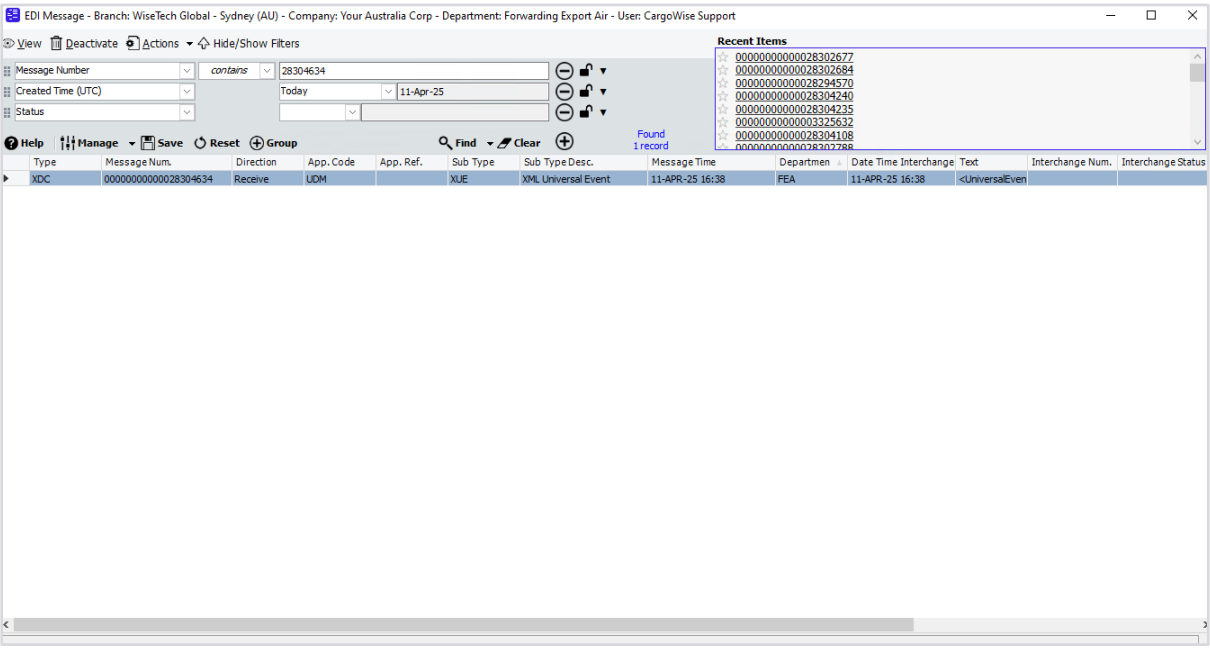


When you select a valid sample file and click the **Open** button, you will see a system prompt confirming the successful creation of your Universal XML EDI Message:



If you then click on the **Find** button on the EDI Messages module search screen with the appropriate filters you will see the EDI Message you have just created with a Status of QUE - Queued for Processing.



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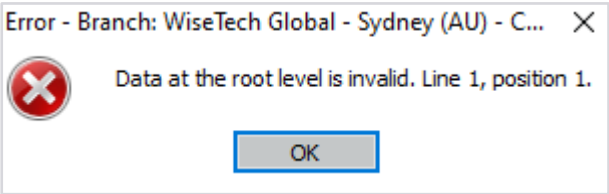
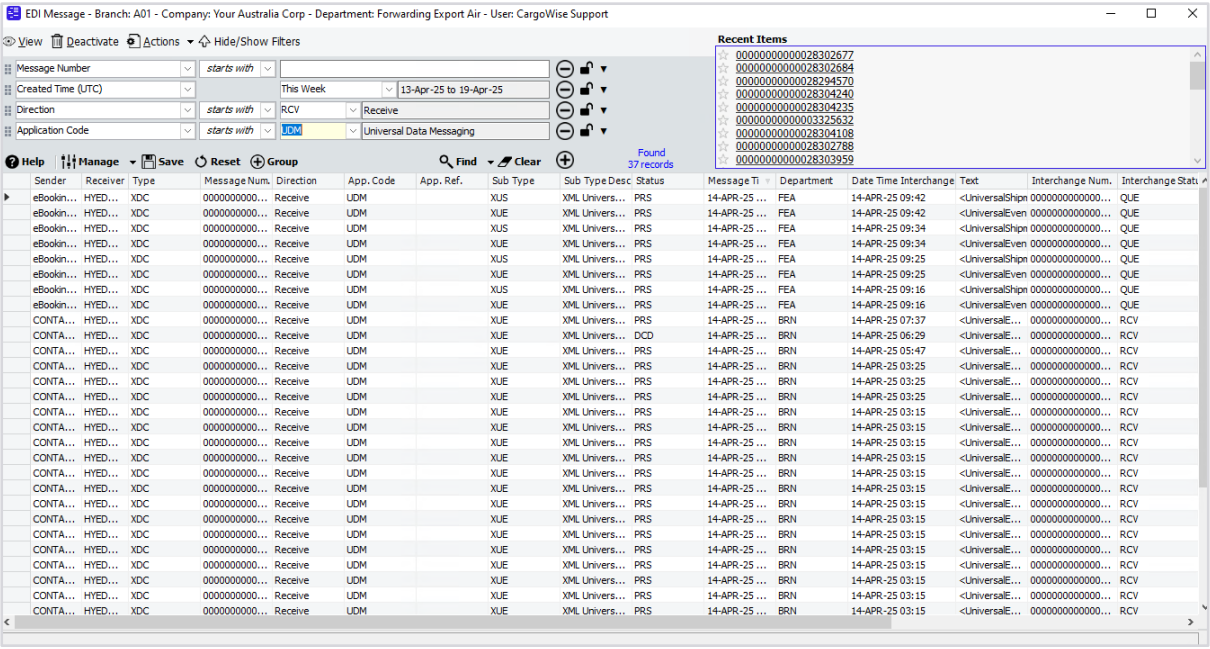
If you try to select a file that does not contain XML with matching opening and closing tags, or a file that does not have a root element matching the signature of a supported Universal XML schema, the system will display an error message similar to the one shown below.



4.4 Logging and processing after import

Universal XML comes into CargoWise via the eAdaptor Web Service Interface from your eAdaptor Web Client implementation. As long as the XML sent in has matching opening and closing tags and is generally well formed, each Universal Interchange will be added into the EDI Interchanges Module under **Maintain > System > EDI Interchanges**.

Each Universal XML Message contained within the Body element will then be extracted out into the EDI Messages module found under **Maintain > System > EDI Messages**. If the content type (determined by the root element type) is recognized, there will be a Message Sub Type code on each EDI Message row identifying the type of Universal XML contained in the message as shown below.



When messages first come in, they will show with a status of QUE - Queued. From this point, the Universal Data Messaging Inbound - UMI Service Task will pick up all queued messages and process them appropriately. The status will then be changed to a code describing the processing result for each incoming message.

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Refer to our learning unit How do I check data imports through EDI messages?

4.5 UMI Service Task

When Universal XML first comes into the EDI Message module, it has a status of QUE - Queued as shown above. From this point, the Universal Data Messaging Inbound - UMI Service Task will pick up all queued messages and process them appropriately.

As messages are processed, the status on each processed message will change to one of the following:

- **PRS – Processed OK** – The message was processed without issue and the data was applied.
- **WAR – Processed with Warnings** – The message imported and was processed with minor non-fatal issues. Examples include over length data provided for elements and out of range individual values.
- **ERR – Processed with Errors** – The message was processed and imported with data loss that should be reviewed. Examples include not being able to import Job Costing data as part of a Shipment and excluding chunks of data due to missing mandatory elements.
- **DCD – Discarded – No Match Found** – Message passed validation and is valid for import but there was no matching job, module or entity that this data could be imported against or into. Examples include not having a job that the supplied references match to, or not having a DataTarget/Recipient Role included that indicates a module to be targeted.
- **REJ – Rejected** – The message was rejected without being processed. The message either failed validation or was not valid for the CargoWise receiving system. Examples include mandatory elements at the core of the XML provided were either missing, critical values being invalid preventing the system from searching for a match, or there is a serious structural problem with the xml provided that stopped our systems from being able to process the data provided.

If you double click on an EDI Message after it has been processed by the UMI Service Task, and go to the Notes tab, you will see a Data Import Log note showing what was done with the imported data:

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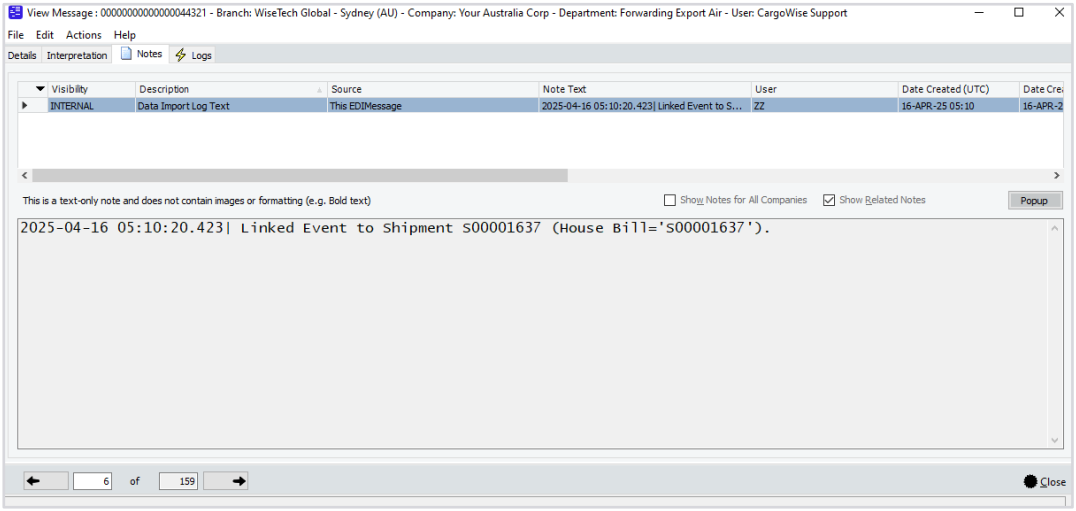
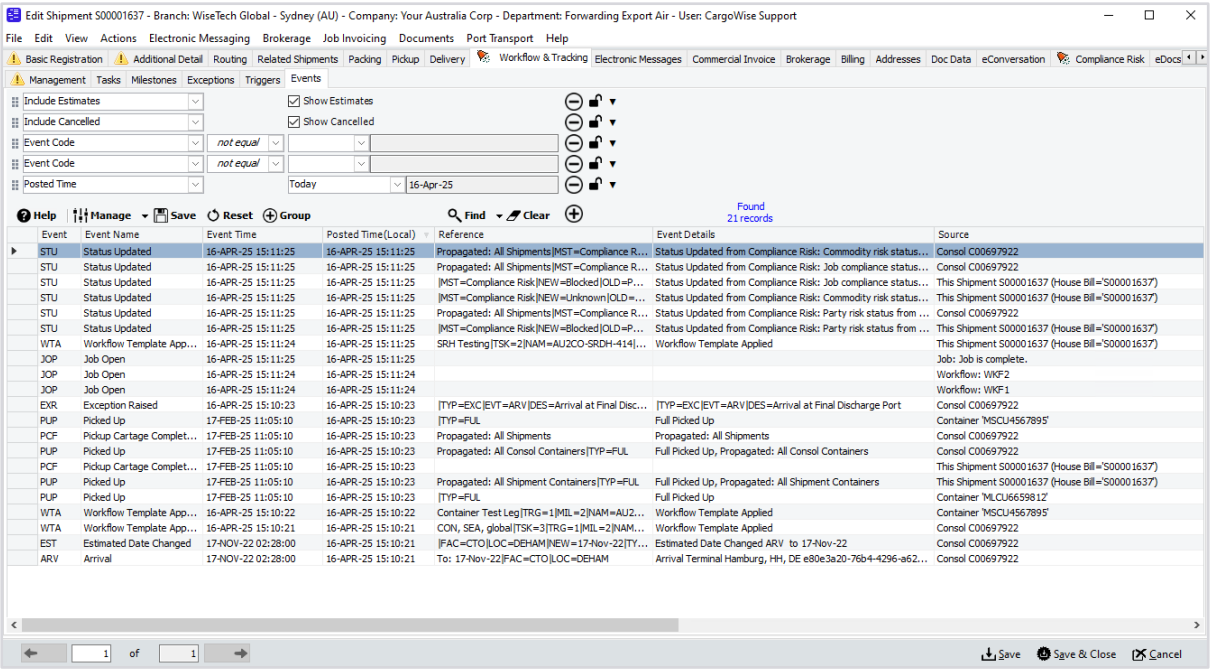


You can see from the above example that Shipment S00001637 was updated with the contents of the imported XML File.

If the XML was missing mandatory segments, or unable to be imported for any reason, any error messages generated will show in this log so that you can see what is wrong with the incoming message.

If you then open the related Business Object the data was applied to, and go to the Workflow & Tracking tab, then the Events sub tab, you will see the Event that was logged as a result of the data import.



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Whenever you select an event that was created from a Universal XML import or export, you will see all Context Information sent with the dataset showing at the bottom of the tab

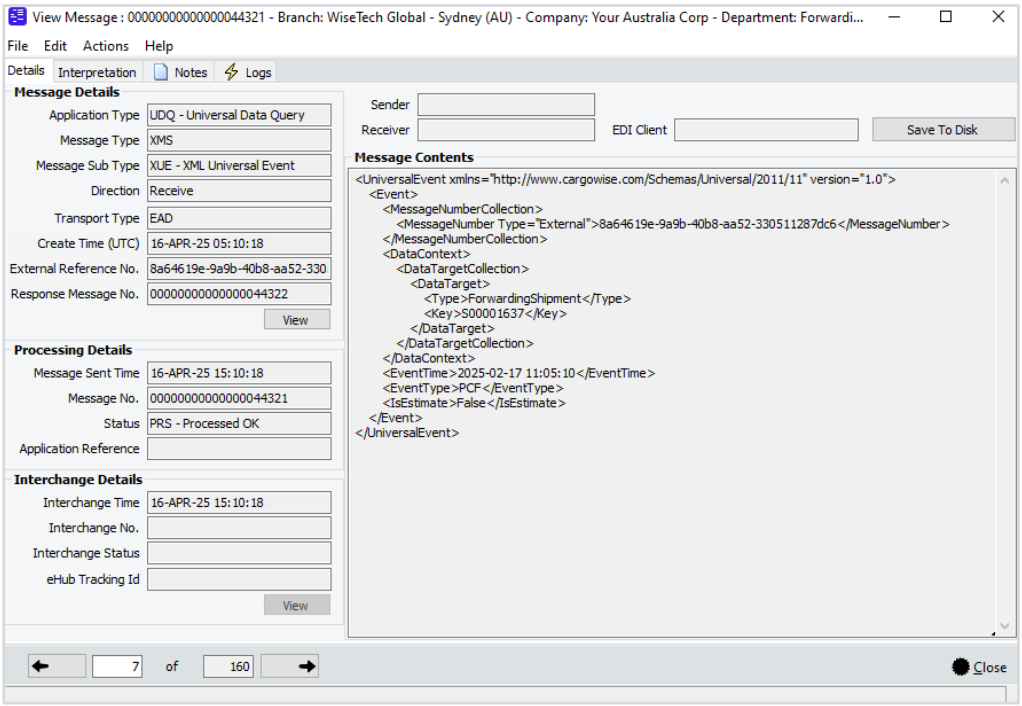
For imported Universal Events, the Event Code on the Event will be the Event Code from the incoming XML and the Context Information includes all items from the ContextCollection in the XML. This is so that when Universal Events are generated from external sources, additional data can be included with the Universal Event to help inform the users exactly what the event relates to.

For all other types of imported Universal XML including Universal Shipments, Activities, and Transactions, the Event Code will be DIM (Data Import), and the Context Information will contain basic information about the data taken from the DataContext element and its contents.

For all Universal XML exported, there will be a DEX (Data Export) event with the same Context Information showing at the bottom of the tab.

For all Events related to incoming and outgoing Universal XML, there is an area to the right of the Context Information that gives basic information about the EDI Message that contains the data that was imported, and a **View** button.

Click on the **View** button, which opens the View Message window containing the XML Message itself along with information about where the message came from, what time it was processed and whether it was transmitted or received.



Refer to our learning unit: 1COR193 - How do I check data imports through EDI messages?

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4.6 Universal Data Buss

When Universal XML comes into CargoWise, as long as basic XML validation is passed and all mandatory top level elements are present, the data will be offered to all modules via the Universal Data Buss. Based on the content of the Universal XML, each module will decide whether the data is applicable to that module, and if it is applicable how it will be used.

If you specify a DataTarget Type, the module described by the DataTarget Type will always try and import the data. For Universal Events this is not always possible as you must be able to find an existing job or entity to apply an Event against. For other Universal XML types like the Universal Shipment, Activity, or Transaction, if a matching job or entity is not found, a new one will be created using the content of the incoming Universal Data.

Even where you specify a DataTarget pointing towards a module or modules, another unrelated module may decide to use the data as well. It might decide to update a job or entity where data relevant to that module is found in the incoming data that it doesn't already have. An example of this is Container Events where a single Universal Event with a Container Number and Master Bill on it can match to jobs in the International Forwarding, Customs and Domestic Transport modules.

4.7 Key matching on import

CargoWise uses various combinations of Keys in a series of fallbacks when trying to find existing jobs or entities that incoming data is related. Keys are relatively unique references that CargoWise matches on, like House Bills, Master Bills, Order Numbers, and Booking Numbers. They can all be used to help match incoming data to existing jobs and entities within CargoWise.

There are three levels of fallback when trying to match to existing jobs or entities. The last level of fallback will use one of two different matching methods depending on a registry setting and what the target module supports. The first two types of matching are Job Number match, then Module Specific Match (where implemented). The last type of matching used will either be Combination Key matching, or Reference and Party ID matching.

Combination Key matching was found to give many false positive matches in Forwarding and Customs modules due to the way references are used in those modules. For this reason, we implemented Reference and Party ID matching. This matching algorithm yields far less false positives as it requires a match on both a reference and its issuing party/organization.

It is recommended that you turn on Reference and Party ID matching in the registry, especially if you are engaging in E2E Messaging with other CargoWise systems.

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4.7.1 Job Number matching

Job number matching is attempted when the EnterpriseID and the ServerID from the DataContext element match the Enterprise ID and the ServerID of the CargoWise installation processing the incoming data. When the Enterprise ID and ServerID match, CargoWise knows that the job numbers, entity codes, and other system specific codes will match to the system data is being imported into.

As an alternative to providing the EnterpriseID and ServerID in the DataContext element, you can force job number matching by supplying the CodesMappedToTarget element with a value of 'true'. This has the same effect as providing a matching EnterpriseID and ServerID in the incoming data. Job numbers, entity codes, and other system specific codes will then be used for matching even though the EnterpriseID and ServerID have not been supplied.

If the above criteria have been met, the DataTarget Type from each element in the DataTargetCollection is used to decide which module to look in, and the Key from the DataTarget provides the Job Number to search with. This means that a DataTarget with a Type of ForwardingShipment and a Key of S00002847 would be matched to a Forwarding Shipment with a Shipment Number of S00002847.

4.7.2 Module Specific matching

Most modules have candidate keys other than the Job Number that are specific to that module. Sometimes they involve multiple parts or crossing multiple modules to get a match. These are coded per module per incoming data type.

A good example of this is Master Bill + House Bill matching across the Forwarding Consolidation and Shipment modules. These matches occur regardless of the value of the EnterpriseID, ServerID and CodesMappedToTarget elements.

4.7.3 Combination Key matching

Entities in each module in CargoWise contain a different set of References specific to the business area the module is designed to service. Combination Key Matching applies three processes in order when trying to find the right entity to match to within a module. It continues through the processes until there is only one matching entity.

Where a process returns no matches, it is skipped, and the next process is engaged. If a process returns multiple matches, the results are passed to the next process. The matching processes are applied in order as follows:

- **Reference Match Count** – Looks for the entity with the most matching references, i.e., three matching references would beat two matching references. No weighting on the type of reference is done at this point. If there are multiple entities that equal the highest match count, those with the highest match count are passed onto the next matching process. If there's only one entity with the highest match count, that's the match made. If there's none at this point, it means that there's no entity in CargoWise with any matching references, so no further matching occurs

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- **Reference Type Weighting** – Some reference types in a given module may be given a better weighting than others. A match on House Bill + Order Number in a given module may get a better rating than two Order Numbers. The weighting works its way through the weighted list of reference types from the strongest reference to the weakest reference excluding any matches where some have the reference type and the others do not. If at any stage in this process there is only one match left, that is returned.
- If no matches exist with that reference type at all, leaving zero matches, that reference type is not used for exclusion. If multiple matches are still present at the end of this process, the reduced list is passed onto the final matching process
- **Context Fallback Matching** – Some incoming data includes values that are not references but provide context to the references provided. Two good examples of this type of data are Origin Port and Destination Port. UNLOCOs are unique references to a particular location, but they are not a reference as there will be many jobs with the same Origin and/or destination in a given CargoWise system. They can however be used to exclude false positives in the rare cases where references do not give a clear match. If only one out of all the remaining entities matches all context fallback values, then that is returned as a match.
- Each Module has its own set of references and context values. Those values are present in different places in different Universal XML schemas, but the matching principles are applied in the same way to those references and context values. Documentation on each module's implementation below gives you the references in weighted order (strongest reference first) and the context values used for Context Fallback Matching

4.7.4 Reference and Party ID matching

The problem with matching by reference numbers alone is that many different parties use the same or overlapping number sequences when issuing reference numbers. The House Bill number S00001234 will identify completely different Shipments when issued by different International Forwarders.

The primary concept behind Reference and Party ID matching is that when you know who the issuing party is for a reference, and you can match the issuing party as well as the reference to a target entity, you have a very reliable match to that target entity. This form of matching is only implemented for Universal Shipment XML as other forms of XML do not commonly contain the OrganizationAddress based information used to identify issuing Parties.

As a secondary concept, where you have a reference provided in the incoming data, and that same type of reference is present on a target entity, but the two values are different, that can be seen as a 'conflict' and reduce the possibility of a match

Reference and Party ID matching is enabled using the **EDI Messaging > Universal XML > Use Combined Reference and Party ID Matching** registry item. It is recommended that you turn this option on, especially if you are engaging in E2E Messaging with other CargoWise systems.

Once enabled, in modules where Reference and Party ID matching has been implemented, the system will try to match using Reference and Party ID pairs. Each module where this has been implemented has its own set of References and Party ID pairs, along with scores for a Match,

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Partial Match (Reference Only) and Conflict (Reference present in both XML and target entity, but different). A sample of how scoring is arranged follows.

Reference Element	Party ID Type	Match	Partial	Conflict
BookingConfirmationReference	Consignor	90	10	0
BookingConfirmationReference	Consignee	90	10	0
InterimReceiptNumber	Consignor	90	10	0
WayBillNumber	SendingForwarder	180	30	-180

References from the Reference Elements that are six characters in length or longer are used to build a list of entities where any of the references match. Scores are calculated for each Reference and Party ID type, and they are added up for each entity. The entity with the highest result is the match, although given the relative uniqueness of the references used, there is generally only one entity matching by these references in a target system.

Not all modules support this method of matching. For modules that do not support Reference and Party ID matching, Combination Key Matching will be used instead. For modules that do support Reference and Party ID matching, you will find a Reference and Party ID scoring table as shown above provided within the Module Mapping Guide for that module.

Module Mapping Guides describe the keys used for matching, give an overview of how the schema is mapped, and provide a list of element level mappings describing each field's location in the GUI. Refer to Module Mapping Guides.

4.8 Verbose Logging

When importing Universal XML, there is a log generated as part of each import describing processing results for many data types and functions. This log can be found on the Notes Tab of an Inbound EDI Message as described in section 3.5 UMI Service Task

There is a registry item, **EDI Messaging > Universal XML > Enable Verbose Logging** that turns on additional logging. Verbose Logging shows why a certain result was achieved, or how the result was

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calculated, rather than logging just the result. This can be useful when trying to diagnose why matches occur when importing Universal XML into CargoWise.

Verbose Logging registry item default behaviour is set to 'No' as it adds a significant amount of additional text to the Data Import Logs. This logging is only useful when you cannot work out why a certain matching result has occurred.

The functionalities that can write Verbose Logs include the following.

- Organization Matching
- Universal Shipment Job Level Matching
- Universal Event Job Level Matching
- Branch Targeting

The Verbose Logs provide an additional insight into how these functionalities work in practice. It is advisable to turn this on in your CargoWise test system while developing new interfaces.

5. Universal Shipments

5.1 Overview

There are many modules in CargoWise that have a similar, common set of data. Each business area or module holds operational details specific to the tasks associated with that business area, but there are common themes across the modules. To make it easier for people to map their shipment related data to each module available in CargoWise, we have created one common Dataset that can be exported or imported from a range of modules across our product.

The following sections in this chapter deal with generic functionality related to the Universal Shipment dataset that is common across all modules. An example of this is that in all areas where the Universal Shipment has been mapped, you can configure Workflow to export Universal Shipments by configuring a trigger action of XML Universal Shipment.

Refer to our Workflow learning units in WiseTech Academy for more information on the Workflow entities mentioned above.

Module Mapping Guides describe the keys used for matching, give an overview of how the schema is mapped, and provide a list of element level mappings describing each field's location in the GUI. Refer to Module Mapping Guides.

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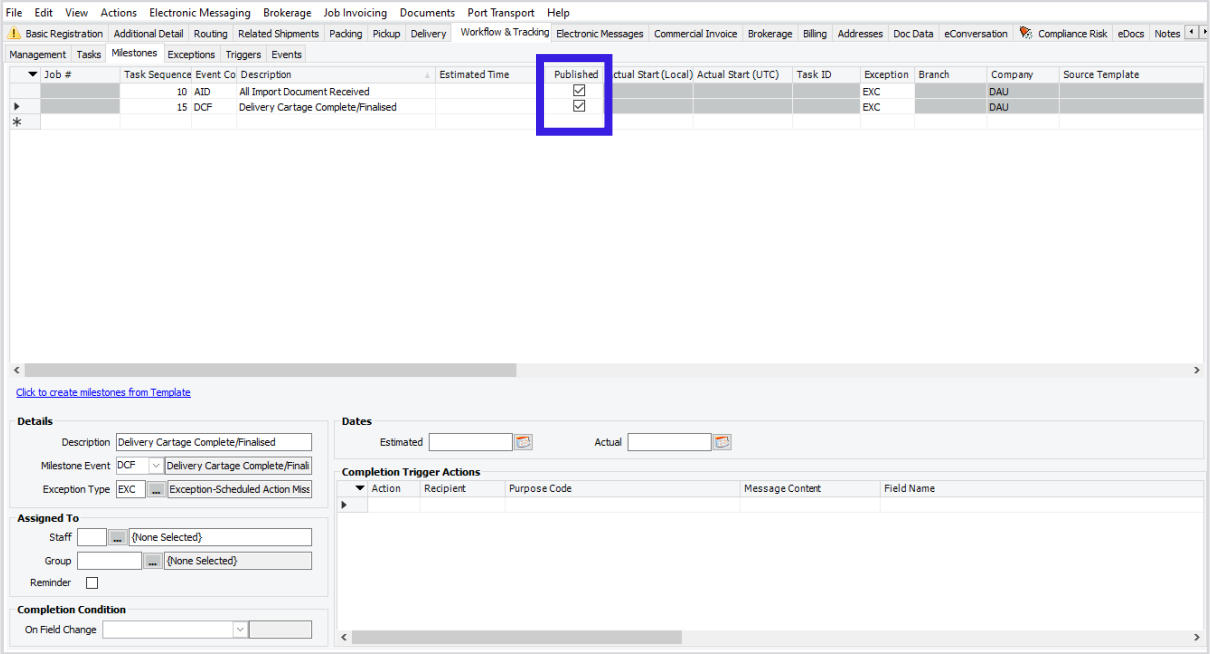
5.2 Milestones

Milestones are included in exported Universal Shipment XML under the MilestoneCollection element. Every module that can send Universal Shipments has Workflow enabled, so this means that Milestones are included from every module when sending a Universal Shipment.

The default behaviour is all Published (non-Internal) Milestones are included in outgoing XML. Each Milestone can have this set individually both on the template and on each instance of a Milestone on each job.



If you want to send both Published and Internal Milestones to a particular recipient, there is an option configurable for each Recipient Organization that allows you to include Internal Milestones as well. You can set this option when configuring the EDI Communications Mode for the sending module for the File Format of XUS (Universal Shipment).

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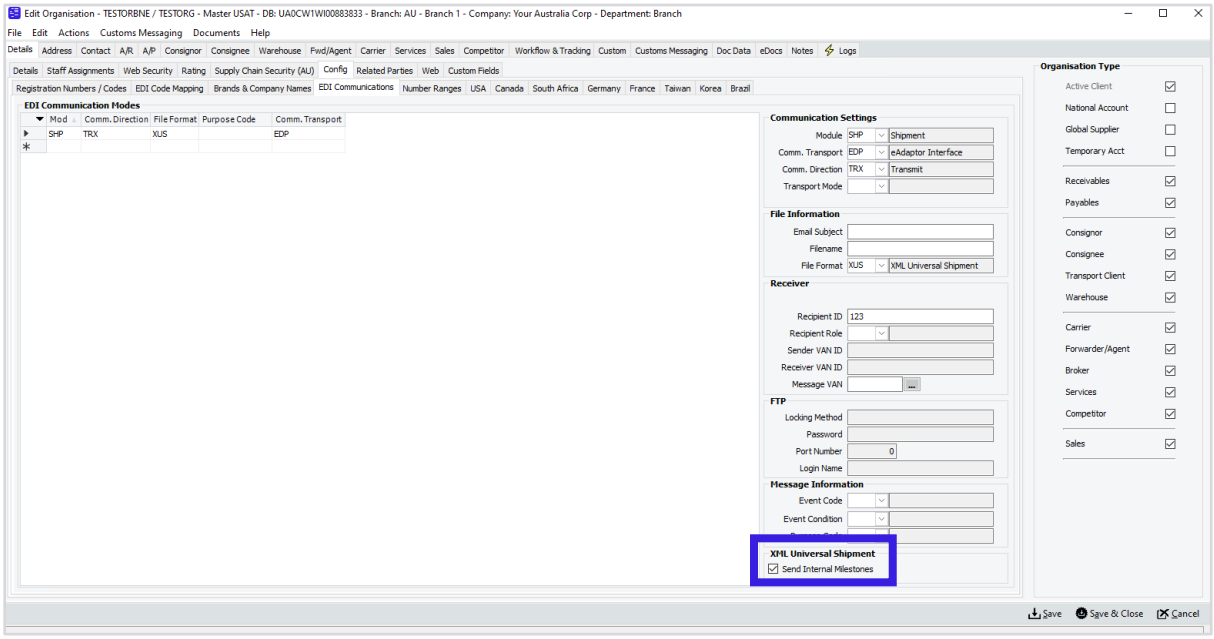

Milestones from Universal Shipment XML are never imported. Workflow is always configured from within a target system. This is to stop a sending system from controlling a receiving system's Workflow configuration along with any associated automations.

If you want to send XML Messages to mark off Milestones in a target CargoWise system, you can send Universal Events with the Event Code and Parameters matching the Milestone that has been met.

Events give a way of sending updates in a timely manner as they are sent as soon as the event itself is recorded in the host system. They will trigger any Trigger Actions configured within the receiving CargoWise system so they cause Workflow to operate, but they will not interfere with the Workflow configuration determined by the recipient.

Milestones are included so that third party systems can see a brief historical summary of the progression of a job where they are not able to handle timely updates via Universal Events. Events are always the preferred option for progress updates as they are a real time push based mechanism providing timely and scalable progress updates.

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5.3 Job Costing (Billing) and Consol Costs

Most operational modules in CargoWise have a Billing Tab which contains a list of charges applied to that job. When you send a Universal Shipment, this job costing information can be included under the JobCosting element in the Universal Shipment XML schema.

In the Forwarding Consol module, there is a Consol Costing tab instead of a Billing tab. This functions differently to the Billing tab. It is used to track consol costs, apportion costs to attached shipments, and for profit sharing. The information from the Consol Costing tab can be included in the Universal Shipment XML under the ConsolCosts element.

The JobCosting and ConsolCosts elements (known collectively as the Costing elements) share some common functionality. When exporting Universal Shipment XML, they are only included when certain conditions are met. When importing Universal Shipment XML, the Charge Lines on the Costing elements must contain ImportMetaData elements that tell the processing CargoWise system how and when to match to existing rows.

This common functionality is described in more detail in the following sections.

5.3.1 Exporting Costing elements and Charge Lines

Costing elements and Charge Lines contain sensitive information. Most companies prefer not to share information about where they make their profit with everyone. For this reason, Costing elements are not included in Universal Shipment XML by default. This prevents users from sending job profit and/or cost price information to third parties.

When sending Universal XML from CargoWise, one of the Recipient Roles you can use is ORP (Org Proxy) which is intended for use when sending Universal XML to your own in-house systems. Because it is a common and reasonable requirement for in-house systems to have access to costing information, the Costing elements are included when you send XML using the Recipient Role ORP.

5.3.2 Importing Costing elements and Charge Lines

The bulk of the information for both kinds of Costing elements is contained in the Charge Line elements. For the JobCosting element, the Charge Lines are contained in the ChargeLineCollection, and for the ConsolCosts elements, the Charge Lines are in the ConsolCostLineCollection.

Each Charge Line element imported must have an ImportMetaData element included with it, or it will be ignored upon import. The ImportMetaData element tells the target CargoWise system how to import the row, and what data to use to match to existing rows where updates or deletes are required. In short it tells the target system how to merge the supplied information along with any existing charge lines.

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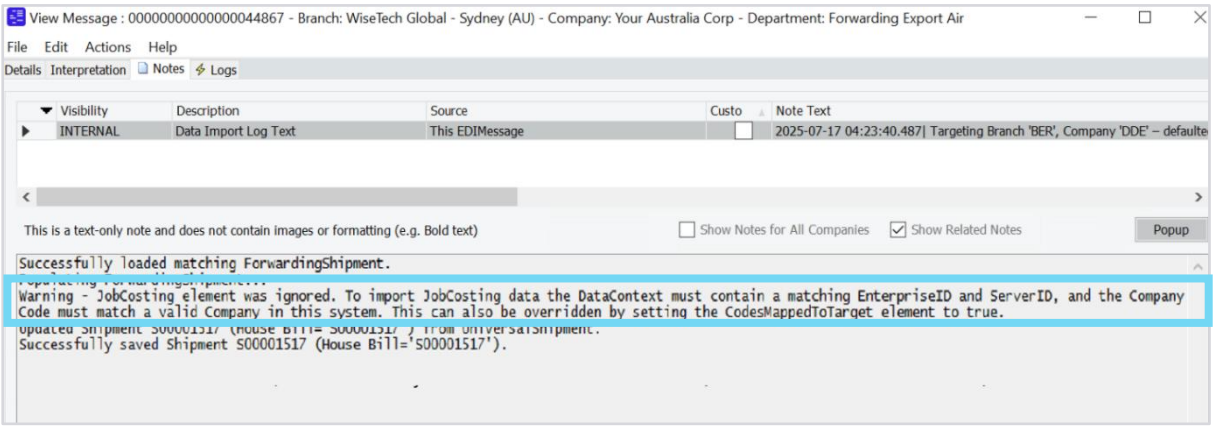
When exporting Charge Lines, CargoWise does not include ImportMetaData elements. These are added by your middleware or business processes when sending data into your CargoWise system only.

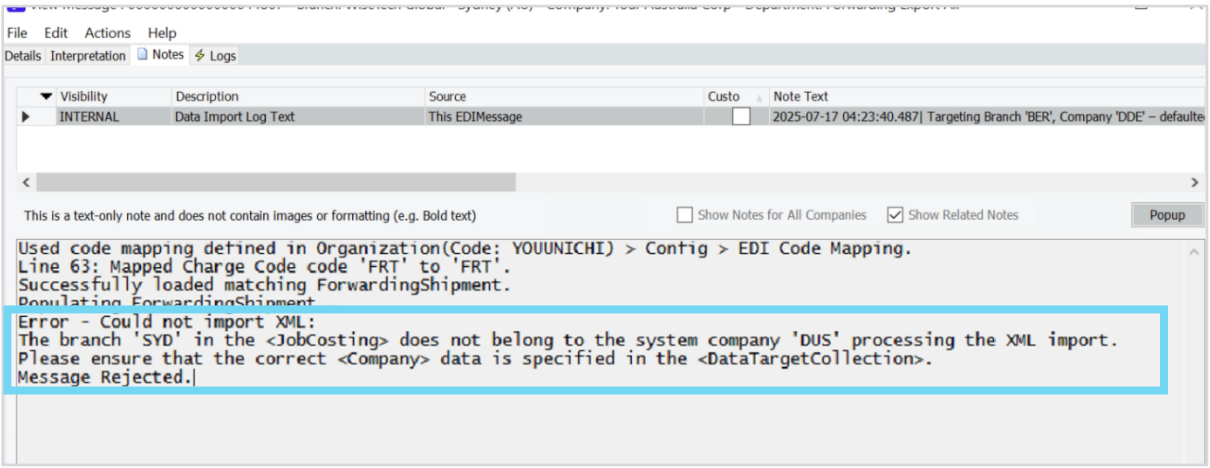
ImportMetaData is not included when exporting from CargoWise for two reasons. Exchanging charges information directly between CargoWise systems without a business process in between yields undesirable effects as your costs are not generally another party's costs. You also don't know how the recipient wants to match and account for costs in their system, so there must be some sort of business process in place to manage this kind of transfer. These are usually added by middleware.

5.3.2.1 Requisite Data Context elements

To import ConsolCosts and/or JobCosting data, as well as having a valid DataTarget, the DataContext must contain a matching EnterpriseID and ServerID, and the CompanyCode must match a valid Company in the system. In the event where any of these values are missing or invalid, the ConsolCosts and JobCosting data elements will be ignored.



In the event where the branch specified in the JobCosting data element does not belong to the company processing the XML import (i.e., CompanyCode), the EDIMessage will be rejected.



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5.3.3 Adding ImportMetaData elements

When importing Charge Lines, CargoWise needs instructions on how to import each line, depending on the instruction, matching criteria to be used when trying to find an existing line to update or delete.

There are four possible instructions that can be specified in the Instruction element within the ImportMetaData element.

- **Insert** – Add a new charge.
- **Update** – Update existing charge using matching criteria
- **UpdateAndInsertIfNotFound** – Update existing row using matching criteria if one exists, otherwise add new charge
- **Delete** – Delete existing charges using matching criteria

In the event where the ImportMetaData element is not included, the ChargeLine/ConsolCostLine element will not be imported. It will be skipped by the import process.

5.3.3.1 Matching criteria

Unless you are using the INSERT instruction, you must also supply matching criteria. MatchingCriteria elements are included in the MatchingCriteriaCollecitON element on the ImportMetaData elements as shown following.

```
<ImportMetaData>
  <Instruction>UpdateAndInsertIfNotFound</Instruction>
  <MatchingCriteriaCollection>
    <MatchingCriteria>
      <FieldName>ChargeCode</FieldName>
      <Value>FSC</Value>
    </MatchingCriteria>
  </MatchingCriteriaCollection>
</ImportMetaData>
```

Each MatchingCriteria element contains a single FieldName and Value that must match to an existing row for it to be considered a match. Where there are multiple MatchingCriteria elements provided, all criteria must be met to obtain a match to a target row.

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The following list of FieldNames can be used in MatchingCriteria.

- ChargeCode
- Creditor
- CostAPInvoiceNumber
- SupplierReference
- CostOSCurrency
- CostOSAmount
- CostLocalAmount
- CostInvoiceDate
- CostDueDate
- CostExchangeRate
- ApportionmentMethod *
- PrepaidCollectFilter *

**Applicable to ConsolCosts Only.*

When using the Update Instruction, it's important to include a set of criteria that will uniquely identify the row you want to update. Where multiple matches exist for Update Instructions provided, an error will result looking like "Error - whilst importing ... multiple charges found when updating charge line. No changes were made due to the above errors. Please fix the errors and try again." For Delete Instructions, all rows matching will be considered a match.

For this reason, sometimes it's best to include separate Delete and Insert instructions. A common case is where all charge lines containing costs from a particular supplier are Deleted, then Insert Instructions are used to add replacement lines from that supplier. Be aware that in this case, any other changes made to the rows being deleted between updates will be lost as you are throwing away the rows when you replace them.

5.3.3.2 Using the Insert instruction

The Insert instruction can be used to insert a new charge line/consol cost. It does not require any MatchingCriteria as no matching is performed.

```
<ImportMetaData>
  <Instruction>Insert</Instruction>
</ImportMetaData>
```


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5.3.3.3 Using the Update instruction

The Update instruction can be used to update an existing matching charge line/consol cost.

In this case, one or more matching criteria can be specified for the system to locate the charge line consol cost to be updated.

```
<ImportMetaData>
  <Instruction>Update</Instruction>
  <MatchingCriteriaCollection>
    <MatchingCriteria>
      <FieldName>ChargeCode</FieldName>
      <Value>FRT</Value>
    </MatchingCriteria>
    <MatchingCriteria>
      <FieldName>CostOSCurrency</FieldName>
      <Value>AUD</Value>
    </MatchingCriteria>
  </MatchingCriteriaCollection>
</ImportMetaData>
```

In the above example, the first line with a Charge Code of FRT and a Cost OS Currency of AUD would be updated. If no matching line was found, the message would be rejected as the update could not be made.

5.3.3.4 Using the UpdateAndInsertIfNotFound instruction

The UpdateAndInsertIfNotFound instruction can be used to update an existing matching charge line/consol cost. In the event where a matching charge line/consol cost cannot be found, a new line will be added.

In this case, one or more matching criteria can be specified for the system to locate the charge line/consol cost to be updated.

```
<ImportMetaData>
  <Instruction>UpdateAndInsertIfNotFound</Instruction>
  <MatchingCriteriaCollection>
    <MatchingCriteria>
      <FieldName>ChargeCode</FieldName>
      <Value>FRT</Value>
    </MatchingCriteria>
    <MatchingCriteria>
      <FieldName>Creditor</FieldName>
      <Value>FREAIRCMH</Value>
    </MatchingCriteria>
  </MatchingCriteriaCollection>
</ImportMetaData>
```

In the above example, the first line with a Charge Code of FRT and a Creditor Code of FREAIRCMH would be updated. If no matching line was found, a new charge line would be added.

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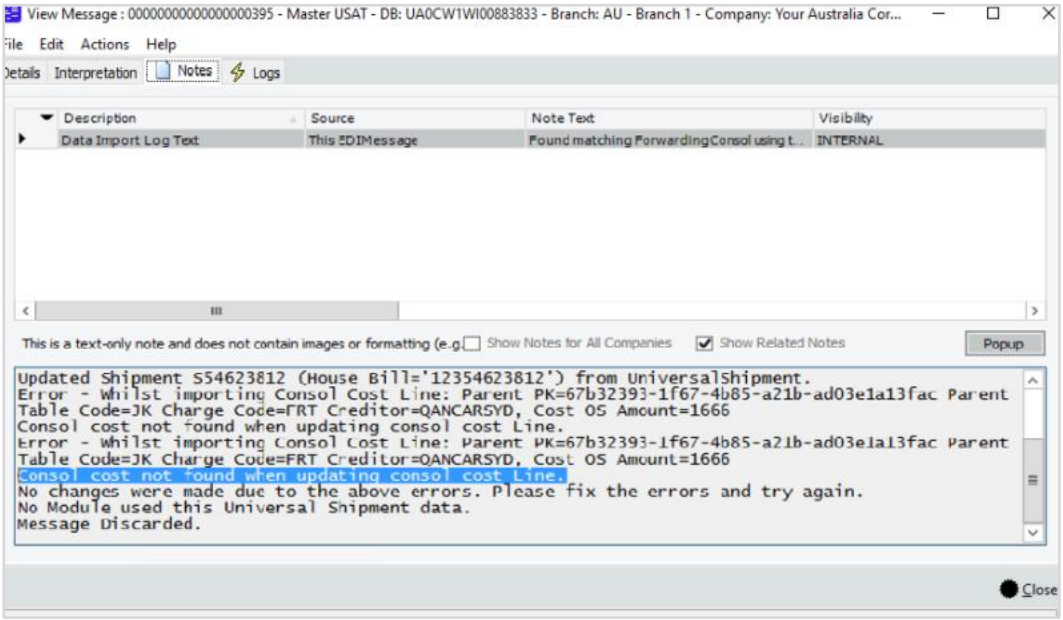
5.3.3.5 Using the Delete instruction

The Delete instruction can be used to delete an existing matching charge line/consol cost.

In this case, one or more matching criteria can be specified for the system to locate the charge line/consol cost to be updated.

```
<ImportMetaData>
  <MatchingCriteriaCollection>
    <Instruction>Delete</Instruction>
    <MatchingCriteria>
      <FieldName>Creditor</FieldName>
      <Value>FREAIRCMH</Value>
    </MatchingCriteria>
  </MatchingCriteriaCollection>
</ImportMetaData>
```

In the above example, any lines with a Creditor Code of FREAIRCMH would be deleted. If no matching lines were found, an error will be logged in the Data Import Log for the message as shown below, and the message will be discarded.



5.3.3.6 Further examples

Single Matching Criteria

```
<ImportMetaData>
  <Instruction>UpdateAndInsertIfNotFound</Instruction>
  <MatchingCriteriaCollection>
    <MatchingCriteria>
      <FieldName>ChargeCode</FieldName>
      <Value>FSC</Value>
    </MatchingCriteria>
  </MatchingCriteriaCollection>
</ImportMetaData>
```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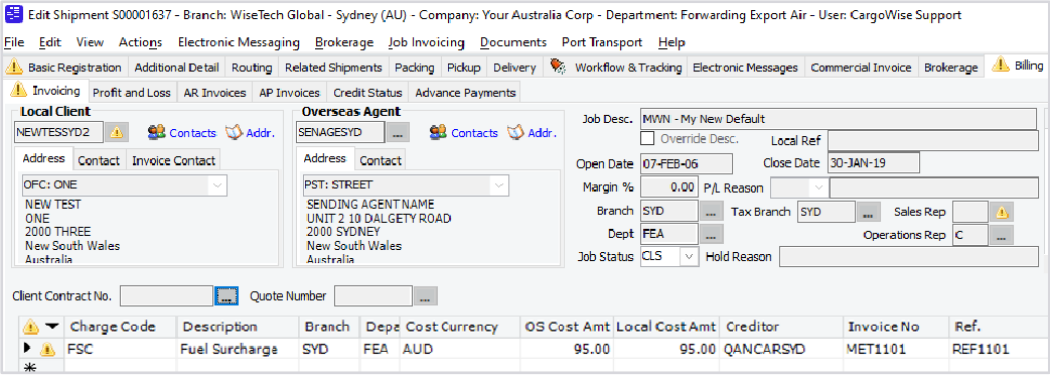
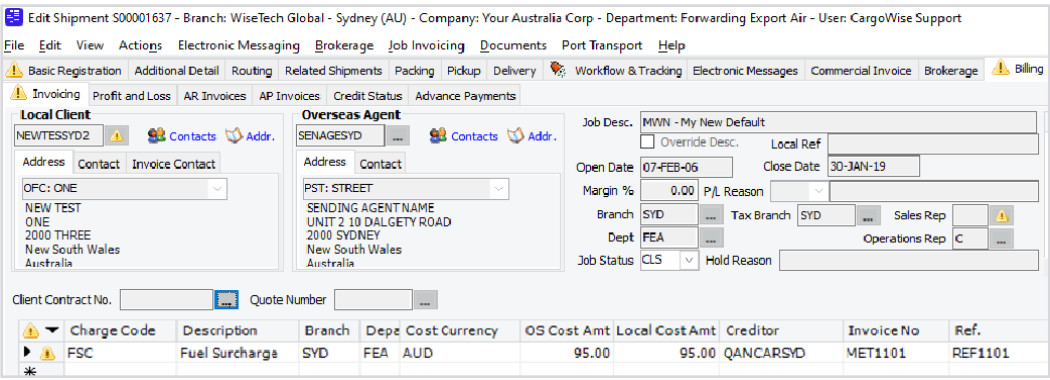
With the above matching criteria, the system will locate existing charge line with Charge code = FSC and update it if found, add a new charge if not found.



Multiple matching criteria

```
<ImportMetaData>
  <Instruction>Update</Instruction>
  <MatchingCriteriaCollection>
    <MatchingCriteria>
      <FieldName>Creditor</FieldName>
      <Value>QANCARSYD</Value>
    </MatchingCriteria>
    <MatchingCriteria>
      <FieldName>CostAPInvoiceNumber</FieldName>
      <Value>MET1101</Value>
    </MatchingCriteria>
    <MatchingCriteria>
      <FieldName>ChargeCode</FieldName>
      <Value>FSC</Value>
    </MatchingCriteria>
  </MatchingCriteriaCollection>
</ImportMetaData>
```

With the above matching criteria, the system will locate existing charge line with Creditor = QANCARSYD Invoice number = MET1101 and Charge code = FSC. If the charge line exists, it will update.



If a matching charge line is not found, an error will be logged in the Data Import Log for the message, and the message will be discarded. For this reason, when using Update, it's important to know what the current state of the data is.

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Combination Consol Cost and Shipment Billing updates

The previous samples demonstrate the use of the ImportMetaData element only. Following is a complete and more complicated sample including the Universal Shipment and multiple charge lines at both the Consol and Shipment levels.

```
<UniversalShipment xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
  <Shipment>
    <DataContext>
      <DataTargetCollection>
        <DataTarget>
          <Type>ForwardingConsol</Type>
          <Key>C300001703</Key>
        </DataTarget>
      </DataTargetCollection>
      <Company>
        <Code>DAU</Code>
      </Company>
      <EnterpriseID>WTG</EnterpriseID>
      <ServerID>XX1</ServerID>
    </DataContext>
    <ConsolCosts>
      <ConsolCostLineCollection>
        <ConsolCostLine>
          <ImportMetaData>
            <Instruction>Insert</Instruction>
          </ImportMetaData>
          <ApportionmentMethod>CHG</ApportionmentMethod>
          <ApportionToSubShipments>false</ApportionToSubShipments>
          <ChargeCode>
            <Code>FRT</Code>
            <Description>International Freight</Description>
          </ChargeCode>
          <ChargeCodeGroup>
            <Code>FRT</Code>
            <Description>Freight</Description>
          </ChargeCodeGroup>
          <CostExchangeRate>1</CostExchangeRate>
          <CostLocalAmount>1555</CostLocalAmount>
          <CostOSAmount>1555</CostOSAmount>
          <CostOSCurrency>
            <Code>AUD</Code>
            <Description>Australian Dollar</Description>
          </CostOSCurrency>
          <Creditor>
            <Type>Organization</Type>
            <Key>QANCARSYD</Key>
          </Creditor>
        </ConsolCostLine>
      </ConsolCostLineCollection>
    </ConsolCosts>
    <SubShipmentCollection>
      <SubShipment>
        <DataContext>
          <DataTargetCollection>
            <DataTarget>
              <Type>ForwardingShipment</Type>
              <Key>S54623816</Key>
            </DataTarget>
          </DataTargetCollection>
        </DataContext>
        <JobCosting>
          <ChargeLineCollection>
            <ChargeLine>
              <ImportMetaData>
                <Instruction>Update</Instruction>
                <MatchingCriteriaCollection>
```


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```
        <MatchingCriteria>
          <FieldName>ChargeCode</FieldName>
          <Value>FSC</Value>
        </MatchingCriteria>
        <MatchingCriteria>
          <FieldName>Creditor</FieldName>
          <Value>KELTRASYD</Value>
        </MatchingCriteria>
      </MatchingCriteriaCollection>
    </ImportMetaData>
    <Branch>
      <Code>SYD</Code>
      <Name>Cargowise EDI - Sydney (AU)</Name>
    </Branch>
    <ChargeCode>
      <Code>FSC</Code>
      <Description>Fuel Surcharge</Description>
    </ChargeCode>
    <CostLocalAmount>112.5000</CostLocalAmount>
    <CostOSAmount>112.5000</CostOSAmount>
    <CostOSCurrency>
      <Code>AUD</Code>
      <Description>Australian Dollar</Description>
    </CostOSCurrency>
    <Creditor>
      <Type>Organization</Type>
      <Key>KELTRASYD</Key>
    </Creditor>
    <Debtor>
      <Type>Organization</Type>
      <Key>AAASHISYD</Key>
    </Debtor>
    <Department>
      <Code>FEA</Code>
      <Name>Forwarding Export Air</Name>
    </Department>
    <Description>Fuel Surcharge</Description>
    <SellLocalAmount>135.0000</SellLocalAmount>
    <SellOSAmount>135.0000</SellOSAmount>
    <SellOSCurrency>
      <Code>AUD</Code>
      <Description>Australian Dollar</Description>
    </SellOSCurrency>
  </ChargeLine>
  <ChargeLine>
    <ImportMetaData>
      <Instruction>UpdateAndInsertIfNotFound</Instruction>
      <MatchingCriteriaCollection>
        <MatchingCriteria>
          <FieldName>ChargeCode</FieldName>
          <Value>ODOC</Value>
        </MatchingCriteria>
      </MatchingCriteriaCollection>
    </ImportMetaData>
    <Branch>
      <Code>SYD</Code>
      <Name>Cargowise EDI - Sydney (AU)</Name>
    </Branch>
    <ChargeCode>
      <Code>ODOC</Code>
      <Description>Origin Documentation Fee</Description>
    </ChargeCode>
    <CostLocalAmount>15.0000</CostLocalAmount>
    <CostOSAmount>15.0000</CostOSAmount>
    <CostOSCurrency>
      <Code>AUD</Code>
      <Description>Australian Dollar</Description>
    </CostOSCurrency>
    <Debtor>
```


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```
            <Type>Organization</Type>
            <Key>AAASHISYD</Key>
          </Debtor>
          <Department>
            <Code>FEA</Code>
            <Name>Forwarding Export Air</Name>
          </Department>
          <Description>Origin Documentation Fee</Description>
          <SellLocalAmount>25.0000</SellLocalAmount>
          <SellOSAmount>25.0000</SellOSAmount>
          <SellOSCurrency>
            <Code>AUD</Code>
            <Description>Australian Dollar</Description>
          </SellOSCurrency>
        </ChargeLine>
        <ChargeLine>
          <ImportMetaData>
            <Instruction>Delete</Instruction>
            <MatchingCriteriaCollection>
              <MatchingCriteria>
                <FieldName>ChargeCode</FieldName>
                <Value>OCART</Value>
              </MatchingCriteria>
              <MatchingCriteria>
                <FieldName>Creditor</FieldName>
                <Value>KELTRASYD</Value>
              </MatchingCriteria>
              <MatchingCriteria>
                <FieldName>SupplierReference</FieldName>
                <Value>REF1101</Value>
              </MatchingCriteria>
            </MatchingCriteriaCollection>
          </ImportMetaData>
          <Branch>
            <Code>SYD</Code>
            <Name>Cargowise EDI - Sydney (AU)</Name>
          </Branch>
          <ChargeCode>
            <Code>OCART</Code>
            <Description>Pick Up Cartage</Description>
          </ChargeCode>
          <CostLocalAmount>128.0000</CostLocalAmount>
          <CostOSAmount>128.0000</CostOSAmount>
          <CostOSCurrency>
            <Code>AUD</Code>
            <Description>Australian Dollar</Description>
          </CostOSCurrency>
          <Debtor>
            <Type>Organization</Type>
            <Key>AAASHISYD</Key>
          </Debtor>
          <Department>
            <Code>FEA</Code>
            <Name>Forwarding Export Air</Name>
          </Department>
          <Description>Pick Up Cartage</Description>
          <SellLocalAmount>138.0000</SellLocalAmount>
          <SellOSAmount>138.0000</SellOSAmount>
          <SellOSCurrency>
            <Code>AUD</Code>
            <Description>Australian Dollar</Description>
          </SellOSCurrency>
        </ChargeLine>
      </ChargeLineCollection>
    </JobCosting>
  </SubShipment>
</SubShipmentCollection>
</Shipment>
</UniversalShipment>
```


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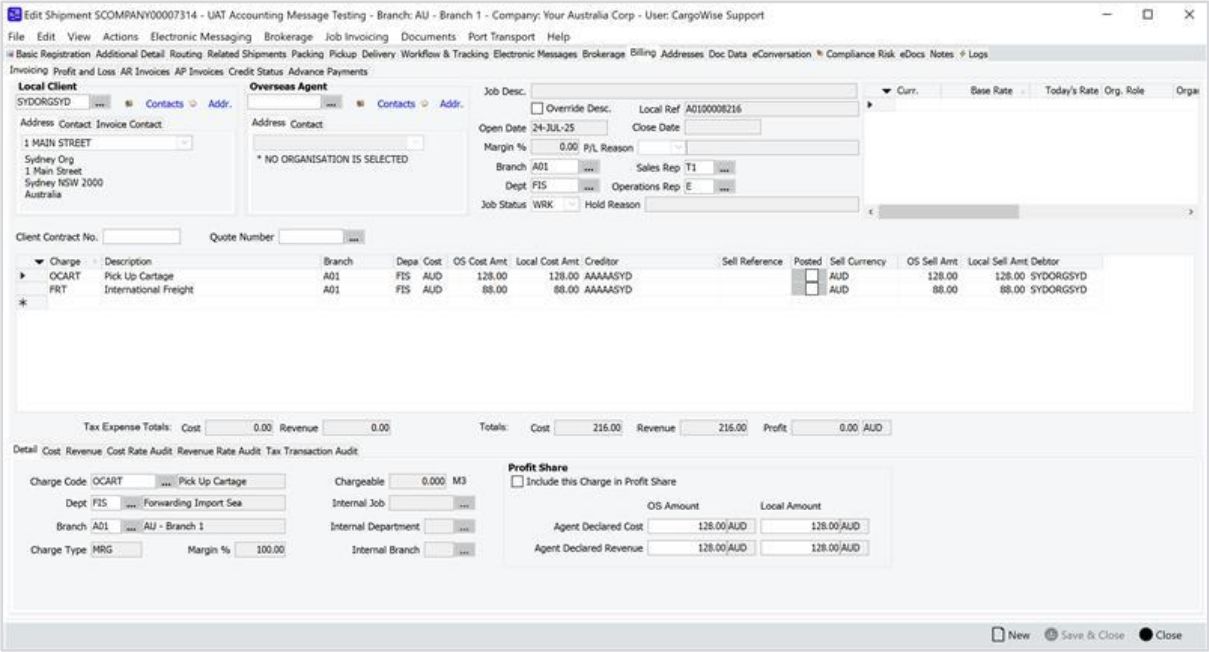
The preceding sample will target Forwarding Consol C00003103 and Forwarding Shipment SCOMPANY00007314 then:

- **Insert** a new "FRT" Consol Cost to Consol C00003103 and apportion to the shipments based on CHG – Chargeable Units
- **Update** "FSC" Shipment SCOMPANY00007314 Job Charge using matching criteria: ChargeCode 'FSC' and Creditor 'KELTRASYD'
- **UpdateAndInsertIfNotFound** "ODOC" Shipment SCOMPANY00007314 Job Charge using matching criteria: ChargeCode 'ODOC'
- **Delete** "OCART" Shipment SCOMPANY00007314 Job Charge using matching criteria: ChargeCode 'OCART', Creditor 'KELTRASYD' and SupplierReference 'REF1101'

The sample above was imported into a system where the Forwarding Consol C00003103 had no charges on the Consol Costs tab and Forwarding Shipment SCOMPANY00007314 had charges on the Billing tab as shown below.



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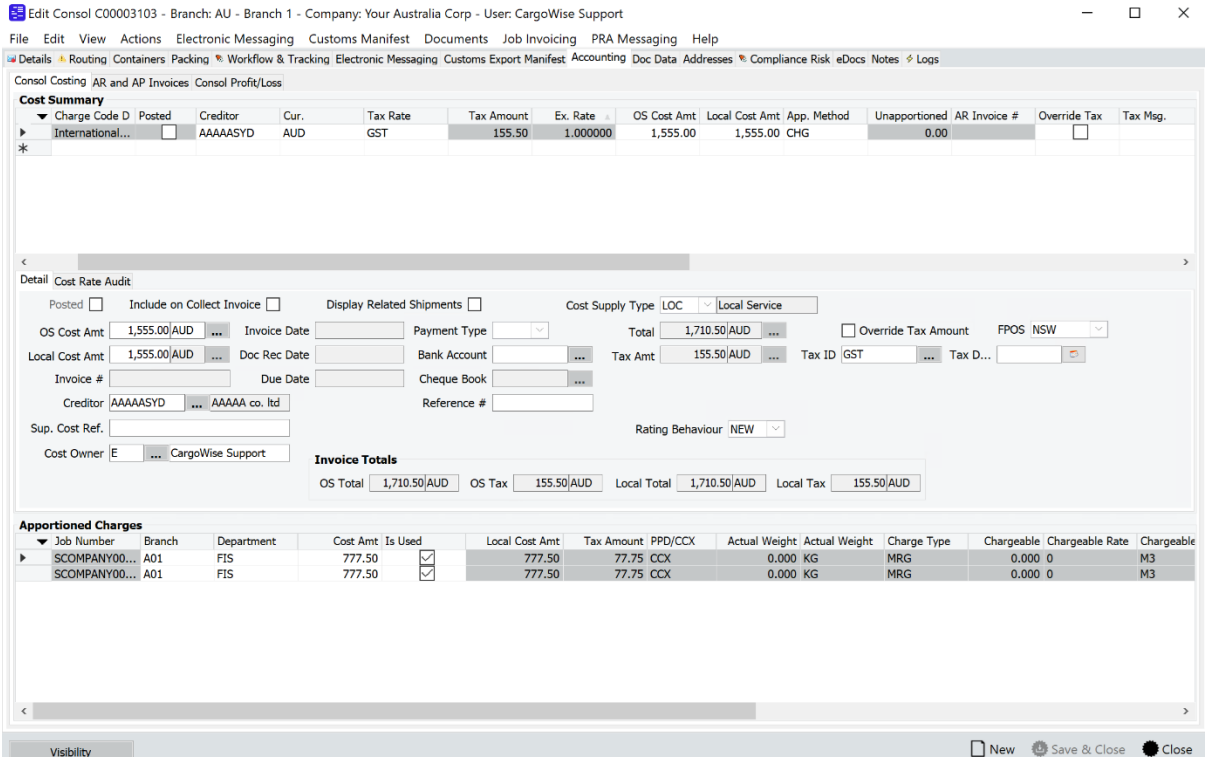
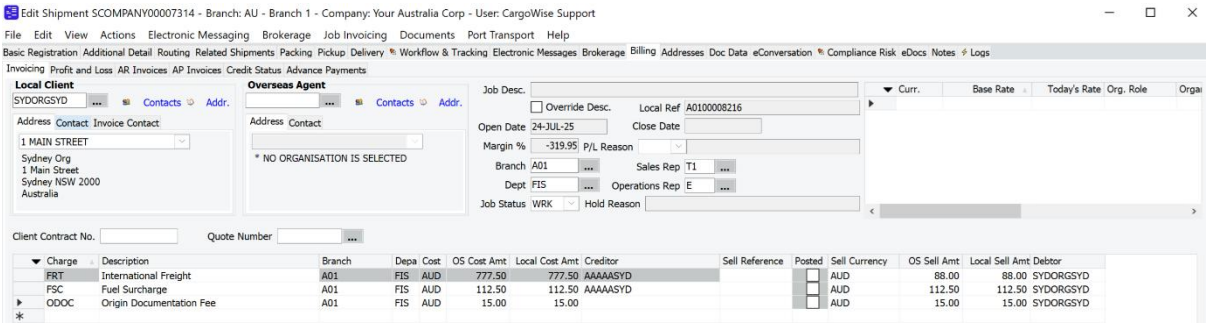
After the import, the Consol had an FRT Consol Cost inserted.



Forwarding Shipment SCOMPANY00007314 is updated as follows:

- FSC Charge Line was updated as a match was found
- ODOC Charge Line was inserted as no match was found
- OCART Charge Line was deleted as a match was found



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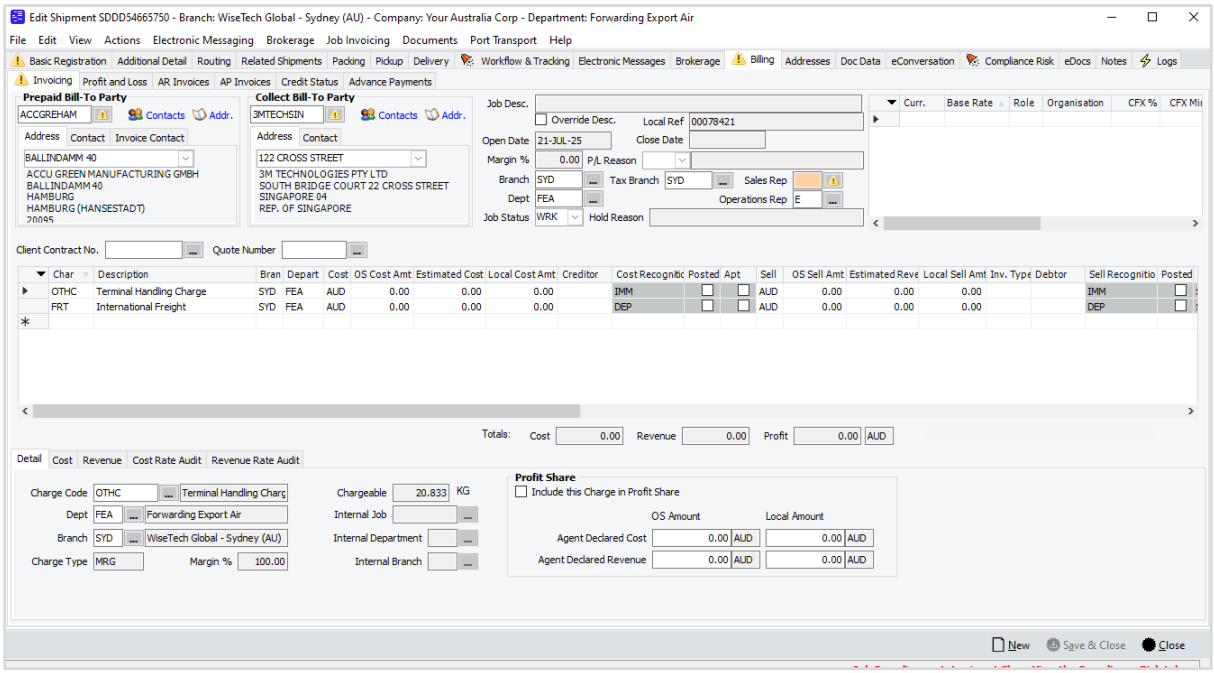
5.4 SendersLocalClient and LocalClient Organization elements

Most operational modules within CargoWise show a Billing Tab. This tab contains charge lines used for billing, along with the Branch, Department, Exchange Rates, Local Client and sometimes a secondary billing party like the Overseas Agent in the case of a Forwarding Shipment.



This information is included in the Universal Shipment XML under the JobCosting element except for the Organizations. The Local Client Organization often shows on the main job tab (as shown following on the Forwarding Shipment), so to keep things consistent the Local Client Organization was included directly under the main Universal Shipment.

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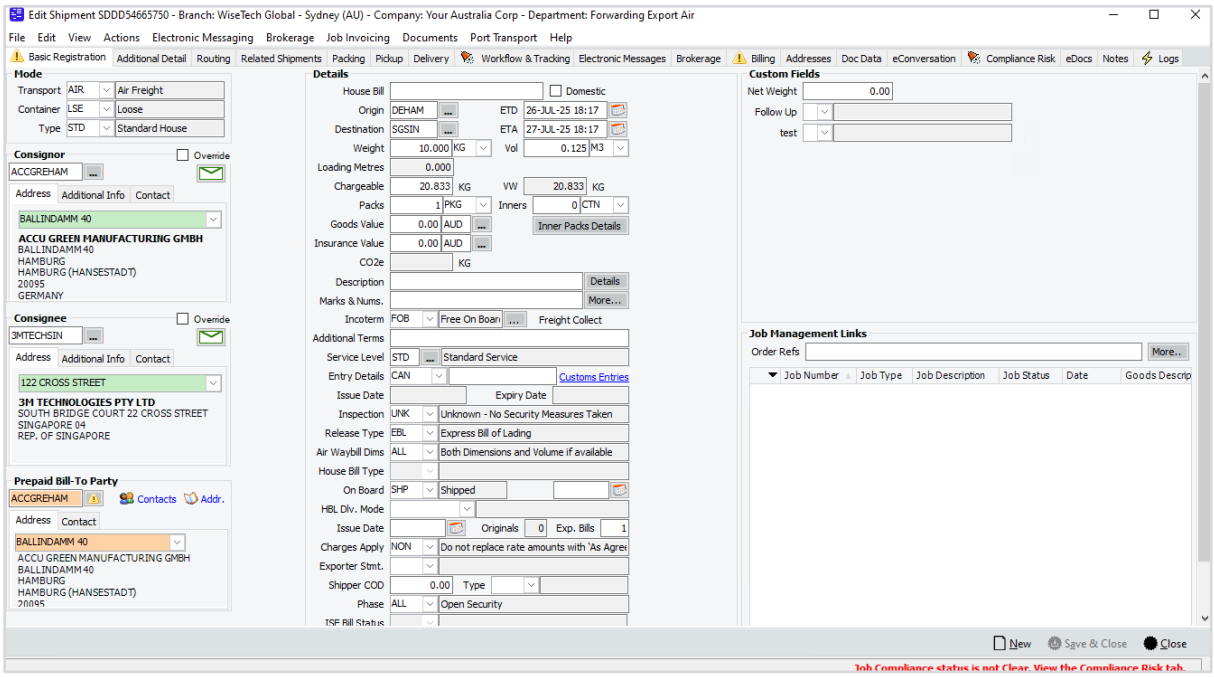

5.4.1 Exporting the Local Client

Organizations on a Universal Shipment are found within the OrganizationAddressCollection as OrganizationAddress elements. Each OrganizationAddress has an AddressType. When you send a Universal Shipment, the AddressType used for the Local Client is SendersLocalClient as shown in the sample following.

```
<OrganizationAddress>
  <AddressType>SendersLocalClient</AddressType>
  <Address1>8770 W BRYN MAWR AVE STE 1300</Address1>
  <AddressShortCode>8770 WEST BRYN MAWR AVE,</AddressShortCode>
  <City>CHICAGO</City>
  <CompanyName>UNITED ENTERPRISES (US) CORPORATION</CompanyName>
  <Country>
    <Code>US</Code>
    <Name>United States</Name>
  </Country>
  <Email>main@unitedenterprises.com</Email>
  <Fax>+18736412232</Fax>
  <OrganizationCode>UNIENTORD</OrganizationCode>
  <Phone>+18736412122</Phone>
  <Port>
    <Code>USORD</Code>
    <Name>O'Hare Apt/Chicago</Name>
  </Port>
  <Postcode>60631</Postcode>
  <State>IL</State>
</OrganizationAddress>
```


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5.4.2 Importing the Local Client

When importing a Universal Shipment, OrganizationAddress elements with an AddressType of SendersLocalClient are ignored. If you want to set the Local Client when importing Universal Shipment XML, you can use the AddressType LocalClient to do so. This means that when Universal Shipments are exported/sent, they use an AddressType of SendersLocalClient, but use LocalClient when they are imported/received.

5.4.3 Local Clients are Different in Different Business Contexts

When using Universal Shipments to transfer job data from one CargoWise system to another, it is very rare that the Local Client in one system will end up being the same as the Local Client in the other system. For the Sending Forwarder the client will be the Consignor, for the Receiving Forwarder, the client will be the Consignee. Where domestic transport is involved, the client may be a Forwarder, Consignor, Consignee, or even a Customs Broker.

These Organizations are already included in the Universal Shipment XML and are used by the receiving system to set the Local Client upon XML Import using the context of the sent Shipment and target module/recipient role. The Sender's Local Client is included in case it is of interest to the receiving system but is not used to set the Local Client in the target system.

5.4.4 Setting the Local Client Directly

Some integrations within house business management systems require the ability to set the Local Client directly. To meet this requirement, you can include an OrganizationAddress with an AddressType of LocalClient which will be stored directly in the Local Client field.

When you do this, you must make sure that you provide a Branch, or that the Branch on the Billing tab has already been set. You must also have the Department provided, or enough information in the incoming XML to be able to default the Department. The reason for this is that the Branch and Department fields are mandatory and must be filled in before any fields on the Billing tab can be set.

If there is not enough information provided to save the Billing/Job Costing information, you will see an error in the Message Processing Log like the following.

```
Error - Could not save Local Client Organization - Must have an Origin, Destination and Transport Mode to be able to calculate a Department for a Job Costing record, as the Local Client is saved on the Job Costing record.
```


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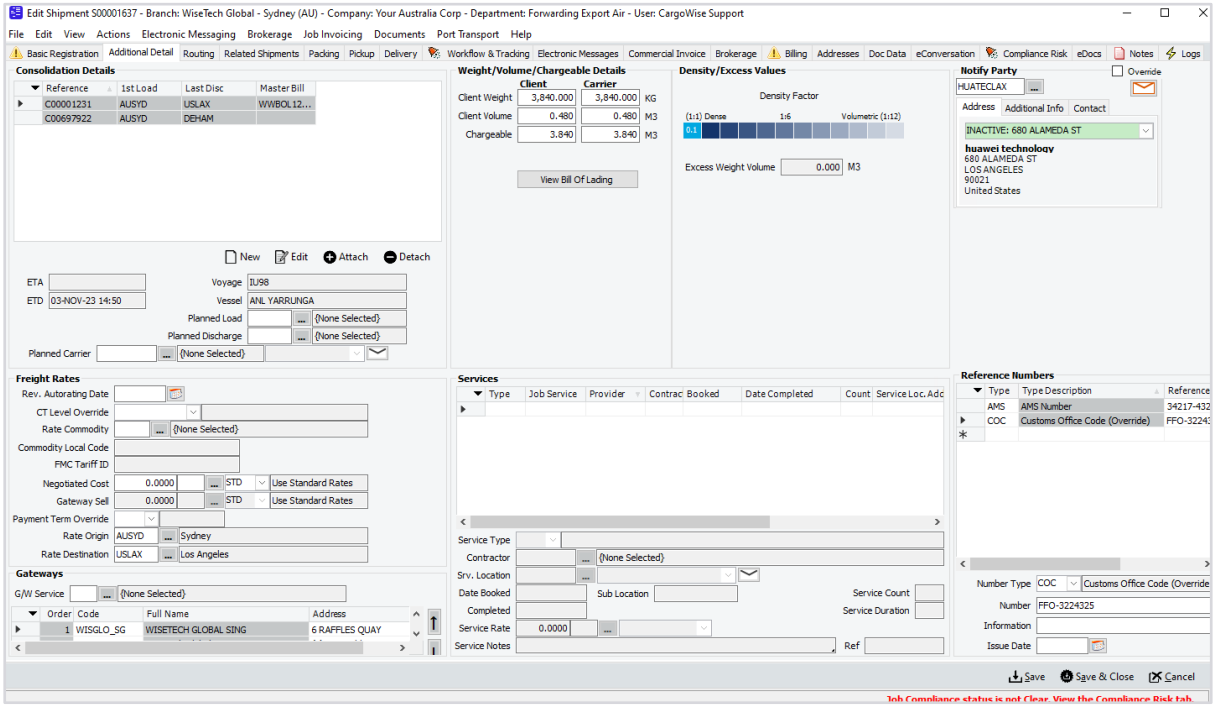
5.5 Additional References

Many modules in CargoWise have a list of Additional References where users can pre-define a list of reference types, they want to be able to track, and enter as many of these references as they like. In modules where these are available, they are included in the AdditionalReferenceCollection when exporting a Universal Shipment. One example of Additional References is the Reference Numbers grid on the Additional Detail tab on a Forwarding Shipment.



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5.5.1 Exporting

When a Universal Shipment is sent from an entity with Additional References, the Additional Reference Code is sent in the Type.Code element, and the description associated with that code is sent in the Type.Description element. The reference number itself is included in the ReferenceNumber element.

This means that if a Universal Shipment was sent from the Forwarding Shipment shown above in the previous sample, the AMS Number and Customs Office Code Override would be included as shown:

```
<UniversalShipment xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
  <Shipment>
    <DataContext>
      <DataSourceCollection>
        <DataSource>
          <Type>ForwardingShipment</Type>
          <Key>S00043996</Key>
        </DataSource>
      </DataSourceCollection>

      <EnterpriseID>CWE</EnterpriseID>
      <ServerID>PRD</ServerID>
      <Company>
        <Code>CWS</Code>
        <Name>WiseTech Global edi Pty Ltd</Name>
      </Company>
    </DataContext>
    <!-- Content Removed for Brevity -->
    <AdditionalReferenceCollection>
      <AdditionalReference>
        <Type>
          <Code>AMS</Code>
          <Description>AMS Number</Description>
        </Type>
        <ReferenceNumber>34217-812349</ReferenceNumber>
      </AdditionalReference>
      <AdditionalReference>
        <Type>
          <Code>COC</Code>
          <Description>Customs Office Code Override</Description>
        </Type>
        <ReferenceNumber>FFG3278197</ReferenceNumber>
      </AdditionalReference>
    </AdditionalReferenceCollection>
    <!-- Content Removed for Brevity -->
  </Shipment>
</UniversalShipment>
```


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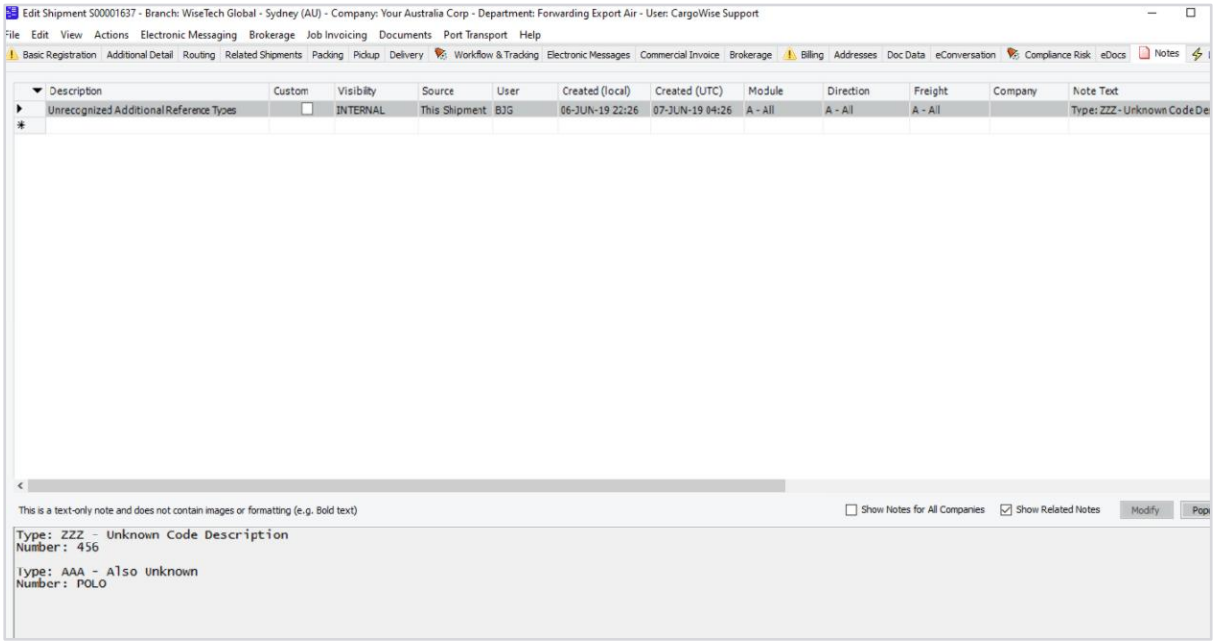
5.5.2 Importing

When Universal Shipments come back into CargoWise, any AdditionalReference elements with a Type.Code value matching a valid Additional Reference Type as defined in the registry are imported as Additional Reference Numbers.

If Additional References are sent with Type.Code values that are not present in the registry, these are recorded in a read-only Note against the updated entity with the description, 'Unrecognised Additional Reference Types'. Users can then see these references and decide whether they need to be entered in another place or can be safely ignored.



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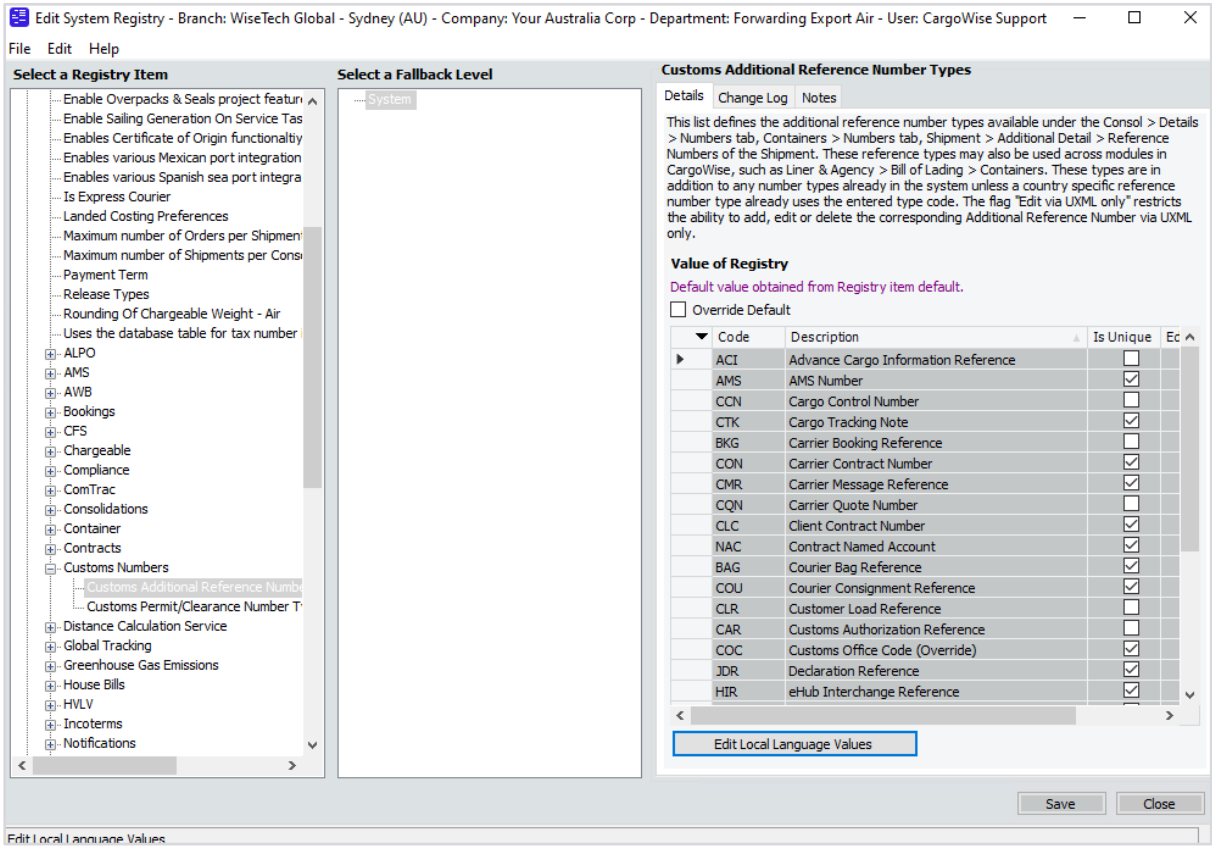
5.5.3 Registry settings

Additional Reference Types are defined in the CargoWise registry with a Code and Description as shown below:



The registry item shown above is used to define the Additional Reference Types available on a Forwarding Shipment. For more information on how to setup Additional Reference Types in a given module you should refer to the documentation for that module. For modules where the Registry settings are company specific, the Company from the DataContext element is used to load the right set of Additional Reference Types from the registry.

5.5.4 Matching on Import

If an appropriate set of Additional Reference Types is loaded from the registry, any Context Values where the Type matches one of the existing Additional Reference Types may be used for matching as part of the Combination Key Match for that module if Combination Key Matching is enabled.

Information on which modules implement Additional References, where they appear in those modules and how they are used for matching is available in the Module Mapping Guides.

If the Additional Reference code provided cannot be found in the appropriate registry item, the Context Value will not be used for Additional Reference matching.

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Module Mapping Guides describe the keys used for matching, give an overview of how the schema is mapped, and provide a list of element level mappings describing each field's location in the GUI. Refer to Module Mapping Guides.

6. Importing and Exporting Notes

Notes in Universal XML are represented by Content attribute within a NoteCollection. They appear within a number of datasets as they are commonly implemented within CargoWise.

6.1 Collection Content Attribute

All notes that are input through a universal XML are by default merged with pre-existing notes. However, the Collection Content attribute within the NoteCollection, allows for the specification of either:

- Partial – where the changes to matching notes are updated, changes to new notes are added and unmatched notes are left unchanged. A sample can be seen below:
- <NoteCollection Content="Partial">
- Complete – where the changes to matching notes are updated, changes to new notes are added and unmatched notes are deleted'. A sample can be seen below:
- <NoteCollection Content="Complete">

When importing universal XMLs the default, where not specified, is "Partial".

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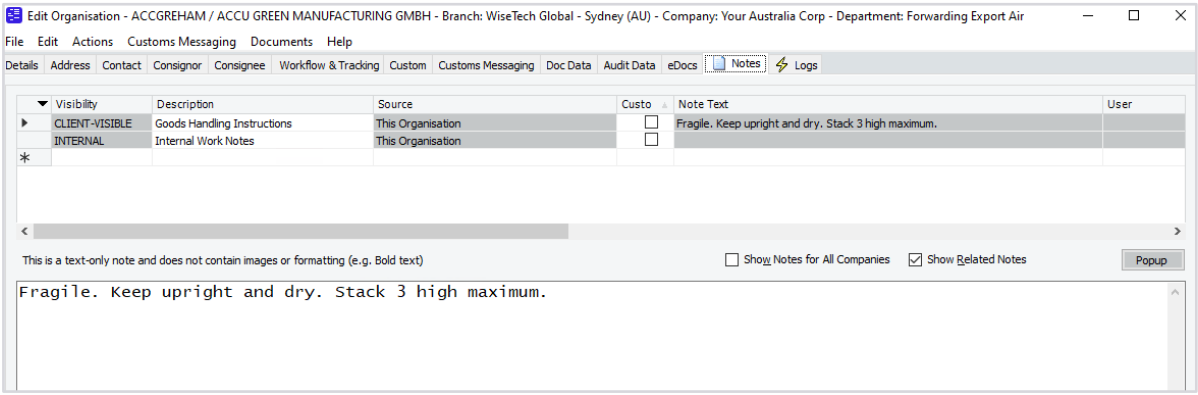
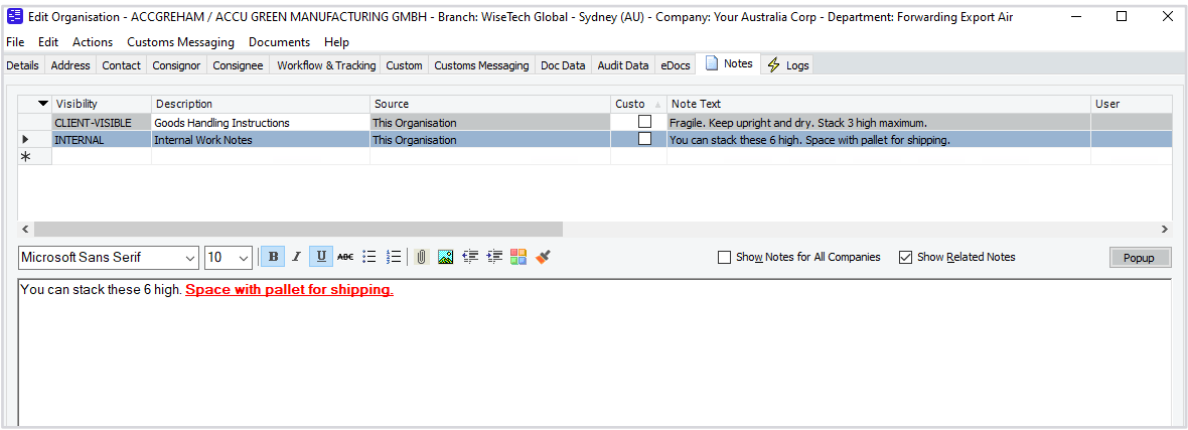
6.2 Note Types In The GUI

Looking at the Notes tab on the Organizations form following you can see examples of three different Note Types. The first one is Goods Handling Instructions.



When that is selected, you will see that above the text box where you type your note, it says the note is Text Only. This is the simplest way to tell which Note Types/Descriptions are Text Only. You will see that there is no formatting available, and all you can enter is plain text.

When you select the Internal Work Notes note, the text entry section at the bottom of the form changes to allow you to select fonts, colors, styles, and insert pictures.



This means that Internal Work Notes is a Rich Text note type. Both of these note types are pre-defined system note types, the last type of notes are Custom notes.

When you turn on the Custom flag on a note, you are allowed to enter any Description/Note Type you like as long as it doesn't match a system defined note type. All Custom Notes are automatically Rich Text notes.

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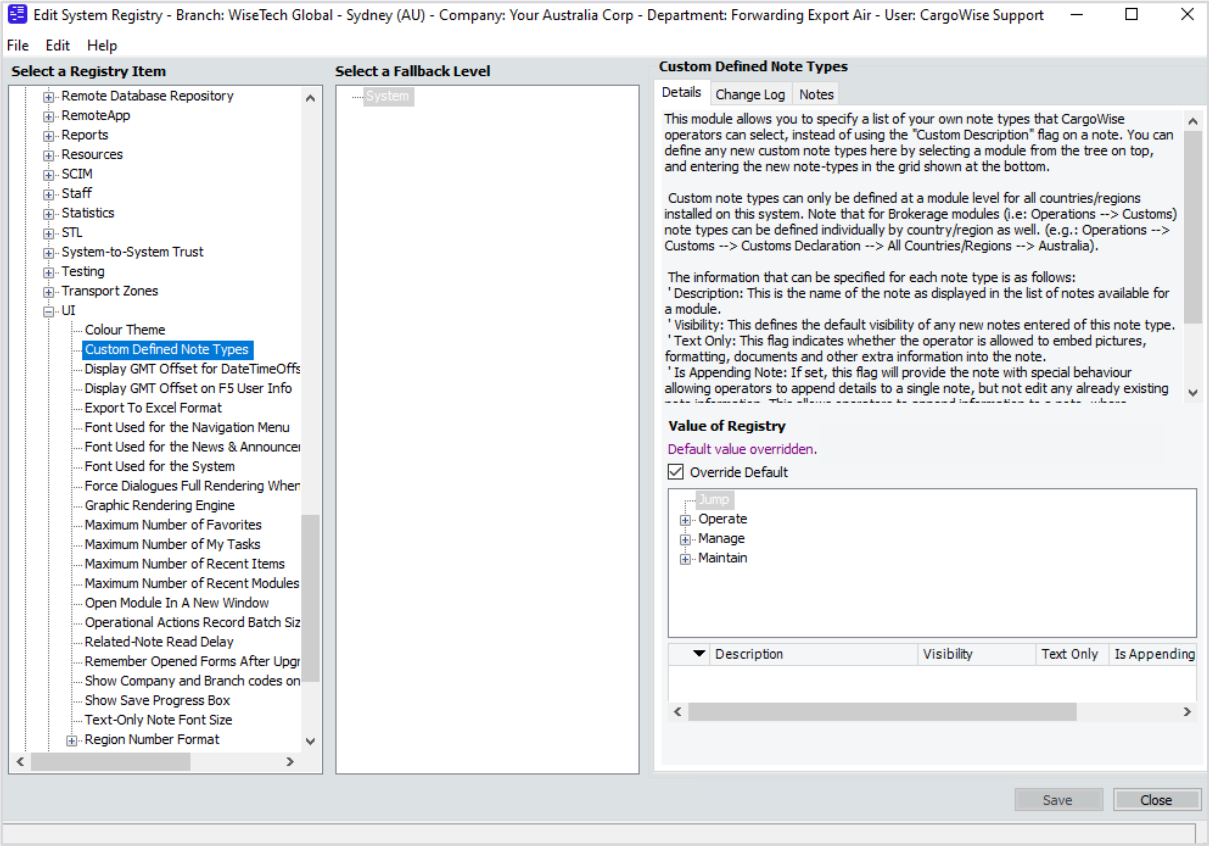
6.3 Note types Registry setting

The Note Types for each module are configured in the Custom Defined Note Types registry item

Go to **Maintain > System > Registry** then open up the sub-tree **System > UI > Custom Defined Note Types.**



When a note is imported via Universal XML, the <Description> element is used to define the Note type, which then attempts to get matched against a Custom Defined Note Types registry item, or a pre-existing system defined note type.

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6.4 Importing and Exporting Plain and Rich Text Notes via Universal XML

There are two types of notes you can have in CargoWise, Text Only and Rich Text Notes. Universal XMLs do not support Rich Text Notes. Rich text notes cannot be imported through a Universal XML and when exported they do not retain their original formatting.

An example of a Universal XML export of a Rich Text Note containing "**Bold**", "*Italic*" and "Underlined" characters can be seen in the sample below.

```
    <NoteCollection Content="Partial">
      <Note>
        <Description>Note test</Description>
        <IsCustomDescription>false</IsCustomDescription>
        <NoteText>The Bold
The Italic
The Underline</NoteText>
        <NoteContext>
          <Code>AAA</Code>
          <Description>Module: A - All, Direction: A - All, Freight: A - All</Description>
        </NoteContext>
        <Visibility>
          <Code>PUB</Code>
          <Description>CLIENT-VISIBLE</Description>
        </Visibility>
      </Note>
    </NoteCollection>
```

As seen in this sample the "**Bold**", "*Italic*" and "Underlined" characters have been reformatted into plain text and the pre-existing formatting from the job notes is not retained.

7. Universal Events

7.1 Overview

Universal Events are an XML representation of a single Event that you can see on the Events sub tab of any Workflow & Tracking tab in most operational modules throughout CargoWise. Universal Events can be sent to a CargoWise installation via the eAdaptor Web Service Interface and will cause Event logs to be added on any entity that the incoming Universal Event is matched to.

The eHub Gateway service can be used to send Universal Events to your CargoWise installation via the eAdaptor interface. This enables you to create your own Web Service based Service Bus using a tool like Microsoft BizTalk to communicate directly with the eAdaptor interface.

As Universal Events come in, they trip any Milestones and Triggers configured on the Workflow tab of the entity they are linked to and can update dates and other fields with data specified in the Universal Event on that entity.

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You can also push events out via the eAdaptor Interface using Workflow Triggers. In all areas where the Workflow & Tracking tab appears, you can configure Workflow to export Universal Events by configuring a trigger action with an Action Type of XUE Send XML Universal Event. These Universal Events are sent from CargoWise via the eAdaptor interface to either our eHub Gateway or your own internal Web Service.

Only field attached to the Business Object Schema can be updated via an XUE. Any other fields that need to be updated (For example. Customs fields) should be done via an XUS.

Refer to our Workflow learning units in WiseTech Academy for more information on the Workflow entities mentioned above.

7.2 Common implementation aspects

7.2.1 Overview

Following is a description of what you should expect to see in each element in a Universal Event when populated:

- **EventType** (mandatory) -This is the Three-character event code. Describes what the nature of the event. You will have seen some of these already on the **Workflow & Tracking > Events** tab on various Business Entities (Shipment, Consol, Declaration etcetera.) throughout CargoWise
- **EventDescription** (optional) - Human readable description of the event. May contain supplementary information to clarify the scope of the event, but this will not affect how the event is processed. Is imported 'as is' upon consumption so the user can identify the event easily
- **EventTime** (mandatory) - This is the time the event occurred, or in the case of 'Estimate' events, the time at which the event is estimated to occur. Please note that this is taken from the Event Time, not the Posted Time in CargoWise
- There is no time zone directly associated with the Event Time; it is seen as the Local Time for the location at which the event occurred. For a Departure event it would be the time zone at the original, for an Arrival event, it would be the time zone at the destination and so on. This is how most industry systems supply event data, so this is how we work with it ourselves
- **CreatedTime** (optional) - This is the time at which the event information was generated or supplied. Only used in the case where the Event Time is different from the time the event was generated. In most cases, the event should be generated at the same time the event occurred so this field will not be included, but for cases like Estimate events, this would be used so that you can tell when the estimate was made
- **DataProvider** (mandatory) - Describes the Communications Service Providers (CSP) providing the event information
- **IsEstimate** – Set to indicate an event is an estimate for a future event

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- **IsPartial** – Set when the event only relates to part of the Business Entity identified by the collection of Keys. Used for things like split masters where the airlines split one master across two flights to meet loading constraints. For example, a partial delivery would have the 'IsPartial' flag set on it, and the last and final partial delivery would not have the flag set showing that the event is fully complete
- **IsCancelled** – Not implemented
- **ContextCollection** (conditional) - Contexts have a Type and Value. These are used to narrow down the context in which the event applies to the Business Entity identified by the Keys above, and to provide additional information about the Event itself where a field for the information does not already exist on the Universal Event. The Type may optionally have a Description attribute where the Type is an abbreviation or short code representing the Type of information being sent
- **AddtionalFieldsToUpdateCollection** (optional) – Contains a list of Field Names to update (Type) along with values to update those fields with. Field Names are expected in the form of TableName.FieldName. Whenever a Universal Event is linked to an entity, if the underlying table name of that entity matches the table name specified in an AdditionalFieldToUpdate and the Field identified by name exists, that field will be updated with the Value specified. This means that things like critical dates and signatory names on PODs can be updated using Events

7.2.2 Context Values

Context Values in the ContextCollection are used by CargoWise to work out where the event should be applied throughout the system. The combination of Values provided determines what type or types of Business Entity the Event will be applied. For example, where an event had:

- **Master Bill only** – Link to a Consolidation with that Master Bill
- **Master Bill and Flight Number**– Link to a Consolidation Transport Leg. with that Flight Number
- **Master Bill and House Bill** – Link to a Shipment or Declaration with both matching
- **Master Bill and Container Number** – Link to a Container against a Consolidation. Event shown on the Consolidation
- **Master Bill, House Bill and Container Number** – Link to a Container against a Shipment or Declaration. Event shown on the Shipment or Declaration

Context Values are only sent where they are required to uniquely identify the entity a given event applies. This means that something like a Flight Number would only be included where the Event related to that flight, or a particular Transport Leg. If the Event related to the entire Consolidation, the Flight Number will not be included.

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Furthermore it is possible to specify the company and Branch selection rules while importing a Universal Event xml message. Below is a sample XML of how this may look:

```
<UniversalEvent xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.0">
    <Event>
        <DataContext>
            <DataTargetCollection>
                <DataTarget>
                    <Type>ForwardingShipment</Type>
                    <Key>SSPAR00000003</Key>
                </DataTarget>
            </DataTargetCollection>
            <Company>
                <Code>DUK</Code>
                <Country>
                    <Code>GB</Code>
                    <Name>United Kingdom</Name>
                </Country>
            </Company>
            <ServerID>PRD</ServerID>
            <EventBranch>
                <Code>LON</Code>
                <Name>Cargowise EDI - London (GB)</Name>
            </EventBranch>
        </DataContext>

        <EventTime>2023-03-08 11:11</EventTime>
        <EventType>Z21</EventType>
        <EventReference>Test Event Z36</EventReference>
        <IsEstimate>False</IsEstimate>
    </Event>
</UniversalEvent>
```

For further information on the rules for selecting the Company and Branch while importing a Universal Event XML message refer to Universal Event XML – Branch and Company

Another common usage of a Context Values is to specify the place the event occurred. A Departure event might apply to the 'first' or 'second leg.' on a Consolidation where multiple legs exist. A location based Context Value then allows us to identify which leg. has just been embarked on.

These are the Context Value Type codes we currently use for identifying entities within CargoWise:

Air Only	Multi-Modal	Other Mode Specific
MAWBNumber	MBOLNumber	VesselName (Sea)
MAWBOriginIATAAirportCode	MBOLOriginUNLOCO	VoyageNumber (Sea, Road, Rail)
MAWBDestinationIATAAirportCode	MBOLDestinationUNLOCO	

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Air Only	Multi-Modal	Other Mode Specific
IATAAirportCode	UNLOCO	
FlightDate	LegDepartureDate	
FlightNumber	LegArrivalDate	
ULDIdentification	ContainerNumber	
HAWBNumber	HBOLNumber	
HAWBOriginIATAAirportCode	HBOLOriginUNLOCO	
HAWBDestinationIATAAirportCode	HBOLDestinationUNLOCO	
	CarriersBookingReference	
	ShippersReference	
	OrderNumber	
	InterimReceipt	
	CFSReference	

These may or may not be used in each module depending on the information available within that module. For more information on which Context Value Types are included in each module when exporting and importing Universal Events, refer to the Module Mapping Guides.

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Module Mapping Guides describe the keys used for matching, give an overview of how the schema is mapped, and provide a list of element level mappings describing each field's location in the GUI. They can be found along with this document on WiseTech Academy Integration Options - Mapping Guides

You can also include any additional information you like in the ContextCollection by making up your own Context Value Type codes. Although CargoWise will only use the list above to work out which entities the Event applies to, you can add any Context Type and Value you like using Pascal Casing on the Type.

When you do this, CargoWise will show every Type and Value provided in a human readable form by inserting spaces into the key where indicated by the Pascal Naming capitalizations. There is a screenshot of how this appears in CargoWise in section 5.5.4 Events Showing in CargoWise .

Context Value Type elements can optionally include a Description attribute. This is used where either a short code or abbreviation has been used in the Type value so that a full human readable description can be specified. This description is used instead of the description generated from Pascal Naming wherever it is used. One place the Description attribute is used is where modules in CargoWise allow users to enter a list of Additional References.

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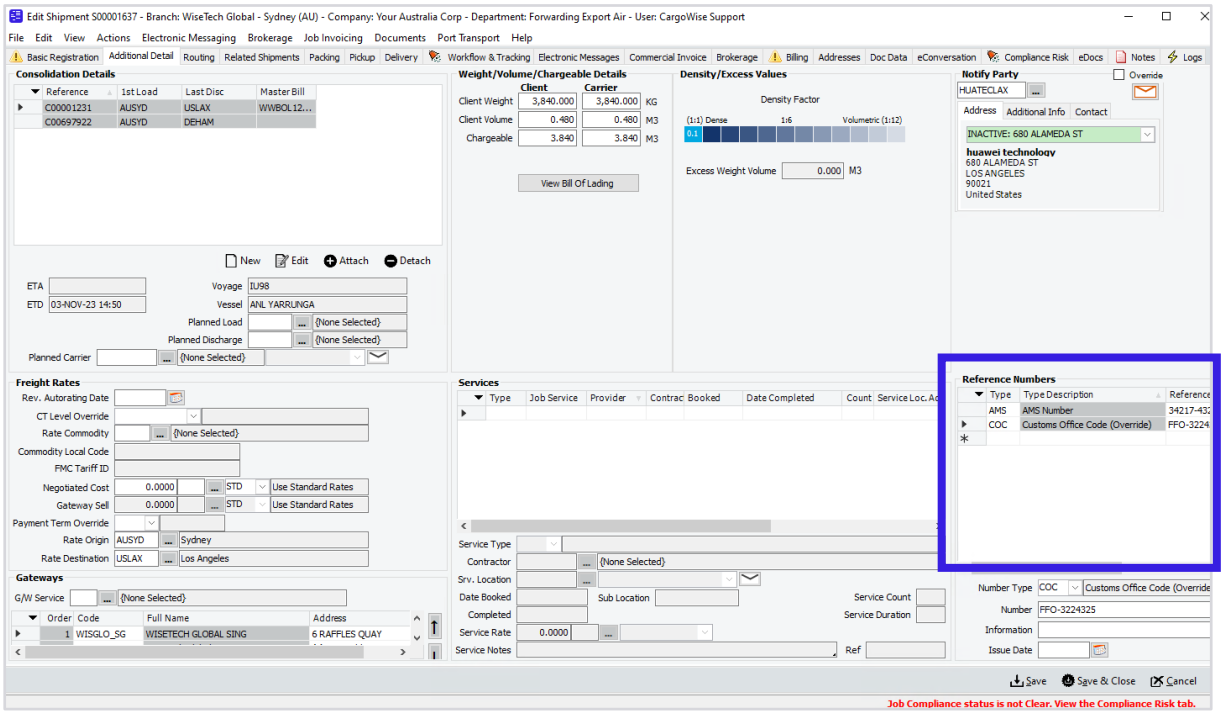
7.2.3 Additional References

Many modules in CargoWise have a list of Additional References where users can pre-define a list of reference types, they want to be able to track and enter as many of these references as they like. In modules where these are available, they are included in the Context Values when exporting a Universal Event. One example of Additional References is the Reference Numbers grid on the Additional Detail tab on a Forwarding Shipment.



When a Universal Event is sent from an entity with Additional References, the Additional Reference Code is sent in the Type element, and the description associated with that code is sent in the Description attribute on the Type element. The reference number itself is included in the Value element of the Context item.

This means that if a Universal Event was sent from the Forwarding Shipment shown above in the previous sample, the AMS Number and Customs Office Code Override would be included as shown:

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```xml
<UniversalEvent xmlns="http://www.cargowise.com/Schemas/Universal/2011/11"
version="1.0">
    <Event>
        <DataContext>
            <EnterpriseID>CWE</EnterpriseID>
            <ServerID>PRD</ServerID>
            <Company>
                <Code>CWS</Code>
                <Name>WiseTech Global edi Pty Ltd</Name>
            </Company>
            <DataProvider>CW</DataProvider>
        </DataContext>
        <EventTime>2011-06-08T19:26:18.96</EventTime>
        <EventType>ATH</EventType>
        <ContextCollection>
            <Context>
                <Type>MAWBNumber</Type>
                <Value>081-11111111</Value>
            </Context>
            <Context>
                <Type>HAWBNumber</Type>
                <Value>HAF0003278137</Value>
            </Context>
              <Context>
                <Type Description="AMS Number">AMS</Type>
                <Value>34217-812349</Value>
            </Context>
            <Context>
                <Type Description="Customs Office Code Override">COC</Type>
                <Value>FFG3278197</Value>
            </Context>
        </ContextCollection>
    </Event>
</UniversalEvent>
```

When Universal Events come back into CargoWise if the EnterpriseID and ServerID of the incoming Universal Event match the Enterprise ID and ServerID of the CargoWise installation, any Context Values with a three-character Type will used for matching to Additional References.

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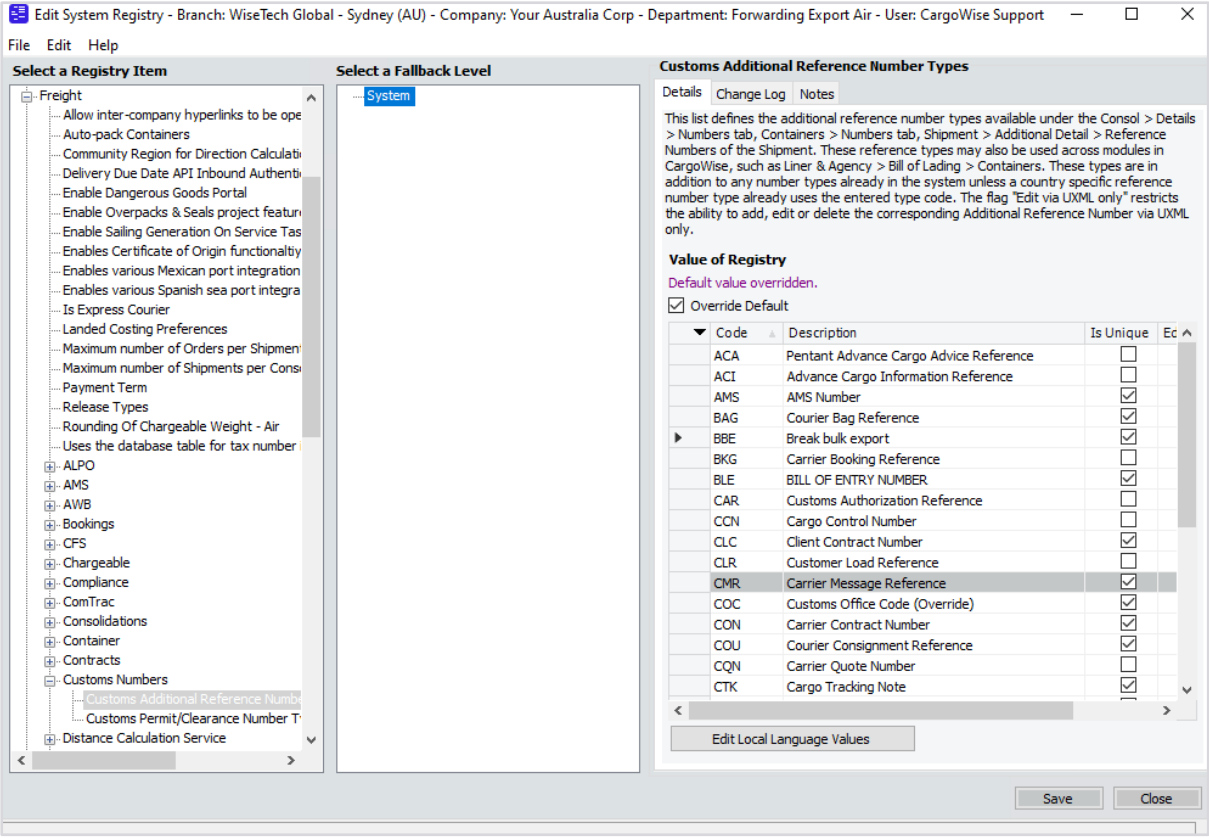
Additional Reference Types are defined in the CargoWise registry with a Code and Description as shown below:



The registry item shown above is used to define the Additional Reference Types available on a Forwarding Shipment. For more information on how to setup Additional Reference Types in a given module you should refer to the documentation for that module. For modules where the Registry settings are company specific, the Company from the DataContext element is used to load the right set of Additional Reference Types from the registry.

If an appropriate set of Additional Reference Types is loaded from the registry, any Context Values where the Type matches one of the existing Additional Reference Types will be used for matching as part of the Combination Key Match for that module. Information on which modules implement Additional References, where they appear in those modules and how they are used for matching is available in the Module Mapping Guides.

If the Additional Reference code provided cannot be found in the appropriate registry item, the Context Value will not be used for Additional Reference matching.

Module Mapping Guides describe the keys used for matching, give an overview of how the schema is mapped, and provide a list of element level mappings describing each field's location in the GUI. Refer to Module Mapping Guides.

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7.2.4 AdditionalFieldsToUpdate Collection

Sometimes when events occur, there is additional information used to supplement or substantiate the event itself. Examples of this kind of information are the name of a signatory when delivery is signed for, or a date related to the action that is being chronicled by the event.

This functionality is designed for small supplementary updates of fields directly related to the entity the Event is being applied against. It has a simple syntax, which does not allow you to update child entities by design. If you want to update more than one or two additional fields or any information on child entities, it is recommended that you use the Universal Shipment instead.

Each AdditionalFieldsToUpdate element contains the following sub elements:

- **Type**: Identifies the Type and Field name you want to update using the value specified.
- **Value**: The value to store in the field identified by the Type element.

Following is an example of a Universal Event with an Event Type of DCF (Delivery Cartage Finalised) including an Additional Field Update for the Delivery Cartage Completed Date:

```
<UniversalEvent xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
  <Event>
    <DataContext>
      <DataTargetCollection>
        <DataTarget>
          <Type>ForwardingShipment</Type>
          <Key>S00006294</Key>
        </DataTarget>
      </DataTargetCollection>
    </DataContext>
    <EventTime>2016-12-05T12:12:00</EventTime>
    <EventType>DCF</EventType>
    <AdditionalFieldsToUpdateCollection>
      <AdditionalFieldsToUpdate>
        <Type>ForwardingShipment.DocsAndCartage.JP_DeliveryCartageCompleted</Type>
        <Value>2016-12-05T12:12:00</Value>
      </AdditionalFieldsToUpdate>
    </AdditionalFieldsToUpdateCollection>
  </Event>
</UniversalEvent>
```

7.2.4.1 Component Parts of the Type

You will notice that the Type on the AdditionalFieldsToUpdate element has three parts to it with a period (.) in between each component part:

- **ForwardingShipment**: The first part is the Entity Type that this update will apply. Universal Events can be consumed by many modules, and the field names are different in each module. When linking a Universal Event to an Entity, field updates will only be attempted when the Entity Type matches the type of Entity the Event is being linked
- **DocsAndCartage**: The second part in this case identifies a related entity that the field being updated actually sits. This is optional and can only be used where there is one and only one related entity identified by the identifier specified. You cannot use this facility to update a value in related child entities as there is no way to know which child row should be updated. If you want to update information on child entities, it is recommended that you use the

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Universal Shipment instead. If we were updating a field directly on the ForwardingShipment, this part and its separating period (.) would not be included.

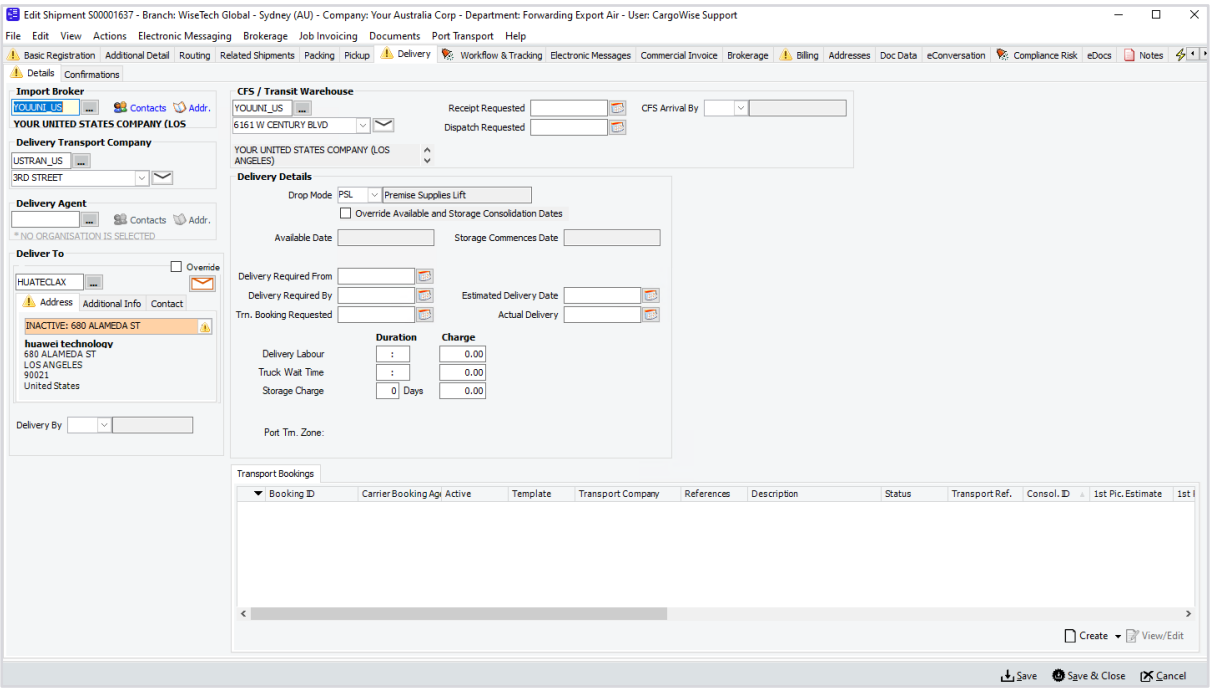
- **JP_DeliveryCartageCompleted**: The last part is always the name of the field being updated

7.2.4.2 Discovering the Type for a field

Now that you know what component parts make up the Fields Update Type, you need to know how to work out what those values are. You can work out what these are from existing samples where they are available, or you can use some of the developer tools available from within CargoWise itself.

For Universal Event sample files, see the eAdaptor Sample Interfaces Pack available from Academy Integration Options – eAdaptor Guides.

You can calculate the Field Update Type from the binding information available from the CargoWise GUI interface. To show the binding information for a field, first navigate to that field in the GUI. For this example, we will open a Forwarding Shipment, then go to the Delivery tab, and select the **Actual Delivery** field.



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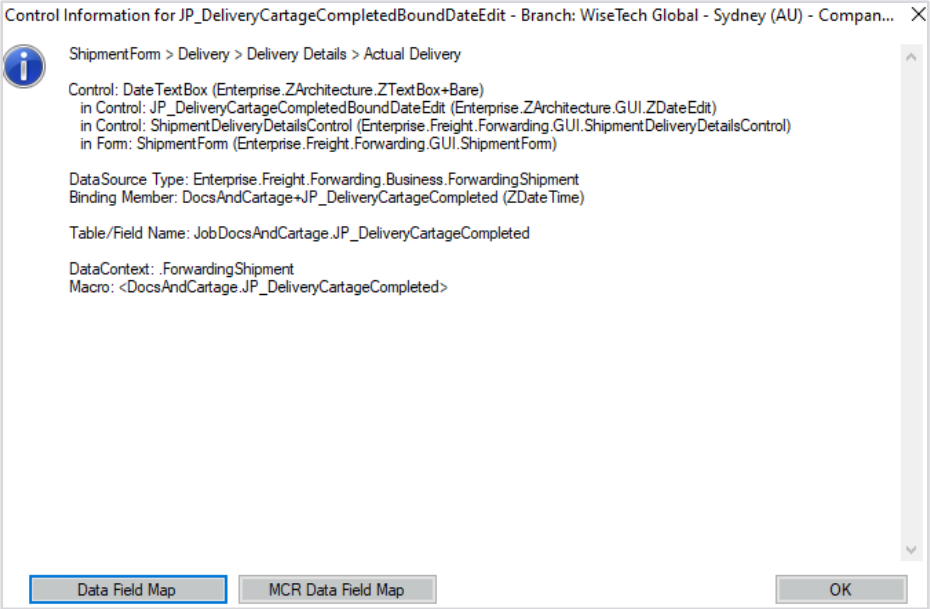
Hold down the **CTRL** and **SHIFT** keys, then press **R**. A dialog will display showing the Binding Information for the selected edit box.



The last DataSource Type and BindingMember shows the Entity Type and field binding information that is used to obtain the FieldToUpdate Type as follows.

- **DataSource type:** Enterprise.Freight.Forwarding.Business.ForwardingShipment
- **BindingMember:** DocsAndCartage+JP_DeliveryCartageCompleted

The first part of the AdditionalFieldsToUpdate Type is obtained from the last part of the DataSource type. In this case, that's ForwardingShipment. The rest of it comes from the BindingMember using a period (.) delimiter between each part. In this case that means we insert a period between the first part and the binding member, then replace the '+' from the binding member with a period. The result is then:

ForwardingShipment.DocsAndCartage.JP_DeliveryCartageCompleted

7.2.4.3 Formatting values for the Value element

The data you provide will be converted to the format required by the field you are trying to update. Although the type of the Value element is a string in the XSD, you should provide the content as though the Value element was actually defined as the XML type corresponding to the type of the field you are updating. This means that if you are trying to update a date field, you should use the XML Date format when formatting your content. The same applies for booleans, integers and decimals.

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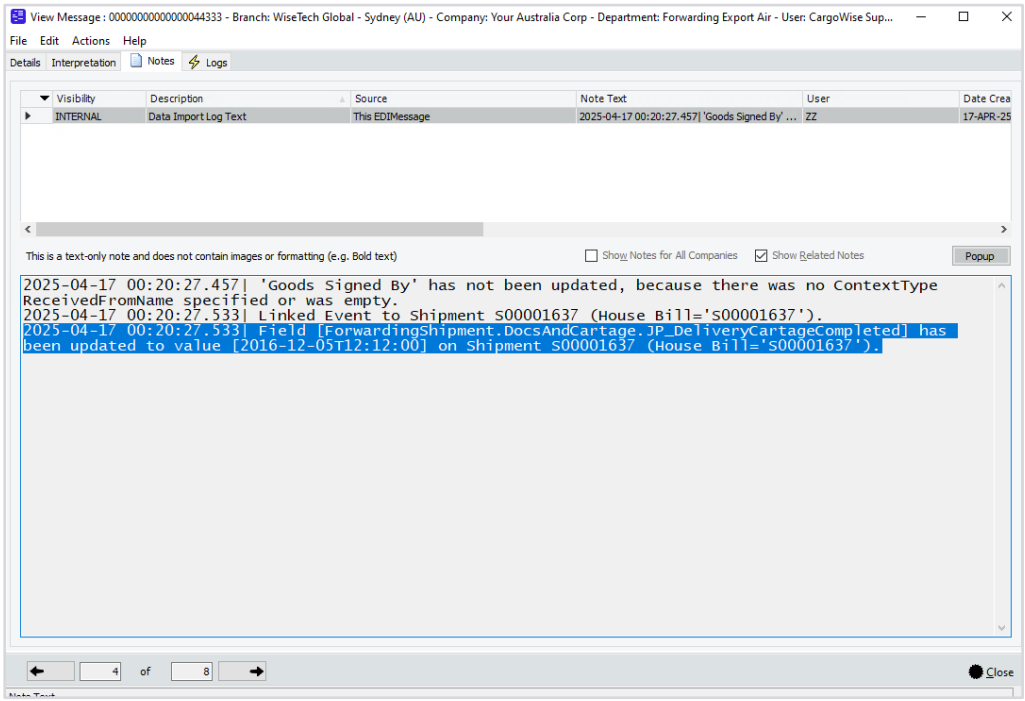
7.2.4.4 Checking that the update occurred

When field updates are successfully made using the AdditionalFieldsToUpdate elements, this is logged in the Data Import Log Text note on the EDI Message itself.



You will see from the message in the example above that the field update was successful. If the first part of the AdditionalFieldsToUpdate Type matches the module being targeted but the field does not exist, or the content in the Value element is not of the right type to be stored in the target field, an error will be shown.

If the first part of the AdditionalFieldsToUpdate Type does NOT match the module being updated, no error will be shown as updates are only attempted when the module being updated matches the type.

When the update is a success, go to the Forwarding Shipments module, open the Shipment being updated and click on the Delivery tab. You will see the Actual Delivery field has been updated with the date from the Value element in the AdditionalFieldsToUpdate element.

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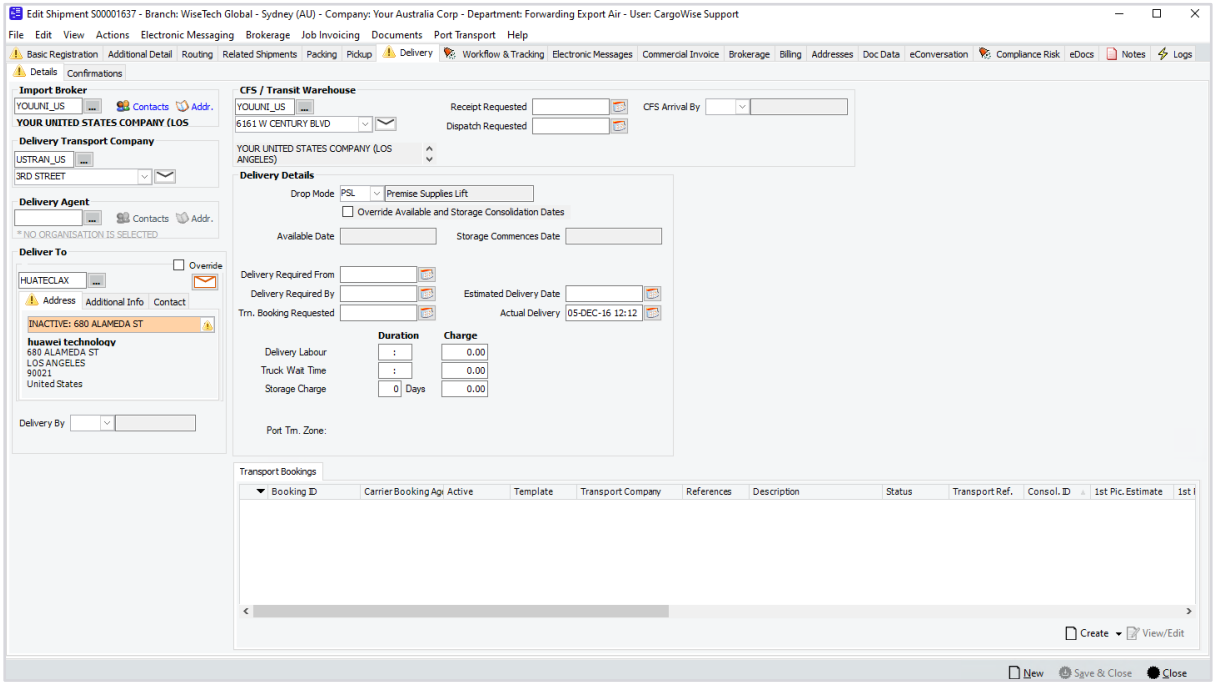

7.3 Events Showing in CargoWise

After the initial configuration work is done, Events flow into the EDI Messages module via the eAdaptor Interface from eHub or your own internal Gateway. CargoWise automatically parses each incoming Universal Event and attempts to locate the appropriate Business Entity to link the Event to.

If a matching Business Entity is found, the Event is added to the list of Events showing on the Events sub tab under the Workflow and Tracking tab on that Business Entity. An example of this is shown below for an FSU/RCS message translated to a CargoWise Universal Event with a type of DLV –Delivered:

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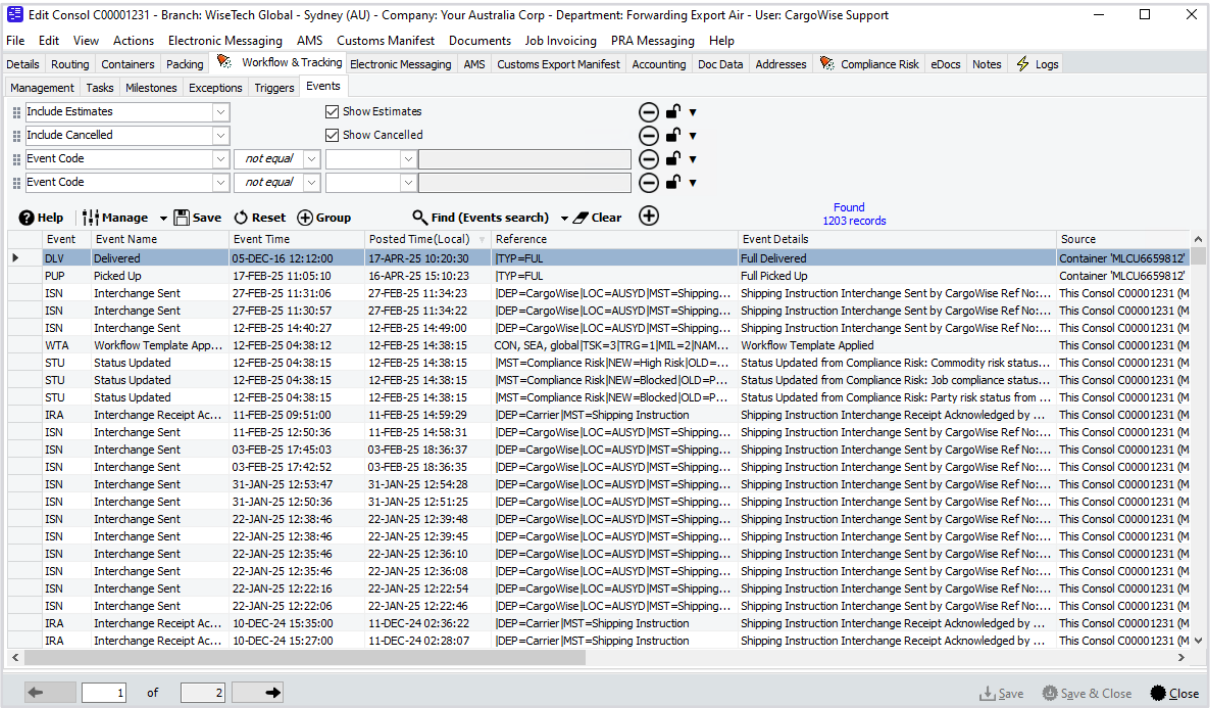

You will note that when you select an Event that was created from an external Universal Event, the Event Context Information grid appears showing all of the Context information that came along with the Event from the Event Provider.

For Business Entities such as Containers that always reside on a parent Business Entity like a Consolidation or a Declaration, the Event will appear on the Events sub tab of the parent Business Entity with the Source column identifying the container the event relates.

7.3.1 Effects an Event has

Once the event has been matched and added, it will be fired so that any Key Date Updates that are supposed to respond to that event will be fired, and any Triggers or Milestones that have been setup for that event will respond.

7.3.2 Event Cascading

When an Event is fired on a Parent Business Entity, the effect of that event on Milestones and Triggers will cascade down to any Milestones and Triggers defined on any Child Business Entities where the Workflow & Tracking tab appears. An example of this would be on the Forwarding Consolidation where any Events raised on the Consolidation (parent job) will now trip Milestones and Triggers on any Shipment (child job) related to that Consolidation. This is called Event Cascading.

When you go to the Workflow & Tracking tab, Events sub tab within a Shipment linked to a Consolidation, you would already have seen Events from the Consolidation showing up on the Shipment. Previously these would not have tripped any Triggers or Milestones on the Shipment.

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Now any events raised on the Consolidation will cascade down to trip Triggers and Milestones defined on the Shipment so that as well as showing in the log, they can also trigger Actions and complete Milestones on the Shipment.

This means that when a DEP - Departed event is raised on the Consolidation, each Shipment linked to the Consolidation is checked for Triggers and Milestones, and if any of those are configured to respond to the Departed event, they will be tripped and completed as is appropriate. If there are events that you do not want to Cascade, you can apply a filter for that Event Type as described above.

At this point Event Cascading only works with Consolidation Events cascading down to Shipment level Workflow, and Shipment Events cascading down to Sub-Shipments, but this will be implemented in other areas of CargoWise over time. Where applicable, Events will continue to cascade down wherever there are more related child objects that have Workflow on them to respond to cascaded events.

7.3.3 Event propagating

When an Event is fired on a Child Business Entity, the effect of that event on Milestones and Triggers will propagate up to any Milestones and Triggers defined on any Parent Business Entities where the Workflow & Tracking tab appears.

This is like the Event Cascading described above, but instead of the Events cascading down, they are propagating up. The main difference is that when propagating up, Milestones and Triggers and only tripped when ALL Child Business Entities on the Parent have raised that same event.

This means that if an Unpacked Event was raised on a Shipment, the Shipment will notify the Consolidation to say that it has had the Unpacked event raised. The Consolidation will then check all of the other Shipments linked to that Consolidation and if they ALL have the Unpacked event on them, will raise an Unpacked event against the Consolidations with a description showing:

Internal Event: All Shipments: Unpacked

Propagation will work in the following places:

- **Shipment Events > Parent Consolidation**
- **Container Events > Parent Shipment**
- **Container Events > Parent Consolidation**

At this point Event Propagation will only go one tier up from the place the Event is raised. This is to stop Container level Events getting double propagated to Consolidations via the Shipments and direct to the Consolidations as well.

Propagation will be implemented in other modules of CargoWise over time. In some of those areas Events may propagate up multiple levels if there's a business requirement to do so but at this point there's only been a need to propagate up one level.

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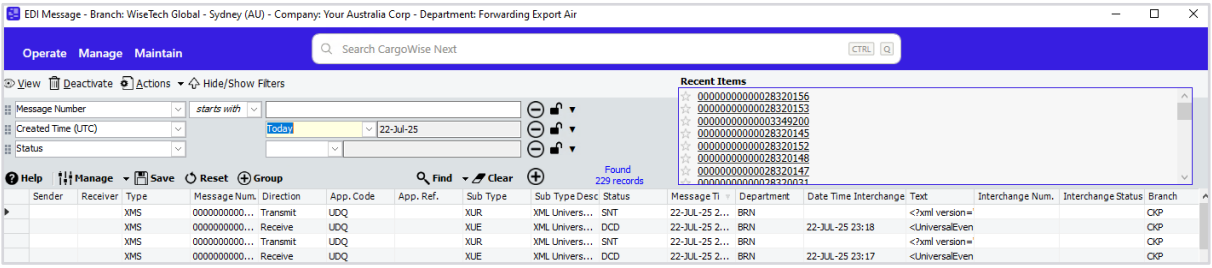
7.3.4 Monitoring Events coming In

All incoming Universal Events show under **Maintain > System > EDI Messages,** so you can see an overview of all Events coming in.



There are also three registry items you can configure that turn on email notifications when eHub XML events arrive in CargoWise. These would generally only be used for debugging purposes while configuring your own Service Bus.

eHub > Notifications > Incoming Event Matched Recipient will trigger an email to all parties specified by this registry item when an Event is successfully Matched to a Business Entity. Options are:

- **None** (default setting) – No email as long as the Event is Matched
- **Operator** – Email sent to the Operator of the Matched Business Entity
- **Email Group** – Email sent to a Specified Group
- **Operator and Email Group** – Email sent to a Specified Group and the Operator of the Matched Business Entity

CargoWise determines who the Operator is by looking at the Business Entity to see if there's an explicit Operator field (e.g., Declaration has 'Broker') then falling back to the last person to edit the Business Entity where nothing more explicit is available.

An Event Matched email contains:

- The Business Entity Types and Reference Numbers that the event was linked (e.g., "Consol COOOO3427" or "Container CRXU9378472")
- A URL Link to any Business Entities showing the event on the Workflow & Tracking tab Events sub tab. For Containers this would be the parent Consol or Declaration
- Information from the XML Event itself including all Fields, Keys, and Contexts

eHub > Notifications > Incoming Event Matched Group

- Allows you to specify a Group, used when Group is one of the options selected in the registry item above. The default setting is PMG - Post Masters

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eHub > Notifications > Incoming Event Not Matched Group

- Will trigger an email to the Group specified by this registry item when an Event arrives from eHub but is not matched to any known Business Entity. The default setting is PMG - Post Masters
- If you want to turn off this notification, simply remove the group from the registry item

An Event Not Matched email contains all information from the XML Event itself including Fields, Keys and Contexts, but no other information as no match was found.

Refer to How can I get email notifications for Inbound Universal XML failures?

7.4 Sending eDocs to DocManager using Universal Event XML

Documents can be uploaded to the eDocs of records by specifying "DocManager" as the DataTarget type of a Universal Event. As with importing documents using the DocManager Import Service task, documents can be allocated based on filename allocation details or barcodes within the document.

UniversalEvents Messages supports functionality to upload documents to the eDocs tab using type of DocManager. The document allocation details (i.e., Document Type, Job Number) for DocManager can either be provided from barcodes within the base64 encoded document or from the details provided in the `<FileName>` element. When using barcodes for allocation, the barcodes can only be processed from TIF or PDF files. They must also be on separate pages and scanned in the order of: *Job Type barcode > Doc Type barcode > document to allocate*.

It is possible to send multiple documents in a single UniversalEvent message. When sending multiple documents, each document will reside within a separate `<AttachedDocument>` element in the `<AttachedDocumentCollection>` element.

In below example below, there are two documents being sent in the UniversalEvent message. The first document contains barcodes within the base64 encoded document that provide the allocation details. The second document contains the allocation details in the filename.

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```xml
<UniversalEvent xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
    <Event>
        <DataContext>
            <DataTargetCollection>
                <DataTarget>
                    <Type>DocManager</Type>
                </DataTarget>
            </DataTargetCollection>
            <Company>
                <Code>DAU</Code>
            </Company>
            <EnterpriseID>WUT</EnterpriseID>
            <ServerID>UAT</ServerID>
        </DataContext>
        <EventTime>2016-12-05T12:12:00</EventTime>
        <EventType>DDI</EventType>
        <AttachedDocumentCollection>
            <AttachedDocument>
                <FileName>Client Agreement.tif</FileName>
                <ImageData>SUkqAB5AAACAPe --- base64 encoded binary stream --- kKJSVFT0YK</ImageData>
                <IsPublished>true</IsPublished>
                <VisibleCompanyCode> </VisibleCompanyCode>
                <VisibleBranchCode> </VisibleBranchCode>
                <VisibleDepartmentCode> </VisibleDepartmentCode>
            </AttachedDocument>
        </AttachedDocumentCollection>
        <AttachedDocumentCollection>
            <AttachedDocument>
                <FileName>[SHP BKC S00001493]Client Agreement.tif</FileName>
                <ImageData>SUkqAB5AAACAPe --- base64 encoded binary stream --- kKJSVFT0YK</ImageData>
                <IsPublished>true</IsPublished>
                <VisibleCompanyCode> </VisibleCompanyCode>
                <VisibleBranchCode> </VisibleBranchCode>
                <VisibleDepartmentCode> </VisibleDepartmentCode>
            </AttachedDocument>
        </AttachedDocumentCollection>
    </Event>
</UniversalEvent>
```


Document Attributes available are:

- **FileName** – Provide the Filename of the Document. Also, using filename, details for allocation can be included within this element.
  For example: `<FileName>[SHP BKC S00001493]Client Agreement.tif</FileName>`
  This will allocate the document in Shipment # S00001493 as Document Type "BKC".
- **ImageDate** – base64 encoded document content.
- **IsPublished** – Flag the document as Published or not.
- **VisibleCompanyCode** – If the CompanyCode provided, the document will be allocated Company Specific and will only be visible for Specified Company.
- **VisibleBranchCode** – If the BranchCode provided, the document will be allocated Branch Specific and will only be visible for Specified Branch.
- **VisibleDepartmentCode** – If the DepartmentCode provided, the document will be allocated Department Specific and will only be visible for Specified Department.

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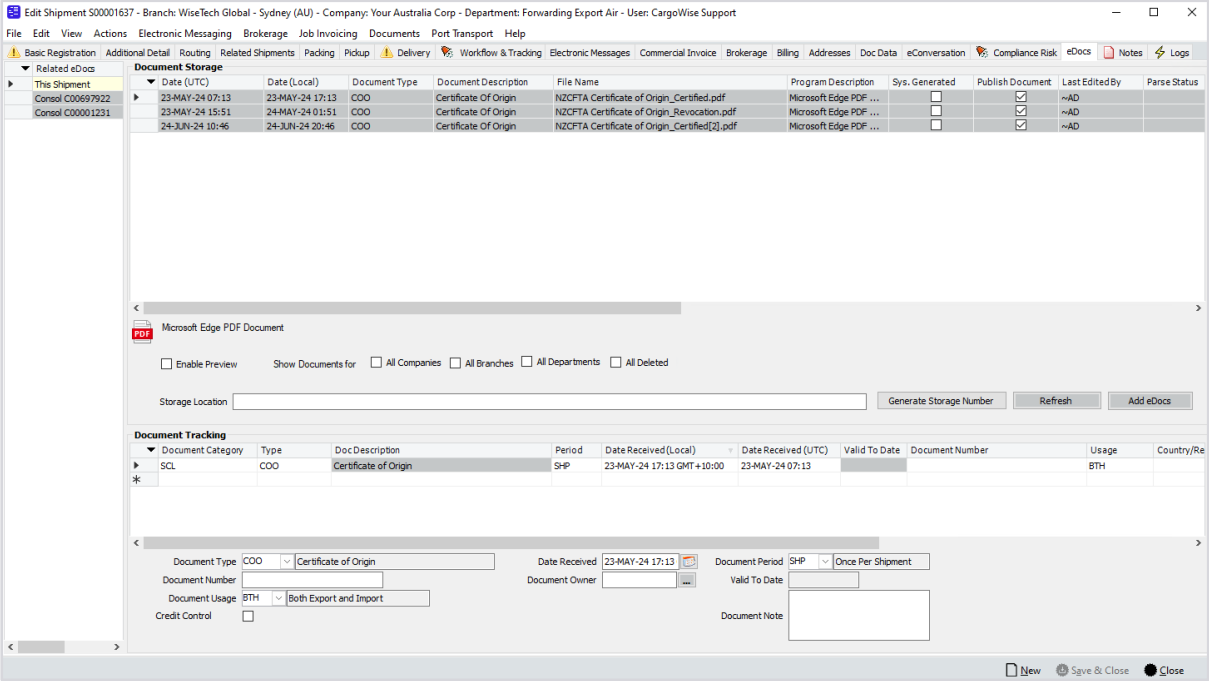
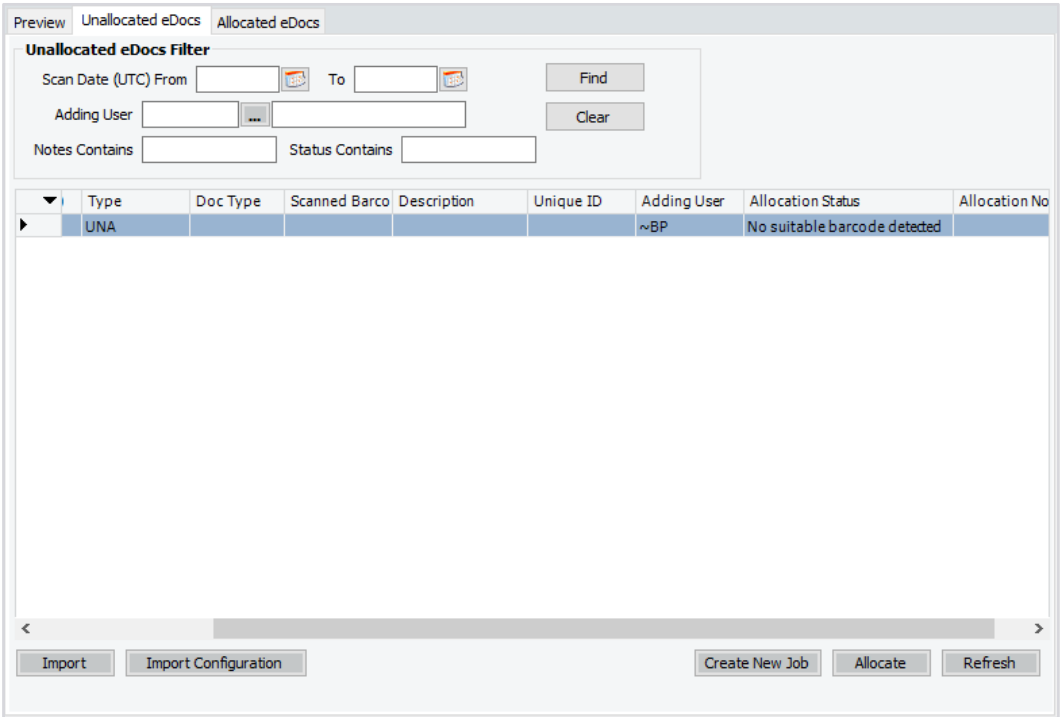
When a UniversalEvent message containing DataTarget Type as “DocManager” and attached documents are received, imported documents will appear in the eDocs tab of the matching job if Document Allocation details provided are matched.



If the Document Allocation does not matched, the document will appear in the Allocate eDocs module on the Unallocated eDocs tab. The Allocation Status column will contain the reason why the document could not be allocated. Once the allocation details are provided, the document can be allocated.



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8. Universal Query

8.1 Overview

Universal queries allow users to send Universal requests intended to query CW and return Universal Response XML. When a successful and well-formatted Universal Request is sent it is expected that the Universal response XML will contain either the desired XML object or a failure status and indication of the reason why the object could not be returned.

There are multiple Universal Queries which can be utilized currently including:

- Universal Shipment Query, and
- Universal Document Query.

8.2 Universal Shipment Query

To use the Universal Shipment Request component, you send a UniversalShipmentRequest XML message and expect a UniversalResponse XML message in return. The UniversalResponse will either contain the requested UniversalShipment or a failure status and a reason why a UniversalShipment could not be returned.

8.2.1 Universal Shipment Request

The UniversalShipmentRequest is designed to query for and return a single UniversalShipment matching the parameters enclosed in the message. Every module that supports exporting UniversalShipment XML works with the UniversalShipmentRequest interface.

Each 'well formed' UniversalShipmentRequest sent (opening and closing tags matching/present and no mandatory elements missing) will get a UniversalResponse XML back.

Following is a sample query for a Customs Declaration:

```
<UniversalShipmentRequest xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
  <ShipmentRequest>
    <DataContext>
      <DataTargetCollection>
        <DataTarget>
          <Type>CustomsDeclaration</Type>
          <Key>B00001000</Key>
        </DataTarget>
      </DataTargetCollection>

      <Company>
        <Code>BEN</Code>
      </Company>
      <EnterpriseID>HYE</EnterpriseID>
      <ServerID>BEN</ServerID>
    </DataContext>
  </ShipmentRequest>
</UniversalShipmentRequest>
```

The UniversalShipmentRequest content is used to Target a particular data source entity in a particular module which is then used to generate the Universal Shipment XML that is returned.

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8.2.1.1 Targeting a Data Source Entity using DataContext

The following content is used to target a particular data source entity:

- DataTarget Type, and
- Key

The DataTarget Type (CustomsDeclaration) determines which module you want to retrieve the data from. The DataTarget Type is mandatory. If you do not include it, you will get an Entity Not Found result no matter what else you include.

The DataTarget Key identifies the job or entity within the module you are querying. Each module has a designated Key field of fields that uniquely identify rows within that module, usually a job number like the Shipment, Consol, and Declaration Numbers.

The DataTarget Type and Key contain, the same values you see in the DataSource of a UniversalShipment or UniversalEvent exported from any module supporting the UniversalShipmentRequest.

EnterpriseID, ServerID and Company Code

Some modules like Customs are Company Specific in that each job is tied to a particular Company. For those modules, you will need to provide the EnterpriseID, ServerID, and Company Code elements to identify the Company you are querying from. If you do not provide this information where it is required, you will get an Entity Not Found response. Most Customs modules are Company specific which is also how they are linked to a given Country (as each company is tied to a country).

Where a module is not Company Specific (like Forwarding Consol or Forwarding Shipment), you can still send the EnterpriseID, ServerID, and Company Code where known. This will not affect which entity is returned, but it will affect the filtering of content that is Company Specific, like Job Costing or Billing information. If you are not using these sections of the returned XML, you do not need to provide these elements in the query.

You can tell if a module is Company Specific or not by logging into the same system under two different companies. If you can see the same set of jobs in a given module, it is not Company Specific.

Demo Companies can be used to send messages within CargoWise. However, when sending data externally Demo Companies cannot be used. For more information on Demo Companies refer to What is the purpose of the Demo Company?

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8.2.1.2 Querying by Reference and Party ID

At this point, only querying by DataTarget Key in the DataContext is implemented. In upcoming releases. The DataTarget Key will become optional, and queries by other references will be possible. This is planned to be done using the Reference and Party ID matching facility already available when importing Universal Shipment XML.

This works by following the premise that References issued by a given Party are globally unique where you can identify the issuing Party. This means that you can query by things like House Bill number where you can identify the Issuing (Sending) Forwarder, and query by Order Number where you can identify the Buyer.

Each module has a list of Reference and Party ID pairs that are known to be good identifiers in that module's context. Examples in use right now are:

Forwarding Consol

- Master Bill + Shipping Line
- Booking Reference + Shipping Line
- Agent's Reference + Sending Forwarder
- Agent's Reference + Receiving Forwarder

Forwarding Shipment

- House Bill + Sending Forwarder
- Interim Receipt + Consignor
- Booking Reference + Consignor
- Booking Reference + Consignee

Customs Declaration

- Owner's Reference + Consignor
- Owner's Reference + Consignee

This set will be added to as required where truly unique references can be identified for each module.

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8.2.2 Universal Shipment Response

Each 'well formed' UniversalShipmentRequest sent with opening and closing tags matching/present and no mandatory elements missing will get a UniversalResponse XML back. The UniversalResponse contains:

- Status - Either ERR for Error or PRS for Processed Ok
- ProcessingLog - Contains Errors and Warnings that were detected during processing of the sent query. Even for Processed Ok there may be warnings regarding processing the content of the incoming query XML. (e.g., Over length element values being truncated on import)
- Data - When the status is Processed Ok, contains the resulting UniversalShipment XML.

Following is a sample of what you would expect to see in the HTTP+XML Sample Client on a successful query:

```
Posting...
Begin POST to https://xxxtstservices.wisegrid.net/eAdaptor
        <<<------------------------------------------------ Begin Message Body ----------------------
---------------------------->>>
<UniversalShipmentRequest xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
  <ShipmentRequest>
    <DataContext>
      <DataTargetCollection>
        <DataTarget>
          <Type>ForwardingConsol</Type>
          <Key>C00001000</Key>
        </DataTarget>
      </DataTargetCollection>

      <Company>
        <Code></Code>
      </Company>
      <EnterpriseID></EnterpriseID>
      <ServerID></ServerID>
    </DataContext>
  </ShipmentRequest>
</UniversalShipmentRequest>
        <<<------------------------------------------------ End Message Body ----------------------
---------------------------->>>

Waiting Response...

Response Received, Status:- 200 - OK
        <<<------------------------------------------------ Be.g.in Response Body -------------------
------------------------------>>>
<?xml version="1.0" encoding="utf-8"?>
<UniversalResponse xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
  <Status>PRS</Status>
  <Data><UniversalShipment xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
  <Shipment>
    <DataContext>
      <DataSourceCollection>
        <DataSource>
          <Type>ForwardingConsol</Type>
          <Key>C00001000</Key>
        </DataSource>
      </DataSourceCollection>
... Content Removed for Brevity...
    </DataContext>
```


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```
    <AgentsReference></AgentsReference>
    <AWBServiceLevel>
      <Code>STD</Code>
      <Description>Standard</Description>
    </AWBServiceLevel>
    <BookingConfirmationReference></BookingConfirmationReference>
    <ChargeableRate>0.0000</ChargeableRate>
    <ContainerCount>1</ContainerCount>
... Content Removed for Brevity ...
        <TransportMode>Sea</TransportMode>
        <VesselLloydsIMO>9159646</VesselLloydsIMO>
        <VesselName>BUNGA TERATAI 3</VesselName>
        <VoyageFlightNo>343</VoyageFlightNo>
      </TransportLeg.>
    </TransportLegCollection>
  </Shipment>
</UniversalShipment>
</Data>
  <ProcessingLog />
</UniversalResponse>
        <<<------------------------------------------------- End Response Body ----------------------
----------------------------->>>

HTTP POST Complete.
Using the Universal Document Request
```

Each 'well formed' UniversalShipmentRequest sent (opening and closing tags matching/present and no mandatory elements missing) will get a UniversalResponse XML back.

A malformed request will return an HTTP '**400-Bad Request'** response with nothing in the body.

8.3 Universal Document Query

The Universal Document Query allows you to retrieve documents from jobs by sending a UniversalDocumentRequest XML. A UniversalResponse XML will be sent in return, containing the requested documents as base64 encoded binary content inside a Universal Event. In the case of failures to retrieve the documents a failure status and reason will be provided in place of the requested documents.

8.3.1 Universal Document Request

The UniversalDocumentRequest is designed to query for and return documents via a UniversalEvent containing documents matching the parameters and filters enclosed in the message. Every module that supports exporting UniversalEvent XML works with the UniversalDocumentRequest interface.

Each 'well formed' UniversalDocumentRequest sent (opening and closing tags matching/present and no mandatory elements missing) will get a UniversalResponse XML back.

The UniversalDocumentRequest has two main sections, the DataContext and the FilterCollection. Information from the DataContext is used to Target a particular data source entity or job in a particular module. The eDocs against that job is then queried using information in the FilterCollection and any eDocs matching all of the supplied Filters are returned in the resulting Universal Event XML.

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Following is a sample query/request for all documents with a document type of ARN from Customs Declaration B00001000:

```
<UniversalDocumentRequest xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
  <DocumentRequest>
    <DataContext>
      <DataTargetCollection>
        <DataTarget>
          <Type>CustomsDeclaration</Type>
          <Key>B00001000</Key>
        </DataTarget>
      </DataTargetCollection>
      <Company>
        <Code>CWI</Code>
      </Company>
      <EnterpriseID>WTG</EnterpriseID>
      <ServerID>SYD</ServerID>
    </DataContext>
    <ReturnDocumentDescriptionsOnly>true</ReturnDocumentDescriptionsOnly>
    <FilterCollection>
      <Filter>
        <Type>DocumentType</Type>
        <Value>ARN</Value>
      </Filter>
    </FilterCollection>
  </DocumentRequest>
</UniversalDocumentRequest>
```

8.3.1.1 Document Retrieval

Documents can be retrieved from eDocs across various modules using a new UniversalEvent type called 'DocManager'.

It is possible to access documents from the eDocs tab across different modules. The new solution uses a type of 'DocManager' and a key in the format 'Module code Job number'.

For more information, refer to Enhancements for eDocs document retrieval using UniversalDocumentQuery.

8.3.1.2 Targeting a Data Source Entity using DataContext

The following content is used to target a particular data source entity:

- **DataTarget Type:** The DataTarget Type is used to determine which module you want to retrieve the data from. The DataTarget Type is mandatory. If you do not include it, you will get an Entity Not Found result no matter what else you include.
- **DataTarget Key:** The DataTarget Key is used to identify the job or entity within the module you are querying. Each module has a designated Key field of fields that uniquely identify rows within that module, usually a job number like the Shipment, Consol and Declaration Numbers.

The DataTarget Type and Key contains the same values you see in the DataSource of a UniversalShipment or UniversalEvent exported from any module supporting the UniversalShipmentQuery.

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An example of these two fields can be seen in the below extract:

```
<DataTargetCollection>
  <DataTarget>
    <Type>CustomsDeclaration</Type>
    <Key>B00001000</Key>
  </DataTarget>
</DataTargetCollection>
```

- EnterpriseID, ServerID and Company Code

Some modules like Customs are 'Company Specific' in that each job is tied to a particular Company. For those modules, you will need to provide the EnterpriseID, ServerID and Company Code elements to identify the Company you are querying from. If you do not provide this information where it is required, you will get an Entity Not Found response. Most Customs modules are Company specific which is also how they are linked to a given Country. (As each company is tied to a country)

Where a module is not 'Company Specific' (like Forwarding Consol or Forwarding Shipment), you can still send the EnterpriseID, ServerID and Company Code where known. This will not affect which entity is returned, but it will affect filtering of content that is Company Specific like Job Costing or Billing information. If you are not using these sections of the returned XML, you do not need to provide these elements in the query.

You can tell if a module is 'Company Specific' or not by logging into the same system under two different companies. If you can see the same set of jobs in a given module, it is not Company Specific.

8.3.1.3 Selecting eDocs using the FilterCollection

The FilterCollection is an optional section that you use when you want to return only some of the eDocs present on a given Job/Entity. If the FilterCollection is not included or contains no elements, all eDocs from the matching parent Job/Entity will be returned.

Each Filter element contains two sub elements, Type to indicate what type of filter to and Value to provide the value that filter is going to use. Where a Filter is not included for any given type, the default behavior is to return all eDocs. This means that for a filter like CompanyCode, default behavior when this type is not specified is to return all documents regardless of company affiliation.

Where multiple filter elements are provided with the same Type, these are treated as an 'Or' style filter so any document matching any of the values presented will be returned. Filter elements with different Types are treated as an 'And' style filter where all values must match on eDocs being returned.

Some filters only work with one per request, these are noted in the definitions following.

Filter types available are:

- **DocumentType** – Retrieve only documents matching the specified document type.
- **IsPublished** – Retrieve only published or un-published documents. Possible values for this filter are: True and False. This can only be specified once.

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- **SaveDateUTCFrom** – Retrieve only documents that were added or modified on or after the specified date/time (provided in UTC time). This can only be specified once.
- **SaveDateUTCTo** – Retrieve only documents that were added or modified on or before the specified date/time (provided in UTC time). This can only be specified once.
- **CompanyCode** – Retrieve only documents related to the specified company or non-company specific. The default behavior without this Type being filtered is to return all documents regardless of company affiliation.
- **BranchCode** – Retrieve only documents related to the specified branch code.
- **DepartmentCode** – Retrieve only documents relevant to specified department code.
- **FileName** – Retrieve only document matching with the specified File Name.
- **DocumentID** – Retrieve only document matching with specified DocumentID (Primary Key – GUID).
- **Related eDocs** – Retrieve documents attached any Related eDocs which are specified. For example, it will return the eDocs under the related CustomsEntry of a Declaration in addition to the eDocs in the Declaration itself.

Refer to Documents attached on Related eDocs can now be retrieved via a UniversalDocumentRequest for more information about how to use the related eDocs filter.

Additionally, list of eDocs files can be fetched using below request element for all (or filtered) eDocs on a job without bulky content.

<ReturnDocumentDescriptionsOnly>true</ReturnDocumentDescriptionsOnly>

DocumentID will be now provided with response message for UniversalDocumentRequest and as well from Document related Events (i.e., DDI, DDR, DDD, etc)

Sample Filters

- Using filter: FileName

```
<FilterCollection>
   <Filter>
     <Type>FileName</Type>
     <Value>sample file.docx</Value>
   </Filter>
</FilterCollection>
```

- Using filter: DocumentID

```
<FilterCollection>
   <Filter>
     <Type>DocumentID</Type>
     <Value> BDF6A934-304B-49B6-B9C6-A88E91FF093A</Value>
   </Filter>
</FilterCollection>
```


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- Using filter: Related EDoc

```
<FilterCollection>
   <Filter>
     <Type>RelatedEDoc</Type>
     <Value>Customs Entry</Value>
   </Filter>
</FilterCollection>
```

Sample Request XML

```
<UniversalDocumentRequest xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
  <DocumentRequest>
    <DataContext>
      <DataTargetCollection>
        <DataTarget>
          <Type>CustomsDeclaration</Type>
          <Key>B00001000</Key>
        </DataTarget>
      </DataTargetCollection>

      <Company>
        <Code>DAU</Code>
      </Company>
      <EnterpriseID>EDI</EnterpriseID>
      <ServerID>82A</ServerID>
    </DataContext>

    <ReturnDocumentDescriptionsOnly>true</ReturnDocumentDescriptionsOnly>

    <FilterCollection>
      <Filter>
        <Type>DocumentID</Type>
        <Value>BDF6A934-304B-49B6-B9C6-A88E91FF093A</Value>
      </Filter>
    </FilterCollection>
  </DocumentRequest>
</UniversalDocumentRequest>
```

8.3.1.4 Using Multiple Filter Elements

Where multiple filter elements are provided with the same Type, these are treated as an 'Or' style filter so any document matching any of the values presented will be returned. Filter elements with different Types are treated as an 'And' style filter where all values must match on eDocs being returned.

Example filters – multiple filters in one filter collection: Only return document type is 'ARN' And file name is 'sample file.docx' Or 'sample file 2.docx'

```
<FilterCollection>
   <Filter>
     <Type>FileName</Type>
     <Value>sample file.docx</Value>
   </Filter>
   <Filter>
     <Type>FileName</Type>
     <Value>sample file 2.docx</Value>
   </Filter>
   <Filter>
     <Type>DocumentType</Type>
     <Value>ARN</Value>
```


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```
    </Filter>
  </F1ilterCollection>
```

8.3.2 Universal Response

Each 'well formed' UniversalDocumentRequest sent with opening and closing tags matching/present and no mandatory elements missing will get a UniversalResponse XML back. The UniversalResponse contains:

- **Status** – Either ERR for Error or PRS for Processed Ok
- **ProcessingLog** – Contains Errors and Warnings that were detected during processing of the sent query. Even for Processed Ok there may be warnings regarding processing the content of the incoming query XML (e.g., Over length element values being truncated on import)
- **Data** – When the status is Processed Ok, contains the resulting UniversalEvent XML

Following is a sample of what would come back from the sample UniversalDocumentRequest shown earlier in this section assuming there was a matching Customs Declaration, and that entity/job had an eDoc attached with a Document Type of ARN:

```
<UniversalResponse xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
  <Status>PRS</Status>
  <ProcessingLog />
  <Data>
    <UniversalEvent xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
      <Event>
        <DataContext>
          <DataSourceCollection>
            <DataSource>
              <Type>CustomsDeclaration</Type>
              <Key>B00001000</Key>
            </DataSource>
          </DataSourceCollection>
          <Company>
            <Code>CWI</Code>
          </Company>
          <EnterpriseID>WTG</EnterpriseID>
          <ServerID>SYD</ServerID>
        </DataContext>
        <EventTime>2016-06-08T09:48:13</EventTime>
        <EventType>DEX</EventType>
        <IsEstimate>false</IsEstimate>
        <ContextCollection>
          <Context>
            <Type>HBOLNumber</Type>
            <Value>HB432178972345</Value>
          </Context>
        </ContextCollection>
        <AttachedDocumentCollection>
          <AttachedDocument>
            <FileName>ArrivalNotice.pdf</FileName>
            <DocumentID>99129579-eb3f-4756-a281-cea4b7b493f3</DocumentID>
            <FileSizeInBytes>957</FileSizeInBytes>
            <ImageData>JVBERi0xLjQKJ --- base64 encoded binary stream --- kKJSVFT0YK</ImageData>
            <Type>
              <Code>ARN</Code>
              <Description>Arrival Notice</Description>
            </Type>
```


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```
            <IsPublished>true</IsPublished>
            <SaveDateUTC>2022-12-07T03:11:00</SaveDateUTC>
            <SavedBy>
              <Code>WTG</Code>
            <Name>CargoWise Support</Name>
            </SavedBy>
            <Source>
              <Code></Code>
              <Description></Description>
            </Source>
            <VisibleBranchCode></VisibleBranchCode>
            <VisibleCompanyCode></VisibleCompanyCode>
            <VisibleDepartmentCode></VisibleDepartmentCode>
          </AttachedDocument>
        </AttachedDocumentCollection>
      </Event>
    </UniversalEvent>
  </Data>
</UniversalResponse>
```

The `<AttachedDocument>` contains:

- **FileName** – Name of this document File.
- **DocumentID** – Primary Key of this document – GUID.
- **FileSizeInBytes** - File Size in Bytes of this document.
- **ImageData** - Document contains as a base64 encoded binary stream. The sample above was truncated for brevity.
- **Type** - Document type
- **IsPublished** – "True" if published or "False" – for an un-published documents.
- **SaveDateUTC** – UTC time when this document gets saved in CW
- **SavedBy** – User `<Code>` and `<Name>` who saved this document.
- **Source** – The source of the Document `<Code>` and `<Description>`
- **VisibleBranchCode / VisibleCompanyCode / VisibleDepartmentCode** - Return the Branch / Company/Department code if the visibility is set to specific Branch / Company / Department.

> The `<ImageData>` element will not be returned if `<ReturnDocumentDescriptionsOnly>` filter is set to "True"

If there is a failure matching to a parent entity/job using information from the requested DataContext, or no eDocs are found matching the filters specified, you will get a UniversalResponse back with a status of ERR for Error and a reason in the Processing Log like so:

```
<UniversalResponse xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
  <Status>ERR</Status>
  <ProcessingLog>
    CustomsDeclaration B00001000 was found, but no eDocs were found matching the filters specified.
  </ProcessingLog>
  <Data />
</UniversalResponse>
```


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A malformed request will return an **HTTP '400-Bad Request'** response with nothing in the body.

8.3.3 Using DocManager

Additionally, it is possible to retrieve documents from the eDocs tab across modules using <DataTarget> as 'DocManager' and <DataTarget> key as 'Module code Job number'.

> For more information on <DataTarget> type of 'DocManager' refer to **Enhancement for eDocs document retrieval using UniversalDocumentQuery**

Below is a sample query/request for documents in the Product Warehouse module using the <DataTarget> type of 'DocManager':

```
<UniversalDocumentRequest xmlns="http://www.cargowise.com/Schemas/Universal/2011/11" version="1.1">
  <DocumentRequest>
    <DataContext>
      <DataTargetCollection>
        <DataTarget>
          <Type>DocManager</Type>
          <Key>WTD W000027944</Key>
        </DataTarget>
      </DataTargetCollection>
      <Company>
        <Code>CWI</Code>
      </Company>
      <EnterpriseID>WTG</EnterpriseID>
      <ServerID>SYD</ServerID>
    </DataContext>
  </DocumentRequest>
</UniversalDocumentRequest>
```

9. Universal Activities

9.1 Overview

The modules making up the PAVE productivity tools built into CargoWise integrate their operational data using the Universal Activity XML schema. These include the Customer Service Ticket, Project and Work Item modules.

Just like the other Universal XML schemas, one schema is used across all PAVE modules making it possible to transfer data across modules and systems seamlessly.

9.2 Module Mapping Guide

For further information on the Universal Activity schema, you can find the Universal Activity Module Mapping Guide in our myaccount web service portal under **CargoWise > Technical Guides > Integration Options > eAdaptor**. This gives an overview of how the schema is mapped to all modules implemented and provides a list of element level mappings describing each field's location in the GUI.

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10. Universal Transactions

If you are trying to interface using posted accounting transactions, then you can use the Universal Transaction XML. As is common across all forms of Universal XML, there is only one schema for all transaction types supported. AR and AP invoices use the same schema as Credit Notes, Payments and Journals.

This is done by having all elements for all supported transaction types present in the one schema. Where values are common across multiple transaction types like Posted Date, Local Total and Currency, the same elements are used. Each transaction type uses elements relevant to that transaction type and ignores those not relevant. As an example, you might have a Check Number on a Payment, but not on an AR Invoice.

For unposted job related costs and revenue, you should use the JobCosting element in the Universal Shipment XML mentioned in section 4.3 Job Costing (Billing) and Consol Costs.

11. Application Codes/Sub Types

The two Application Codes for Universal XML are **Universal Data Messaging** (UDM) and **Universal Data Query** (UDQ).

A list of the message Sub Types for Universal XML are as mentioned below:

Universal XML	Sub Type
XmlUniversalActivity	XUA
XmlUniversalActivityRequest	XAR
XmlUniversalDocumentRequest	XDR
XmlUniversalEvent	XUE
XmlUniversalInterchangeRequeueRequest	XIR
XmlUniversalResponse	XUR
XmlUniversalSchedule	XSC

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Universal XML	Sub Type
XmlUniversalShipment	XUS
XmlUniversalShipmentRequest	XSR
XmlUniversalTransaction	XUT
XmlUniversalTransactionBatch	XUB
XmlUniversalTransactionBatchRequest	XBR

Support: Press the **F1** key from anywhere within CargoWise to raise an eRequest

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